Special DPP on Indefinite Integration **DPP # 1**

1. $\int 2^x.e^x \ dx =$

Ans. $\frac{2^x \cdot e^x}{1+\ell n2} + C$

Sol. $\int 2^x \cdot e^x dx = \int (2e)^x \, dx = \frac{(2e)^x}{\ell n(2e)} + C$

$\frac{2^x \cdot e^x}{1+\ell n2} + C$

2. $\int \frac{1+\cos^2 x}{1+\cos 2x} \, dx$

Ans. $\frac{1}{2}(\tan x + x) + C$

Sol. $\int \frac{1+\cos^2 x}{1+\cos 2x} dx = \int \frac{1+\cos^2 x}{2\cos^2 x} dx = \frac{1}{2}\int \sec^2 x dx + \frac{1}{2}\int dx$

$= \frac{1}{2}(\tan x + x) + C$

3. $\int \frac{1-\tan^2 x}{1+\tan^2 x} \, dx$

Ans. $\frac{1}{2}\sin 2x + C$

Sol. $\int \frac{1-\tan^2 x}{1+\tan^2 x} dx = \int \cos 2x dx = \frac{\sin 2x}{2} + C$

4. $\int \frac{1+\tan^2 x}{1+\cot^2 x} \, dx$

Ans. $\tan x - x + C$

Sol. $\int \frac{1+\tan^2 x}{1+\cot^2 x} dx = \int \tan^2 x dx = \tan x - x + C$

5. $\int \frac{e^{5\ell n x} - e^{4\ell n x}}{e^{3\ell n x} - e^{2\ell n x}}\, dx$

Ans. $\frac{x^3}{3} + C$

Sol. $\int \frac{e^{5\ell nx} - e^{4\ell nx}}{e^{3\ell nx} - e^{2\ell nx}} dx = \int \frac{x^5 - x^4}{x^3 - x^2} dx$

$\int x^2 dx = \frac{x^3}{3} + C$

6. $\int (e^{a\ell n x} + e^{x\ell n a}) dx \ (a > 0)$

Ans. $\frac{x^{a+1}}{a+1} + \frac{a^x}{\ell n a} + c$

Sol. $\int \left(e^{a\ell nx} + e^{x\ell na}\right) dx \ ; \ a > 0 = \int \left(x^a + a^x\right) dx$

$= \frac{x^{a+1}}{a+1} + \frac{a^x}{\ell na} + C$

7. $\int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx$

Ans. $-(\cot x + \tan x) + C$

Sol. $\int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx = \int \left(\cos ec^2 x - \sec^2 x\right) dx$

$= -\cot x - \tan x + C$

$= -(\cot x + \tan x) + C$

8. $\int \frac{1 + 2x^2}{x^2\left(1 + x^2\right)} dx$

Ans. $-\frac{1}{x} + \tan^{-1} x + C$

Sol. $\int \frac{1+2x^2}{x^2\left(1+x^2\right)} dx = \int \frac{\left(1+x^2+x^2\right)}{x^2\left(1+x^2\right)} dx$

$= \int \frac{1}{x^2} dx + \int \frac{1}{1+x^2} dx$

$= -\frac{1}{x} + \tan^{-1} x + C$

9. $\int \frac{\cos x - \sin x}{\cos x + \sin x} (2 + 2\sin 2x)\, dx$

Ans. $\sin 2x + C$

Sol. $\int \frac{\cos x - \sin x}{\cos x + \sin x} 2(1 + \sin 2x)\, dx$

Let $\cos x + \sin x = t$

$\Rightarrow (-\sin x + \cos x)dx = dt$

$\int \frac{2(1 + t^2 - 1)}{t} dt = t^2 + C$

$= 1 + \sin 2x + C$

$= \sin 2x + C'$

10. $\int (3\sin x \cos^2 x - \sin^3 x)\, dx$

Ans. $-\frac{\cos 3x}{3} + C$

Sol. $\int (3\sin x \cos^2 x - \sin^3 x) dx$

$= \int (3\sin x - 4\sin^3 x) dx$

$= \int \sin 3x dx = \frac{-\cos 3x}{3} + C$

11. $\int \cos x^\circ\, dx$

Ans. $\frac{180}{\pi} \sin x^\circ + C$

Sol. $\int \cos x^\circ dx = \int \cos\left(\frac{\pi x}{180}\right) dx$ $\quad \{\because 180^\circ = \pi^R\}$

$= \frac{180}{\pi} \sin\left(\frac{\pi x}{180}\right) + C = \frac{180}{\pi} \sin x^\circ + C$

12. $\int \frac{(1 + x)^2}{x(1 + x^2)} dx$

Ans. $\ln x + 2\tan^{-1} x + C$

Sol. $\int \frac{(1+x)^2}{x\left(1+x^2\right)}dx = \int \frac{1+x^2+2x}{x\left(1+x^2\right)}dx = \int \frac{dx}{x} + \int \frac{2}{1+x^2}dx$

$= \ell n|x| + 2\tan^{-1}(x) + C$

13. $\int \frac{x}{2x+1}\,dx$

Ans. $\frac{1}{2}\left[x - \frac{\ell n(2x+1)}{2}\right] + C$

Sol. $\int \frac{x}{2x+1}dx = \frac{1}{2}\int \frac{(2x+1-1)}{2x+1}dx$

$= \frac{1}{2}\left[x - \frac{1}{2}\ell n|2x+1|\right] + C$

14. $\int \frac{\sec 2x - 1}{\sec 2x + 1}\,dx$

Ans. $\tan x - x + C$

Sol. $\int \frac{\sec 2x - 1}{\sec 2x + 1}dx = \int \tan^2 x dx = \tan x - x + C$

15. $\int \frac{2x - 1}{x - 2}\,dx$

Ans. $2x + 3\,\ell n\,(x - 2) + C$

Sol. $\int \frac{2x-1}{x-2}dx = \int\left(\frac{2x-4+3}{x-2}\right)dx = 2\int dx + 3\int \frac{dx}{x-2}$

$= 2x + 3\,\ell n\,|x - 2| + C$

16. $\int \frac{e^{2x} - 1}{e^x}\,dx$

Ans. $e^x + e^{-x} + C$

Sol. $\int \frac{e^{2x}-1}{e^x}dx = \int e^x dx - \int e^{-x}dx$

$= e^x + e^{-x} + C$

17. $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}}\, dx\ (\cos x + \sin x > 0)$

Ans. $x + C$

Sol. $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx \qquad (\sin x + \cos x > 0)$

$= \int \frac{\sin x + \cos x}{|\sin x + \cos x|} dx = \int dx = x + C$

18. $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha}\, dx$

Ans. $2(\sin x + x \cos \alpha) + C$

Sol. $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx = 2\int (\cos x + \cos \alpha)\, dx$

$= 2(\sin x + x \cos \alpha) + C$

19. $\int \frac{x^6 - 1}{x^2 + 1}\, dx$

Ans. $\frac{x^5}{5} - \frac{x^3}{3} + x - 2\tan^{-1} x + C$

Sol. $\int \frac{x^6 - 1}{x^2 + 1} dx = \int \frac{\left(x^2 + 1\right)\left(x^4 - x^2 + 1 - \frac{2}{1 + x^2}\right)}{\left(1 + x^2\right)} dx$

$= \int \left(x^4 - x^2 + 1 - \frac{2}{1 + x^2}\right) dx = \frac{x^5}{5} - \frac{x^3}{3} + x - 2\tan^{-1} x + C$

20. $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx$

Ans. $\sec x - \operatorname{cosec} x + C$

Sol. $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx = \int (\sec x \tan x + \operatorname{cosec} x \cot x)\, dx$

$= \sec x - \operatorname{cosec} x + C$

21. $\int \frac{x^4 + x^2 + 1}{2\left(1 + x^2\right)}\, dx$

Ans. $\frac{1}{2}\left[\frac{x^3}{3}+\tan^{-1}x\right]+C$

Sol. $\int\frac{x^4+x^2+1}{2\left(1+x^2\right)}dx=\int\frac{\left(x^2+1\right)^2-x^2}{2\left(1+x^2\right)}dx$

$=\frac{1}{2}\int\left(x^2+1\right)dx-\frac{1}{2}\int\frac{x^2}{1+x^2}dx$

$=\frac{1}{2}\left[\frac{x^3}{3}+\tan^{-1}x\right]+C$

22. $\int\sqrt{1-\sin 2x}\,dx$

Ans. $(\sin x+\cos x)\,\text{sgn}\,(\cos x-\sin x)+C$

Sol. $\int\sqrt{1-\sin 2x}dx=\int\left|\sin x-\cos x\right|dx$

$$=\begin{cases}\int\left(\sin x-\cos x\right)dx & \text{if} \quad \sin x-\cos x>0\\ \int 0dx & \text{if} \quad \sin x-\cos x=0\\ \int\left(\cos x-\sin x\right)dx & \text{if} \quad \sin x-\cos x<0\end{cases}$$

$$=\begin{cases}-\left(\cos x+\sin x\right)+C & \text{if} \quad \cos x-\sin x<0\\ 0+C & \text{if} \quad \cos x=\sin x\\ \left(\sin x+\cos x\right) & \text{if} \quad \cos x>\sin x\end{cases}$$

$=(\sin x+\cos x)\,\text{sgn}\,(\cos x-\sin x)+C$

23. $\int\frac{\sin^6 x+\cos^6 x}{\sin^2 x\,.\,\cos^2 x}dx$

Ans. $\tan x-\cot x-3x+C$

Sol. $\int\frac{\sin^6 x+\cos^6 x}{\sin^2 x\cos^2 x}dx=\int\left(\frac{1-3\sin^2 x\cos^2 x}{\sin^2 x\cos^2 x}\right)dx$

$=\int\left(\text{cosec}^2x+\sec^2 x-3\right)dx$

$=\tan x-\cot x-3x+C$

24. $\int \frac{\left(\sqrt{x}+1\right)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}}dx$

Ans. $\frac{x^2}{2}-x+C$

Sol. $\int \frac{\left(\sqrt{x}+1\right)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}}dx$

$=\int \frac{(x-1)\left(x\sqrt{x}+x+\sqrt{x}\right)}{\left(x\sqrt{x}+x+\sqrt{x}\right)}dx=\frac{x^2}{2}-x+C$

25. $\int \left[\sin^2\left(\frac{9\pi}{8}+\frac{x}{4}\right)-\sin^2\left(\frac{7\pi}{8}+\frac{x}{4}\right)\right]dx$

Ans. $-\sqrt{2}\cos\frac{x}{2}+C$

Sol. $\int\left[\sin^2\left(\frac{9\pi}{8}+\frac{x}{4}\right)-\sin^2\left(\frac{7\pi}{8}+\frac{x}{4}\right)\right]dx$

$=\int \sin\left(2\pi+\frac{x}{2}\right)\cdot\sin\left(\frac{\pi}{4}\right)dx$

$=\frac{1}{\sqrt{2}}\int \sin\left(\frac{x}{2}\right)dx=-\frac{1}{\sqrt{2}}\frac{\cos\left(\frac{x}{2}\right)}{\frac{1}{2}}+C$

$=-\sqrt{2}\cos\left(\frac{x}{2}\right)+C$

26. $\int \frac{\cos 4x+1}{\cot x-\tan x}\,dx$

Ans. $-\frac{\cos 4x}{8}+C$

Sol. $\int \frac{\cos 4x+1}{\cot x-\tan x}dx=\int \frac{2\cos^2 2x}{2\cot 2x}dx$

$=\frac{1}{2}\int \sin 4x dx=-\frac{1}{8}\cos 4x+C$

27. $\int \left[\sin\alpha \, \sin(x-\alpha) + \sin^2\left(\frac{x}{2} - \alpha\right)\right] dx$

Ans. $\frac{1}{2}(x - \sin x) + C$

Sol. $\int\left(\sin\alpha \sin(x-\alpha) + \sin^2\left(\frac{x}{2}-\alpha\right)\right)dx$

$= \frac{1}{2}\int\left(2\sin\alpha \sin(x-\alpha) + 2\sin^2\left(\frac{x}{2}-\alpha\right)\right)dx$

$= \frac{1}{2}\int\left(-\cos x + \cos(2\alpha - x) + 1 - \cos(x - 2\alpha)\right)dx$

$= \frac{1}{2}\int(1-\cos x)\,dx = \frac{1}{2}(x - \sin x) + C$

1. $\int \frac{\sin 2x + \sin 5x - \sin 3x}{\cos x + 1 - 2\sin^2 2x}\, dx$

Ans. $-2\cos x + C$

Sol. $\int \frac{\sin 2x + \sin 5x - \sin 3x}{\cos x + 1 - 2\sin^2 2x} dx$

$$= \int \frac{\sin 5x + 2\cos\left(\frac{5x}{2}\right)\sin\left(\frac{-x}{2}\right)}{\cos x + \cos 4x} dx$$

$$= \int \frac{2\cos\frac{5x}{2}\left(\sin\frac{5x}{2} - \sin\frac{x}{2}\right)}{2\cos\frac{5x}{2}\cos\frac{3x}{2}} dx$$

$$= \int \frac{2\cos\left(\frac{3x}{2}\right)\sin(x)}{\cos\left(\frac{3x}{2}\right)} dx = -2\cos x + C$$

2. $\int \frac{\cos^4 x - \sin^4 x}{\sqrt{1 + \cos 4x}}\, dx \ (\cos 2x > 0)$

Ans. $\frac{x}{\sqrt{2}} + C$

Sol. $\int \frac{\cos^4 x - \sin^4 x}{\sqrt{1+\cos 4x}} dx$ $(\cos 2x > 0)$

$$= \int \frac{\cos 2x}{\sqrt{2}\cos 2x} dx = \frac{1}{\sqrt{2}}x + C = \frac{x}{\sqrt{2}} + C$$

3. $\int \frac{2x^3 + 3x^2 + 4x + 5}{2x + 1}\, dx$

Ans. $\frac{x^3}{3} + \frac{x^2}{2} + \frac{3x}{2} + \frac{7}{4}\ln(2x+1) + C$

Sol. $\int \frac{2x^3+3x^2+4x+5}{(2x+1)}dx$

$$= \int \frac{(2x+1)\left(x^2+x+\frac{3}{2}\right)+\frac{7}{2}}{(2x+1)}dx$$

$$= \frac{x^3}{3}+\frac{x^2}{2}+\frac{3x}{2}+\frac{7}{4}\ell n|2x+1|+C$$

4. $\int \frac{dx}{\sqrt{9-16x^2}}$

Ans. $\frac{1}{4}\sin^{-1}\frac{4}{3}x+C$

Sol. $\int \frac{dx}{\sqrt{9-16x^2}} = \int \frac{dx}{\sqrt{(3)^2-(4x)^2}}$

let $4x = t$

$\Rightarrow dx = \frac{dt}{4}$

$$= \frac{1}{4}\int \frac{dt}{\sqrt{9-t^2}} = \frac{1}{4}\sin^{-1}\left(\frac{4x}{3}\right)+C$$

5. $\int \frac{dx}{25+4x^2}$

Ans. $\frac{1}{10}\tan^{-1}\frac{2x}{5}+C$

Sol. $\int \frac{dx}{25+4x^2} = \int \frac{dx}{5^2+(2x)^2} = \frac{1}{10}\tan^{-1}\left(\frac{2x}{5}\right)+C$

6. $\int \tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)dx$

Ans. $\frac{\pi x}{2}-\frac{x^2}{4}+C$

Sol. $\int \tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)dx = \int \tan^{-1}\left(\cot\left(\frac{x}{2}\right)\right)dx$

$$= \int \left(\frac{\pi}{2} - \frac{x}{2} \right) dx = \frac{\pi}{2} x - \frac{x^2}{4} + C$$

7. $\int \frac{dx}{1 + \sin x} =$

Ans. $\tan x - \sec x + C$

Sol. $\int \frac{dx}{1+\sin x} = \int \frac{(1-\sin x)}{\cos^2 x} dx$

$= \int \left(\sec^2 x - \sec x \tan x \right) dx$

$= \tan x - \sec x + C$

8. $\int \frac{2 + 3x^2}{x^2 \left(1 + x^2\right)} dx =$

Ans. $\frac{2}{x} + \tan^{-1} x + C$

Sol. $\int \frac{2+3x^2}{x^2\left(1+x^2\right)} dx = \int \frac{\left(2+2x^2+x^2\right)}{x^2\left(1+x^2\right)} dx$

$= \int \frac{2}{x^2} dx + \int \frac{dx}{1+x^2} = -\frac{2}{x} + \tan^{-1} x + C$

9. A function g defined for all positive real numbers, satisfies $g'(x^2) = x^3$ for all $x>0$ and $g(1) = 1$. Compute $g(4)$.

एक फलन g जो कि सभी धनात्मक वास्तविक संख्याओं के लिए परिभाषित है, $g'(x^2) = x^3$, सभी $x>0$ तथा $g(1) = 1$ को सन्तुष्ट करता है। $g(4)$ का मान ज्ञात कीजिए।

Ans. $\frac{67}{5}$

Sol. $g'\left(x^2\right) = x^3, \forall x > 0 \ \& \ g(1) = 1$

$\Rightarrow g'(t) = t^{\frac{3}{2}}$

$\Rightarrow g(t) = \frac{2}{5} t^{\frac{5}{2}} + C$

$\therefore g(1) = 1 \Rightarrow C = +\frac{3}{5}$

$$\Rightarrow g(t)=\frac{2}{5}t^{\frac{5}{2}}+\frac{3}{5}$$

$$\Rightarrow g(4)=\frac{64}{5}+\frac{3}{5}=\frac{67}{5}$$

10. $\int\left[\sin(5x+2)+e^{\ell n(2x)^{1/2}}+2^{(x/2)}\right]$

Ans. $\frac{-\cos(5x+2)}{5}+\frac{(2x)^{3/2}}{3}+\frac{2.2^{x/2}}{\ell n2}+c$

Sol. $\int\sin(5x+2)dx+\int e^{\ell n(2x)^{\frac{1}{2}}}dx+\int 2^{\frac{x}{2}}dx$

$$=-\frac{\cos(5x+2)}{5}+\frac{(2x)^{\frac{3}{2}}}{2}\cdot\frac{1}{2}+\frac{2\cdot 2^{\frac{x}{2}}}{\ell n2}+C$$

$$=-\frac{1}{5}(\cos(5x+2))+\frac{(2x)^{\frac{3}{2}}}{3}+\frac{2\cdot 2^{\frac{x}{2}}}{\ell n2}+C$$

11. $\int 4\sin x\cos\frac{x}{2}\cos\frac{3x}{2}dx$

Ans. $\cos x-\frac{1}{2}\cos 2x-\frac{1}{3}\cos 3x+c$

Sol. $\int 4\sin x\cos\frac{x}{2}\cos\frac{3x}{2}dx$

$$=2\int\sin x(\cos 2x+\cos x)dx$$

$$=\int\sin 2xdx+2\int\sin x\cos 2xdx$$

$$=-\frac{\cos 2x}{2}+\int(\sin 3x-\sin x)dx$$

$$=-\frac{\cos 2x}{2}-\frac{\cos 3x}{3}+\cos x+C$$

12. $\int\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+.....\right)dx$

Ans. e^x+c

Sol. $\int\left(1+x+\frac{x^2}{\lfloor 2}+\frac{x^3}{\lfloor 3}+.....\right)dx$

$=\int e^x dx = e^x + C$

13. $\int\left[\frac{\cot^2 2x - 1}{2\cot 2x} - \cos 8x \cot 4x\right] dx$

Ans. $-\frac{\cos 8x}{8} + C$

Sol. $\int\left(\frac{\cot^2 2x-1}{2\cot 2x}-\cos 8x\cot 4x\right)dx$

$=\int\left(\frac{\cos^2 2x-\sin^2 2x}{2\sin 2x\cos 2x}-\frac{\cos 8x\cos 4x}{\sin 4x}\right)dx$

$=\int\frac{\cos 4x(1-\cos 8x)}{\sin 4x}dx=\int 2\sin 4x\cos 4xdx$

$=\int\sin 8xdx=-\frac{\cos 8x}{8}+C$

14. $\int \frac{\left(x^2 + \sin^2 x\right)\sec^2 x}{1 + x^2}\, dx$

Ans. $\tan x - \tan^{-1} x + c$

Sol. $\int\frac{\left(x^2+\sin^2 x\right)\sec^2 x}{1+x^2}dx$

$=\int\sec^2 xdx-\int\frac{\cos^2 x\cdot\sec^2 x}{1+x^2}dx$

$=\tan x-\tan^{-1}x+C$

15. $\int\frac{\cos 8x-\cos 7x}{1+2\cos 5x}dx$

Ans. $\frac{\sin 3x}{3}-\frac{\sin 2x}{2}+C$

Sol. $$\int \frac{\cos 8x - \cos 7x}{1+2\cos 5x}dx = \int \frac{-2\sin\frac{15x}{2}\sin\frac{x}{2}}{1+2\left(1-2\sin^2\frac{5x}{2}\right)}dx$$

$$= -\int \frac{2\sin\frac{15x}{2}\sin\frac{x}{2}}{\left(3-4\sin^2\frac{5x}{2}\right)}.\frac{\sin\left(\frac{5x}{2}\right)}{\sin\frac{5x}{2}}dx$$

$$= -\int \frac{2\sin\frac{15x}{2}\sin\frac{x}{2}\cdot\sin\frac{5x}{2}}{\sin\frac{15x}{2}}dx$$

$$= -2\int \sin\frac{x}{2}\sin\frac{5x}{2}dx$$

$$= -\int (\cos 2x - \cos 3x)dx$$

$$= \int \cos 3x dx - \int \cos 2x dx$$

$$= \frac{\sin 3x}{3} - \frac{\sin 2x}{2} + C$$

16. $$\int \frac{dx}{\sin(x-a)\cos(x-b)}$$

Ans. $\sec(a-b)\log\frac{\sin(x-a)}{\cos(x-b)} + c$

Sol. $$\int \frac{dx}{\sin(x-a)\cos(x-b)} = \frac{1}{\cos(b-a)}\int \frac{\cos((x-a)-(x-b))}{\sin(x-a)\cos(x-b)}dx$$

$$= \frac{1}{\cos(b-a)}\left[\int(\cot(x-a)+\tan(x-b))dx\right]$$

$$= \frac{1}{\cos(b-a)}\left(\ell n|\sin(x-a)| - \ell n|\cos(x-b)|\right) + C$$

$$= \sec(a-b)\ell n\left|\frac{\sin(x-a)}{\cos(x-b)}\right| + C$$

17. $\int \frac{\cos\ 4x+1}{\cot\ x-\tan\ x}dx$

Ans. $-\frac{1}{8}\cos 4x + c$

Sol. $\int \frac{1+\cos 4x}{\cot x-\tan x}dx = \int \frac{2\cos^2 2x}{2\cot 2x}dx$

$= \int \cos 2x \sin 2x dx = \frac{1}{2}\int \sin 4x dx$

$= -\frac{1}{8}\cos 4x + C$

18. $\int \frac{(x\ +1)\ e^x}{\cos^2\ (x\,e^x)}\ dx$

Ans. $\tan (x\,e^x) + c$

Sol. $\int \frac{(x+1)e^x}{\cos^2(xe^x)}dx$

let $xe^x = t$

$(x + 1)e^x\,dx = dt$

$= \int \sec^2 t dt = \tan t + C\ = \tan (xe^x) + C$

19. $\int \frac{a}{b+ce^x}\ dx$

Ans. $\frac{a}{b}\ \ln\left|\frac{e^x}{b+ce^x}\right| + C$

Sol. $\int \frac{a}{b+ce^x}dx = a\int \frac{e^{-x}dx}{be^{-x}+C}$

let $c + be^{-x} = t$

$\Rightarrow -be^{-x}\,dx = dt$

$= -\frac{a}{b}\int \frac{dt}{t} = -\frac{a}{b}\ell n|t| + C$

$= -\frac{a}{b}\ell n\left|c+be^{-x}\right| + C$

$$= -\frac{a}{b}\ell n\left|\frac{e^x \cdot c + B}{e^x}\right| + C$$

$$= \frac{a}{b}\ell n\left|\frac{e^x}{b + ce^x}\right| + C$$

20. $\int \tan x \sec^2 x \sqrt{1 - \tan^2 x}\ dx$

Ans. $-\frac{1}{3}\ .(1 - \tan^2 x)^{3/2} + c$

Sol. $\int \tan x \sec^2 x\sqrt{1 - \tan^2 x}dx$

let $1 - \tan^2 x = t$

$\Rightarrow -2\tan x \sec^2 x\ dx = dt$

$$= -\frac{1}{2}\int \sqrt{t}dt$$

$$= -\frac{1}{2} \times \frac{t^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{-\left(1 - \tan^2 x\right)^{\frac{3}{2}}}{3} + C$$

21. $\int \frac{dx}{x(x^2 + 1)}$

Ans. $\frac{1}{2}\log \frac{x^2}{x^2 + 1} + c$

Sol. $\int \frac{dx}{x\left(x^2 + 1\right)}$

let $x = \tan\theta$

$dx = \sec^2\theta\ d\theta$

$$= \int \frac{\sec^2 \theta d\theta}{\tan\theta \sec^2 \theta d\theta} = \int \frac{d\theta}{\tan\theta} = \int \frac{\cos\theta d\theta}{\sin\theta}$$

$$= \ell n\left|\sin\theta\right| + C$$

$$= \ell n\left|\sin\left(\tan^{-1} x\right)\right| + C$$

22. $\int \log x \frac{\sin\left\{1+(\log x)^2\right\}}{x} dx$

Ans. $-\frac{1}{2}\cos\{(1+\log x)^2\}+c$

Sol. $\int \log x \frac{\sin\left\{1+(\log x)^2\right\}}{x} dx$

let $1+(\ell nx)^2 = t$

$\Rightarrow \frac{2\ell nx}{x} dx = dt$

$= \int \frac{\sin t}{2} dt = -\frac{\cos t}{2} + C = -\frac{1}{2}\cos\left\{1+(\ell nx)^2\right\} + C$

23. $\int e^{e^{e^{e^x}}} e^{e^{e^x}} e^{e^x} e^x \, dx$

Ans.

Sol. $\int e^{e^{e^{e^x}}} e^{e^{e^x}} e^{e^x} e^x dx$

let $e^{e^{e^{e^x}}} = t$

$\Rightarrow e^{e^{e^{e^x}}} \cdot e^{e^{e^x}} \cdot e^{e^x} . e^x dx = dt$

$= \int dt = t + C$

$= e^{e^{e^{e^x}}} + C$

24. $\int \frac{\left(a + b \sin^{-1} x\right)^n}{\sqrt{1-x^2}} \, dx$

Ans. $\frac{1}{b}\frac{\left(a + b \sin^{-1} x\right)^{n+1}}{(n+1)} + c$

Sol. $\int \frac{\left(a+b\sin^{-1}x\right)^n}{\sqrt{1-x^2}} dx$

let $a + b\sin^{-1}x = t$

$$\Rightarrow \frac{b}{\sqrt{1-x^2}}dx = dt$$

$$= \frac{1}{b}\int t^n dt = \frac{1}{b}\frac{t^{n+1}}{n+1} + C$$

$$= \frac{1}{b}\left[\frac{\left(a + b\sin^{-1}x\right)^{n+1}}{n+1}\right] + C$$

1. $\int 2x \sec^3 (x^2 + 3) \tan (x^2 + 3)\, dx$

Ans. $\frac{1}{3} \sec^3 (x^2 + 3) + c$

Sol. $\int 2x \sec^3\left(x^2+3\right)\tan\left(x^2+3\right)dx$

let $\sec (x^2 + 3) = t$

$\Rightarrow \sec\left(x^2+3\right)\cdot\tan\left(x^2+3\right)\cdot 2xdx = dt$

$= \int t^2 dt = \frac{t^3}{3} + C = \frac{1}{3}\sec^3\left(x^2+3\right) + C$

2. $\int \sec x \,.\, \log (\sec x + \tan x)\, dx$

Ans. $\frac{1}{2}\left\{\log\left|\sec x + \tan x\right|\right\}^2 + c$

Sol. $\int \sec x \log\left(\sec x + \tan x\right) dx$

let $\ell n\left(\sec x + \tan x\right) = t$

$\sec x\, dx = dt$

$= \int t dt = \frac{t^2}{2} + C = \frac{\left\{\ell n\left(\sec x + \tan x\right)\right\}^2}{2} + C$

3. $\int \sqrt{\cos x - \cos^3 x}\, dx$

Ans. $-\frac{2}{3}\cos x\sqrt{\cos x} + c$

Sol. $\int \sqrt{\cos x - \cos^3 x}\, dx$

$= \int \sin x\sqrt{\cos x}\, dx$

let $\cos x = t$

$-\sin x\, dx = dt$

$$= -\int \sqrt{t}dt = \frac{-t^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= -\frac{2}{3}(\cos x)^{\frac{3}{2}} + C$$

4. $\int \frac{dx}{(\sec^{-1} x)\, x \sqrt{x^2 - 1}}$

Ans. $\log(\sec^{-1} x) + c$

Sol. $\int \frac{dx}{\sec^{-1} x \cdot x \cdot \sqrt{x^2 - 1}}$

let $\sec^{-1} x = t$

$$\Rightarrow \frac{1}{x\sqrt{x^2 - 1}} dx = dt$$

$$= \int \frac{dt}{t} = \ell n|t| + C$$

$$= \ell n\left|\sec^{-1} x\right| + C$$

5. $\int \frac{3x^2}{x^6 + 1} dx$

Ans. $\tan^{-1}(x^3) + c$

Sol. $\int \frac{3x^2}{x^6 + 1} dx$

let $x^3 = t \Rightarrow 3x^2 dx = dt$

$$= \int \frac{dt}{1 + t^2} = \tan^{-1} t + C = \tan^{-1}\left(x^3\right) + C$$

6. $\int \frac{\cos x}{1 + \sin x} dx$

Ans. $\log(1 + \sin x) + c$

Sol. $\int \frac{\cos x}{1 + \sin x} dx$

let $1 + \sin x = t$

$\Rightarrow \cos x\, dx = dt$

$$= \int \frac{dt}{t} = \ell n|t| + C = \ell n|1 + \sin x| + C$$

7. $\int \frac{\log(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx$

Ans. $\frac{1}{2}\left[\log(x + \sqrt{1+x^2})\right]^2 + c$

Sol. $\int \frac{\ell n\left(x + \sqrt{1+x^2}\right)}{\sqrt{1+x^2}} dx$

let $\ell n\left(x + \sqrt{1+x^2}\right) = t$

$$\Rightarrow \frac{1}{x + \sqrt{1+x^2}} \cdot \left(1 + \frac{2x}{2\sqrt{1+x^2}}\right) dx = dt$$

$$= \int t dt = \frac{t^2}{2} + C = \frac{1}{2}\left(\ell n\left(x + \sqrt{x^2+1}\right)\right)^2 + C$$

8. $\int \frac{\left(\sin^{-1} x\right)^2}{\sqrt{1-x^2}} dx =$

Ans. $\frac{1}{3}(\sin^{-1} x)^3 + c$

Sol. $\int \frac{\left(\sin^{-1} x\right)^2}{\sqrt{1-x^2}} dx$

let $\sin^{-1} x = t$

$$\Rightarrow \frac{1}{\sqrt{1-x^2}} dx = dt$$

$$= \int t^2 dt = \frac{t^3}{3} + C = \frac{\left(\sin^{-1} x\right)^3}{3} + C$$

9. $\int \frac{dx}{e^x + e^{-x}}$

Ans. $\tan^{-1}(e^x) + c$

Sol. $\int \frac{dx}{e^x + e^{-x}} = \int \frac{e^x dx}{e^{2x} + 1}$

$e^x = t$

$\Rightarrow e^x dx = dt$

$= \int \frac{dt}{1+t^2} = \tan^{-1}(t) + C = \tan^{-1}\left(e^x\right) + C$

10. $\int \frac{\sec^2 x}{\tan x \sqrt{\tan^2 x - 1}} dx$

Ans. $\sec^{-1}(\tan x) + c$

Sol. $\int \frac{\sec^2 x dx}{\tan x \sqrt{\tan^2 x - 1}}$

let $\tan x = t$

$\Rightarrow \sec^2 x\, dx = dt$

$= \int \frac{dt}{t\sqrt{t^2 - 1}} = \sec^{-1}(t) + C = \sec^{-1}(\tan x) + C$

11. $\int \frac{x^{e-1} - e^{x-1}}{x^e - e^x} dx$

Ans. $(1/e)\log(x^e - e^x) + c$

Sol. $\int \frac{x^{e-1} - e^{x-1}}{x^e - e^x} dx$

let $x^e - e^x = t$

$\left(ex^{e-1} - e^x\right) dx = dt$

$= \int \frac{dt}{e \cdot t} = \frac{1}{e} \ell n|t| + C = \frac{1}{e} \ell n\left|x^e - e^x\right| + C$

12. $\int \frac{x^2 \tan^{-1} x^3}{1 + x^6} dx$

Ans. $\frac{1}{6} (\tan^{-1} x^3)^2 + c$

Sol. $\int \frac{x^2 \tan^{-1}\left(x^3\right)}{1 + x^6} dx$

let $x^3 = t$

$\Rightarrow 3x^2\,dx = dt$

$$= \int \frac{\tan^{-1}(t)}{1+t^2}\cdot\frac{dt}{3} = \frac{1}{3}\int \frac{\tan^{-1} t}{1+t^2}\,dt$$

$$= \frac{1}{3}\frac{\left(\tan^{-1} t\right)^2}{2} + C = \frac{1}{6}\left(\tan^{-1} x^3\right)^2 + C$$

13. $\displaystyle\int \frac{(x+1)(x+\log x)^2}{x}\,dx$

Ans. $\frac{1}{3}(x+\log x)^3 + c$

Sol. $\displaystyle\int \frac{(x+1)(x+\ell nx)^2}{x}dx$

let $x + \ell nx = t$

$$\left(\frac{x+1}{x}\right)dx = dt$$

$$= \int t^2 dt = \frac{t^3}{3} + C = \frac{(x+\ell nx)^3}{3} + C$$

14. Integral of $\frac{1}{1+(\log x)^2}$ w.r.t. $\log x$

Ans. $\tan^{-1}(\log x) + C$

Sol. $\displaystyle\int \frac{1}{1+(\ell nx)^2}d(\ell nx) = \tan^{-1}(\ell nx) + C$

15. Integral of $\frac{1}{\sqrt{x^2+4}}$ w.r.t. (x^2+3)

Ans. $2\sqrt{x^2+4} + C$

Sol. $\displaystyle\int \frac{1}{\sqrt{x^2+4}}d\left(x^2+3\right) = \int \frac{1}{\sqrt{x^2+3+1}}d\left(x^2+3\right)$

$$= \int (t+1)^{-\frac{1}{2}}\,dt = 2\sqrt{x^2+4} + C$$

16. $\int \frac{1}{\sqrt{(1-x^2)\left\{9+(\sin^{-1}x)^2\right\}}}\,dx$

Ans. $\log\left|\sin^{-1}x+\sqrt{9+(\sin^{-1}x)^2}\right|+c$

Sol. $\int \frac{1}{\sqrt{(1-x^2)\left\{9+(\sin^{-1}x)^2\right\}}}dx$

$=\int \frac{1}{\sqrt{1-x^2}}\cdot\frac{1}{\sqrt{9+(\sin^{-1}x)^2}}dx$

let $\sin^{-1}x=t$

$\frac{1}{\sqrt{1-x^2}}dx=dt$

$=\int \frac{dt}{\sqrt{9+t^2}}=\ell n\left|t+\sqrt{9+t^2}\right|+C$

$=\ell n\left|\sin^{-1}x+\sqrt{9+(\sin^{-1}x)^2}\right|+C$

17. $\int \frac{dx}{x[(\log x)^2+4\log x-1]}=$

Ans. $\frac{1}{2\sqrt{5}}\log\left[\frac{\log x+2-\sqrt{5}}{\log x+2+\sqrt{5}}\right]+c$

Sol. $\int \frac{dx}{x\left((\ell nx)^2+4\ell nx-1\right)}$

let $\ell nx=t$

$\frac{1}{x}dx=dt$

$=\int \frac{dt}{t^2+4t-1}=\int \frac{dt}{\left(t-\left(-2-\sqrt{5}\right)\right)\left(t-\left(-2+\sqrt{5}\right)\right)}$

$$= \frac{1}{2\sqrt{5}} \int \frac{\left(\left(t+2+\sqrt{5}\right)-\left(t+2-\sqrt{5}\right)\right)}{\left(t-\left(-2-\sqrt{5}\right)\left(t-\left(-2+\sqrt{5}\right)\right)\right)}$$

$$= \frac{1}{2\sqrt{5}} \ell n \left| \frac{t+2-\sqrt{5}}{t+2+\sqrt{5}} \right| + C$$

$$= \frac{1}{2\sqrt{5}} \ell n \left| \frac{\ell nx+2-\sqrt{5}}{\ell nx+2+\sqrt{5}} \right| + C$$

18. $\int \frac{x}{x^4+x^2+1} dx$

Ans. $\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) + c$

Sol. $\int \frac{x}{x^4+x^2+1} dx$

let $x^2 = t$

$\Rightarrow 2xdx = dt$

$$= \frac{1}{2}\int \frac{dt}{t^2+t+1} = \frac{1}{2}\int \frac{dt}{\left(t+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \frac{1}{2}.\frac{1}{\left(\frac{\sqrt{3}}{2}\right)} \tan^{-1}\left(\frac{t+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + C$$

$$= \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right) + C$$

19. $\int \frac{(x \cos x + 1)}{\sqrt{2x^3 e^{\sin x} + x^2}} dx$

Ans. $\ell n \left\{ \frac{\sqrt{2x e^{\sin x} + 1} - 1}{\sqrt{2x e^{\sin x} + 1} + 1} \right\} + c$

Sol. $\int \frac{(x\cos x+1)}{\sqrt{2x^3e^{\sin x}+x^2}}dx$

$= \int \frac{(x\cos x+1)}{x\sqrt{2xe^{\sin x}+1}}dx$

let $xe^{\sin x} = t$

$\left(xe^{\sin x}\cos x + e^{\sin x}\right)dx = dt$

$= \int \frac{dt}{t\sqrt{2t+1}}$

let $2t + 1 = p^2$

$\Rightarrow 2dt = 2pdp$

$= \int \frac{2dp}{p^2-1} = 2\ell n\left|\frac{p-1}{p+1}\right| + C$

$= 2\ell n\left|\frac{\sqrt{2xe^{\sin x}+1}-1}{\sqrt{2xe^{\sin x}+1}+1}\right| + C$

20. $\int \frac{1}{1+\sqrt{x}}\,dx$

Ans. $2\sqrt{x} - 2\log\left|1+\sqrt{x}\right| + c$

Sol. $\int \frac{dx}{1+\sqrt{x}}$

let $\sqrt{x} = t$

$\Rightarrow \frac{1}{2\sqrt{x}}dx = dt$

$= \int \frac{2tdt}{1+t} = 2\int \frac{(1+t-1)}{1+t}dt = 2t - 2\ell n|1+t| + C$

$= 2\sqrt{x} - 2\ell n\left|1+\sqrt{x}\right| + C$

21. $\int \frac{e^{-x}}{16+9\,e^{-2x}}\,dx$

Ans. $-\frac{1}{12}\tan^{-1}\left(\frac{3\,e^{-x}}{4}\right)+c$

Sol. $\int\frac{e^{-x}}{16+9e^{-2x}}dx$

let $e^{-x}=t$

$\Rightarrow -e^{-x}dx=dt$

$=\int\frac{-dt}{16+(3t)^2}=-\frac{1}{4}\tan^{-1}\left(\frac{3t}{4}\right)\cdot\frac{1}{3}+C$

$=-\frac{1}{12}\tan^{-1}\left(\frac{3e^{-x}}{4}\right)+C$

22. $\int\frac{2e^{5x}+e^{4x}-4e^{3x}+4e^{2x}+2e^{x}}{(e^{2x}+4)\ (e^{2x}-1)^2}dx$

Ans. $\tan^{-1}\left(\frac{e^x}{2}\right)-\frac{1}{2(e^{2x}-1)}+c$

Sol. $\int\frac{2e^{5x}+e^{4x}-4e^{3x}+4e^{2x}+2e^{x}}{\left(e^{2x}+4\right)\left(e^{2x}-1\right)^2}dx$

let $e^x=t$

$e^x\,dx=dt$

$=\int\frac{2t^4+t^3-4t^2+4t+2}{\left(t^2+4\right)\left(t^2-1\right)^2}dt$

$=\int\frac{2t^4-4t^2+2+t\left(t^2+4\right)}{\left(t^2+4\right)\left(t^2-1\right)^2}dt$

$=2\int\frac{dt}{t^2+4}+\int\frac{t}{\left(t^2-1\right)^2}dt$

$=\frac{2}{2}\tan^{-1}\left(\frac{t}{2}\right)+\frac{1}{2}\frac{\left(t^2-1\right)^{-2+1}}{-2+1}+C$

$$= \tan^{-1}\left(\frac{t}{2}\right) - \frac{1}{2}\left(t^2 - 1\right) + C$$

$$= \tan^{-1}\left(\frac{e^x}{2}\right) - \frac{1}{2\left(e^{2x} - 1\right)} + C$$

23. $\int \frac{dx}{\sqrt{x} + \sqrt[4]{x}}$

Ans. $2\sqrt{x} - 4\sqrt[4]{x} + 4\ell n\left(1 + \sqrt[4]{x}\right) + c$

Sol. $\int \frac{dx}{\sqrt{x} + (x)^{\frac{1}{4}}}$

let $\left(\sqrt{x}\right)^{\frac{1}{4}} = t \qquad \Rightarrow dx = 4t^3 dt$

$$= \int \frac{4t^3 dt}{r + r^2} = 4\int \frac{t^2}{t+1} dt = 4\int \frac{t^2 + t - t}{t+1} dt$$

$$= 4\left[\int\left(\left(\frac{t^2 + t}{t+1}\right) - \left(\frac{t}{t+1}\right)\right) dt\right]$$

$$= 4\left[\frac{t^2}{2} - t + \ell n\left|t + 1\right|\right] + C$$

$$= 2t^2 - 4t + 4\ell n\left|1 + t\right| + C$$

$$= 2x^{\frac{1}{2}} - 4x^{\frac{1}{4}} + 4\ell n\left|1 + x^{\frac{1}{4}}\right| + C$$

24. $\int \frac{x\ dx}{\sqrt{1 + x^2 + \sqrt{\left(1 + x^2\right)^3}}}$

Ans. $2\sqrt{1 + \sqrt{1 + x^2}} + c$

Sol. $\int \frac{xdx}{\sqrt{1+x^2+\sqrt{\left(1+x^2\right)^3}}}$

let $1 + x^2 = t^2 \Rightarrow 2xdx = 2tdt$

$$\int \frac{tdt}{\sqrt{t^2+\sqrt{t^6}}} = \int \frac{tdt}{\sqrt{t^2+t^3}}$$

$$\int \frac{dt}{\sqrt{1+t}} = 2\left(t+1\right)^{\frac{1}{2}} + C$$

$$= 2\left(1+\sqrt{1+x^2}\right)^{\frac{1}{2}} + C$$

25. $\int \frac{1}{(x+1)\left(x^2+2x+2\right)}\,dx$

Ans. $\log\left|\frac{x+1}{\sqrt{x^2+2x+2}}\right| + c$

Sol. $\int \frac{dx}{\left(x+1\right)\left(x^2+2x+2\right)}$

$$\frac{1}{\left(x+1\right)\left(x^2+2x+2\right)} = \frac{A}{x+1} + \frac{B}{x^2+2x+2}$$

$\Rightarrow A + B = 0,\ 2A + B + C = 0\ \&\ 2A + C = +1$

$\Rightarrow A = 1,\ B = -1,\ C = -1$

$$= \int \frac{dx}{x+1} - \int \frac{\left(x+1\right)dx}{x^2+2x+2}$$

$$= \ell n\left|x+1\right| - \frac{1}{2}\ell n\left|x^2+2x+2\right| + C$$

$$= \ell n\left|\frac{x+1}{\sqrt{x^2+2x+2}}\right| + C$$

26. $\int \frac{\sec^4 x}{\sqrt{\tan x}}\,dx$

Ans. $\frac{2}{5}\sqrt{\tan x}\ (5 + \tan^2 x) + c$

Sol. $\int \frac{\sec^4 x}{\sqrt{\tan x}} dx$

let $\sqrt{\tan x} = t$

$\Rightarrow \frac{1}{2\sqrt{\tan x}} \times \sec^2 x dx = dt$

$= 2\int \left(1 + t^4\right) dt = \frac{2}{5} t^5 + 2t + C$

$= \frac{2}{5}\sqrt{\tan x}\left(5 + \tan^2 x\right) + C$

27. $\int \frac{\tan x}{1 + \tan^2 x} dx$

Ans. $\frac{1}{2}\ \log (\cos^2 x + 2 \sin^2 x) + c$

Sol. $\int \frac{\tan x}{1 + \tan^2 x} dx$

$= \frac{1}{2}\int \sin 2x dx$

$= \frac{-\cos 2x}{4} + C$

28. Integrate : $x^{13/2} . (1 + x^{5/2})^{1/2}$ w. r. t. x

Ans. $\frac{4}{5}\left[\frac{t^7}{7} - \frac{2t^5}{5} + \frac{t^3}{3}\right] + c$ where $t^2 = 1 + x^{5/2}$

Sol. $\int x^{\frac{13}{2}} \sqrt{\left(1 + x^{\frac{5}{2}}\right)} dx = \int x^{\frac{3}{2}} \cdot \left(x^{\frac{5}{2}}\right)^2 \cdot \sqrt{1 + x^{\frac{5}{2}}} dx$

let $1 + x^{\frac{5}{2}} = t^2$

$\Rightarrow \frac{5}{2} x^{\frac{5}{2}-1} dx = 2tdt \Rightarrow x^{\frac{3}{2}} dx = \frac{4}{5} tdt$

$$= \frac{4}{5}\int t^2 \cdot \left(t^2 - 1\right)^2 \, dt$$

$$= \frac{4}{5}\int \left[t^6 - 2t^4 + t^2\right] dt$$

$$= \frac{4}{5}\left(\frac{t^7}{7} - \frac{2t^5}{5} + \frac{t^3}{3}\right] + C; t^2 = 1 + x^{\frac{5}{2}}$$

29. $\int \frac{\cos\ 2x + x + 1}{x^2 + \sin\ 2x + 2x}\, dx$

Ans. $\frac{1}{2}\log\left(x^2 + \sin 2x + 2x\right) + c$

Sol. $\int \frac{x + \cos 2x + 1}{x^2 + \sin 2x + 2x} dx$

let $x^2 + \sin 2x + 2x = t$

$\Rightarrow 2(x + \cos 2x + 1)\, dx = dt$

$$= \frac{1}{2}\int \frac{dt}{t} = \frac{1}{2}\ell n\left|t\right| + C = \frac{1}{2}\ell n\left|x^2 + \sin 2x + 2x\right| + C$$

30. $\int \frac{1 + x^2}{\sqrt{1 - x^2}}\, dx$

Ans. $\frac{3}{2}\sin^{-1}(x) - \frac{1}{2}x\sqrt{1 - x^2} + c$

Sol. $\int \frac{1 + x^2}{\sqrt{1 - x^2}} dx = \int \frac{2dx}{\sqrt{1 - x^2}} - \int \frac{1 - x^2 dx}{\sqrt{1 - x^2}}$

$$= 2\int \frac{dx}{\sqrt{1 - x^2}} - \int \sqrt{1 - x^2}\, dx$$

$$= 2\sin^{-1} x - \left[\frac{x}{2}\sqrt{1 - x^2} + \frac{1}{2}\sin^{-1} x\right] + C$$

$$= \frac{3}{2}\sin^{-1} x - \frac{x}{2}\sqrt{1 - x^2} + C$$

31. $\int \frac{(5x^2 - 12)\ dx}{(x^2 - 6x + 13)^2}$

Ans. $\frac{13x-159}{8\left(x^2-6x+13\right)}+\frac{53}{16}\tan^{-1}\frac{x-3}{2}+c$

Sol. $\int\frac{\left(5x^2-12\right)dx}{\left(x^2-6x+13\right)^2}=\int\frac{\left(5x^2-12\right)dx}{\left((x-3)^2+4\right)^2}$

$=\int\frac{\left(5\left(x^2-6x+13+6x-13\right)-12\right)dx}{\left(x^2-6x+13\right)^2}$

$=5\int\frac{dx}{\left(x^2-6x+13\right)}+\int\frac{30x-65-12}{\left(x^2-6x+13\right)^2}dx$

$=5\int\frac{dx}{(x-3)^2+4}+\int\frac{15(2x-6)+90-77}{\left(x^2-6x+13\right)^2}dx$

$=5\int\frac{dx}{(x-3)^2+4}+15\int\frac{(2x-6)}{\left(x^2-6x+13\right)^2}dx+13\int\frac{dx}{\left(x^2-6x+13\right)^2}$

$=\frac{5}{2}\tan^{-1}\left(\frac{x-3}{2}\right)-\frac{15}{x^2-6x+13}+13\int\frac{dx}{\left((x-3)^2+4\right)^2}$

$\int\frac{dx}{\left((x-3)^2+4\right)^2}=I$ (let)

let $x-3=2\tan\theta$

$dx=2\sec^2\theta\,d\theta$

$\Rightarrow I=\int\frac{2\sec^2\theta d\theta}{16\left(1+\tan^2\theta\right)^2}=\frac{1}{8}\int\frac{d\theta}{\sec^2\theta}=\frac{1}{16}\theta+\frac{1}{32}\sin 2\theta+C$

$=\frac{5}{2}\tan^{-1}\left(\frac{x-3}{2}\right)-\frac{15}{x^2-6x+13}+13I$

$=\frac{5}{2}\tan^{-1}\left(\frac{x-3}{2}\right)-\frac{15}{x^2-6x+13}+\frac{13}{16}\tan^{-1}\left(\frac{x-3}{2}\right)+\frac{13}{32}\sin\left(2\tan^{-1}\left(\frac{x-3}{2}\right)\right)+C$

$$= \frac{53}{16}\tan^{-1}\left(\frac{x-3}{2}\right) - \frac{15}{x^2-6x+13} + \frac{13}{32}\left(\frac{(x-3)}{1+\left(\frac{x-3}{2}\right)^2}\right) + C$$

$$= \frac{53}{16}\tan^{-1}\left(\frac{x-3}{2}\right) - \frac{15}{x^2-6x+13} + \frac{13}{8}\frac{(x-3)}{x^2-6x+13} + C$$

$$= \frac{53}{16}\tan^{-1}\left(\frac{x-3}{2}\right) + \frac{13x-159}{8\left(x^2-6x+13\right)} + C$$

1. $\int \frac{\log \cos x}{\cos^2 x}\, dx$

Ans. $\tan x \cdot \log \cos x + \tan x - x + c$

Sol. $\int \frac{\log \cos x}{\cos^2 x} dx = \int \underbrace{\sec^2 x}_{II} \underbrace{\log \cos x}_{I} dx$

$= \tan x \log \cos + \int \frac{\tan x \cdot \sin x}{\cos x} dx$

$= \tan x \cdot \log \cos x + \int \tan^2 x dx$

$= \tan x \cdot \log \cos x + \tan x - x + C$

2. $\int \tan^{-1} x \ . \ln (1 + x^2)\, dx$

Ans. $x \tan^{-1} x \ . \ln (1 + x^2) + (\tan^{-1} x)^2 - 2x \tan^{-1} x + \ln (1 + x^2) - \left(\ell n \sqrt{1 + x^2}\right)^2 + c$

Sol. $\int \underset{\downarrow}{1}_{II} \cdot \underbrace{\tan^{-1} x \ell n\left(1+x^2\right)}_{I} dx$

$= x \tan^{-1} x \ell n\left(1+x^2\right) - \int x\left(\frac{1}{1+x^2} \ell n\left(1+x^2\right) + \frac{2x}{1+x^2} \tan^{-1} x\right) dx$

$= x \tan^{-1} x \ell n\left(1+x^2\right) - \int \frac{x}{1+x^2} \ell n\left(1+x^2\right) dx - \int \frac{2x^2 \tan^{-1} x}{1+x^2} dx$

$= x \tan^{-1} x \ell n\left(1+x^2\right) - \frac{1}{4}\left(\ell n\left(1+x^2\right)\right)^2 - \underbrace{2\int \frac{x^2 \tan^{-1} x}{1+x^2} dx}_{I \text{ (let)}}$

$I = 2\int \frac{x^2 \tan^{-1} x}{1+x^2} dx$

let $\tan^{-1} x = t$

$\frac{dx}{1+x^2} = dt$

$= 2\int \tan^2 t \cdot t dt = 2\int \left(\sec^2 t - 1\right) t dt$

$$= 2\int \sec^2 t \cdot t\,dt - 2\int t\,dt$$

$$= 2\left[t\tan t - \int \tan t\,dt \right] - t^2 + C$$

$$= 2\left[\tan^{-1} x \cdot x + \ell n\left|\cos t\right| \right] - t^2 + C$$

$$\Rightarrow I = 2x\tan^{-1} x + 2\ell n\left|\cos\left(\tan^{-1} x\right)\right| - \left(\tan^{-1} x\right)^2 + C$$

$$\Rightarrow \int \tan^{-1} x \cdot \ell n\left(1+x^2\right)dx = x\tan^{-1} x \cdot \ell n\left(1+x^2\right) - \left(\ell n\sqrt{1+x^2}\right)^2$$

$$-2x\tan^{-1} x + \left(\tan^{-1} x\right)^2 + \ell n\left(1+x^2\right) + C$$

3. $\int \dfrac{x^2\left(x\sec^2 x + \tan x\right)}{(x\tan x + 1)^2}\,dx$

Ans. $\dfrac{-x^2}{x\tan x + 1} + 2\ell n(x\sin x + \cos x) + c$

Sol. $\int \dfrac{x^2\left(x\sec^2 x + \tan x\right)}{\left(x\tan x + 1\right)^2}dx$

$$= \int \underbrace{x^2}_{I}\underbrace{\left(\frac{x\sec^2 x + \tan x}{\left(x\tan x + 1\right)^2}\right)}_{II}dx$$

$$= x^2\cdot\left(\frac{-1}{x\tan x + 1}\right) + \int \frac{2x}{x\tan x + 1}dx$$

$$= -\frac{x^2}{x\tan x + 1} + \int \frac{2x\cos x}{x\sin x + \cos x}dx$$

$$= \frac{-x^2}{x\tan x + 1} + 2\ell n\left(x\sin x + \cos x\right) + C$$

4. $\int xe^{-x}dx$

Ans. $-(x + 1)\,e^{-x} + c$

Sol. $\int \underbrace{x}_{I}\underbrace{e^{-x}}_{II}dx = \dfrac{xe^{-x}}{-1} + \int e^{-x}dx = -\left(x+1\right)e^{-x} + C$

5. $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}}\, dx$

Ans. $x - \sqrt{1-x^2}\ \sin^{-1} x + c$

Sol. $\int x \cdot \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

let $\sin^{-1} x = t$

$\frac{dx}{\sqrt{1-x^2}} = dt$

$= \int t \cdot \sin t\, dt = -t \cos t + \int \cos t\, dt$

$= -t \cos t + \sin t + C$

$= x - \sqrt{1-x^2} \sin^{-1} x + C$

6. $\int x\ \sin x \sec^3 x\, dx$

Ans. $\frac{1}{2}[x \sec^2 x - \tan x] + c$

Sol. $\int x \sin x \cdot \sec^3 x dx = \int \underset{I}{x} \underbrace{\tan x \sec^2 x}_{II} dx$

$= x \int (\sec x \tan x) \cdot \sec x dx - \int \left(1 \cdot \int \sec^2 x \tan x dx\right) dx$

$= \frac{x \sec^2 x}{2} - \frac{1}{2} \tan x + C$

7. $\int \cos(\log x) dx$

Ans. $\frac{x}{\sqrt{2}}\ \cos(\log x - \pi/4) + c$

Sol. $\int \cos(\log x)\, dx$

let $\log x = t$

$x = e^t$

$dx = e^t dt$

$I = \int e^t \cos t\, dt$

$= e^t \sin t - \int e^t \cos t\, dt$

$$I = \frac{e^t \cdot \sqrt{2}}{2} \cos\left(t - \frac{\pi}{4}\right) + C$$

$$I = \frac{x}{\sqrt{2}} \cos\left(\log x - \frac{\pi}{4}\right) + C$$

8. $\int \frac{x}{1 + \sin x}\, dx$

Ans. $-x \cot\left(\frac{\pi}{4} + \frac{x}{2}\right) + 2 \ln \sin\left(\frac{\pi}{4} + \frac{x}{2}\right) + c$

Sol. $\int \frac{x}{1+\sin x} dx = \int \frac{2x}{\sin^2\left(\frac{x}{2} + \frac{\pi}{4}\right)} dx$

$$= 2\int \underset{\text{I}}{x} \underbrace{\operatorname{cosec}^2\left(\frac{x}{2} + \frac{\pi}{4}\right)}_{\text{II}} dx$$

$$= 2x \cdot \frac{1}{2}\left(-\cot\left(\frac{\pi}{4} + \frac{x}{2}\right)\right) + \frac{2}{2}\int \cot\left(\frac{\pi}{4} + \frac{x}{2}\right) dx$$

$$= -x \cot\left(\frac{\pi}{4} + \frac{x}{2}\right) + \ell n \left|\sin\left(\frac{\pi}{4} + \frac{x}{2}\right)\right| + C$$

9. $\int x \cos^2 x\, dx$

Ans. $\frac{x^2}{4} + \frac{1}{4} x \sin 2x + \frac{1}{8} \cos 2x + c$

Sol. $\int x \cos^2 x dx = \int \frac{x(1+\cos 2x)}{2} dx$

$$= \frac{x^2}{4} + \frac{1}{2}\int \underset{\text{I}}{x} \underbrace{\cos 2x}_{\text{II}}\, dx$$

$$= \frac{x^2}{4} + \frac{1}{4} x \sin 2x - \frac{1}{4}\int \sin 2x dx$$

$$= \frac{x^2}{4} + \frac{x}{4} \sin 2x + \frac{1}{8} \cos 2x + C$$

10. If $I = \int e^x \sin 2x\, dx$, then for what value of k, $kI = e^x(\sin 2x - 2\cos 2x) + \text{constant}$

यदि $I = \int e^x \sin 2x\, dx$ तो k के किस मान के लिए $kI = e^x(\sin 2x - 2\cos 2x) + \text{constant}$ है।

Ans. 5

Sol. $I = \int \underset{II}{e^x} \underset{I}{\sin 2x}\, dx$

$\Rightarrow I = \sin 2x e^x - 2\int \underset{II}{e^x} \underset{I}{\cos 2x}\, dx$

$\Rightarrow I = \sin 2x e^x - 2\cos 2x e^x - 4\int e^x \sin 2x dx + C$

$\Rightarrow 5I = e^x(\sin 2x - 2\cos 2x) + C$

$\Rightarrow k = 5$

11. $\int \frac{3x+1}{2x^2-2x+3} dx$

Ans. $\frac{3}{4}\log(2x^2-2x+3) + \frac{\sqrt{5}}{2}\tan^{-1}\left(\frac{2x-1}{\sqrt{5}}\right) + c$

Sol. $\int \frac{3x+1}{2x^2-2x+3} dx = \int \frac{\frac{3}{4}(4x-2+2)+1}{2x^2-2x+3} dx$

$= \frac{3}{4}\ell n\left|2x^2-2x+3\right| + 5\int \frac{dx}{4x^2-4x+6}$

$= \frac{3}{4}\ell n\left|2x^2-2x+3\right| + 5\int \frac{dx}{(2x-1)^2+\left(\sqrt{5}\right)^2}$

$= \frac{3}{4}\ell n\left|2x^2-2x+3\right| + \frac{5}{\sqrt{5}}\tan^{-1}\left(\frac{2x-1}{\sqrt{5}}\right)\cdot\frac{1}{2} + C$

12. $\int \frac{dx}{\sqrt{2x^2-x+2}}$

Ans. $\frac{1}{\sqrt{2}}\log\left|\frac{4x-1}{\sqrt{15}} + \sqrt{\frac{16x^2-8x+16}{15}}\right| + c$

Sol. $\int \frac{dx}{\sqrt{2x^2-x+2}} = \frac{1}{2}\int \frac{dx}{\sqrt{x^2-\frac{x}{2}+1}}$

$$= \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\left(x - \frac{1}{4}\right)^2 + \left(\frac{\sqrt{15}}{4}\right)^2}}$$

$$= \frac{1}{\sqrt{2}} \ell n \left| \frac{4x-1}{4} + \sqrt{\left(x - \frac{1}{4}\right)^2 + \left(\frac{\sqrt{15}}{4}\right)^2} \right|$$

13. $\int \frac{1}{x} \sqrt{\frac{x-1}{x+1}}$

Ans. $\log \left| x + \sqrt{x^2 - 1} \right| - \sec^{-1} x + c$

Sol. $\int \frac{1}{x} \sqrt{\frac{x-1}{x+1}} dx$

$$= \int \frac{x-1}{x\sqrt{x^2-1}} dx = \int \frac{dx}{\sqrt{x^2-1}} - \int \frac{dx}{x\sqrt{x^2-1}}$$

$$= \log \left| x + \sqrt{x^2 - 1} \right| - \sec^{-1} x + C$$

14. $\int e^{\tan^{-1} x} \left(\frac{1 + x + x^2}{1 + x^2} \right) dx$

Ans. $x e^{\tan^{-1} x} + C$

Sol. $\int e^{\tan^{-1} x} \left(\frac{1 + x + x^2}{1 + x^2} \right) dx$

let $\tan^{-1} x = t$

$$\Rightarrow \frac{dx}{1 + x^2} = dt$$

$$= \int e^t \left(1 + \tan t + \tan^2 t\right) dt$$

$$= \int e^t \left(\tan t + \sec^2 t\right) dt = e^t \tan t + C$$

$$= x e^{\tan^{-1} x} + C$$

15. $\int \frac{(\ell n\ x - 1)\ dx}{\ell n^2\ x}$

Ans. $\frac{x}{\ell n\ x} + c$

Sol. $\int \frac{(\ell nx - 1)}{\ell n^2 x} dx$

let $\ell nx = t$

$\Rightarrow x = e^t$

$\Rightarrow dx = e^t\ dt$

$= \int \left(\frac{t-1}{t^2}\right) e^t dt = \int e^t \left(\frac{1}{t} - \frac{1}{t^2}\right) dt$

$= \frac{e^t}{t} + C = \frac{x}{\ell nx} + C$

16. $\int \{\tan(\log x) + \sec^2(\log x)\}\ dx$

Ans. $x \tan(\log x) + c$

Sol. $\int \left(\tan\left(\ell nx\right) + \sec^2\left(\ell nx\right)\right) dx$

let $\ell nx = t$

$\Rightarrow dx = e^t\ dt$

$= \int \left(\tan t + \sec^2 t\right) e^t dt = e^t \tan t + C$

$= x \cdot \tan(\ell nx) + C$

17. $\int e^{\cot\theta}(\cos\theta - \text{cosec}\ \theta)\ d\theta$

Ans. $e^{\cot\theta} . \sin\theta + C$

Sol. $\int e^{\cot\theta}\left(\cos\theta - \cos ec\theta\right) d\theta$

$= \int e^{\cot\theta}\left(\cos ec\theta - \cos\theta\right) \cdot \frac{\cos ec^2\theta}{-\cos ec^2\theta} d\theta$

$= \int e^{\cot\theta}\left(-\sin\theta + \cot\theta \cdot \sin\theta\right) \cos ec^2\theta d\theta$

let $\cot\theta = t$

$-\text{cosec}^2\theta\ d\theta = dt$

$$= \int -e^t (t-1) \cdot \frac{1}{\sqrt{1+t^2}} \cdot dt$$

$$= \int e^t \left(\underbrace{\frac{1}{\sqrt{1+t^2}}}_{\downarrow f(t)} - \underbrace{\frac{t}{\sqrt{1+t^2}}}_{\downarrow f'(t)} \right) dt$$

$$= \frac{e^t}{\sqrt{1+t^2}} + C = e^{\cot\theta} \cdot \sin\theta + C$$

18. $\int \frac{\sqrt{1 - \sin x}}{1 + \cos x} e^{-x/2} \, dx$

Ans. $- e^{-x/2} \sec(x/2) + c$

Sol. $\int \frac{\sqrt{1-\sin x}}{1+\cos x} e^{-\frac{x}{2}} dx$

$$= \int \frac{e^{-\frac{x}{2}} \left[\cos\frac{x}{2} - \sin\frac{x}{2} \right]}{2\cos^2\frac{x}{2}} dx$$

$$= \int \frac{e^{-\frac{x}{2}}}{2} \left(\sec\frac{x}{2} - \sec\frac{x}{2}\tan\frac{x}{2} \right) dx$$

let $-\frac{x}{2} = t$

$\Rightarrow dx = -2dt$

$= -\int e^t (\sec t + \sec t \tan t) dt$

$= -e^t \sec t + C$

$= -e^{-\frac{x}{2}} \sec\frac{x}{2} + C$

19. $\int e^x \left(\frac{1-x}{1+x^2}\right)^2 dx$

Ans. $e^x \cdot \frac{1}{1+x^2} + c$

Sol. $\int e^x \left(\frac{1-x}{1+x^2}\right)^2 dx$

$$= \int e^x \left(\frac{1+x^2-2x}{\left(1+x^2\right)^2}\right) dx$$

$$= \int e^x \left(\underbrace{\frac{1}{1+x^2}}_{\substack{\downarrow \\ f(x)}} - \underbrace{\frac{2x}{\left(1+x^2\right)^2}}_{\substack{\downarrow \\ f'(x)}} \right) dx$$

$$= \frac{e^x}{1+x^2} + C$$

20. $\int e^x \frac{\left(x^2 + 3x + 3\right)}{(x+2)^2} dx =$

Ans. $e^x \left(\frac{x+1}{x+2}\right) + c$

Sol. $\int e^x \frac{\left(x^2+3x+3\right)}{\left(x+2\right)^2} dx$

$$= \int e^x \left[\frac{(x+1)(x+2)}{\left(x+2\right)^2} + \frac{1}{\left(x+2^2\right)}\right] dx$$

$$= \int e^x \left(\underbrace{\frac{x+1}{x+2}}_{\substack{\downarrow \\ f(x)}} + \underbrace{\frac{1}{(x+2)^2}}_{\substack{\downarrow \\ f'(x)}} \right) dx$$

$$= \frac{e^x (x+1)}{(x+2)} + C$$

21. $\int e^x \left(\frac{1-x}{1+x} \right)^2 dx$

Ans. $e^x \left(\frac{x-3}{x+1} \right) + c$

Sol. $\int e^x \left(\frac{1-x}{1+x} \right)^2 dx = \int e^x \frac{(1+x^2-2x)}{(1+x)^2} dx$

$$= \int e^x \left(\frac{(1+x)(x-3)}{(1+x)^2} + \frac{4}{(1+x)^2} \right) dx$$

$$= \int e^x \left(\frac{x-3}{(1+x)} + \frac{4}{(1+x)^2} \right) dx$$

$$= \frac{e^x (x-3)}{x+1} + C$$

22. $\int e^x \left[\sin^{-1} \frac{x}{a} + \frac{1}{\sqrt{a^2 - x^2}} \right] dx =$

Ans. $e^x \sin^{-1} \frac{x}{a} + c$

Sol. $$\int e^x \left[\underbrace{\sin^{-1}\frac{x}{a}}_{\downarrow f(x)} + \underbrace{\frac{1}{\sqrt{a^2-x^2}}}_{\downarrow f'(x)} \right] dx$$

$$= e^x \sin^{-1}\left(\frac{x}{a}\right) + C \qquad \left\{\because \int e^x \left(f(x) + f'(x)\right) dx = e^x f(x) + C\right\}$$

23. $$\int e^x \frac{1 + n x^{n-1} - x^{2n}}{\left(1 - x^n\right)\sqrt{1 - x^{2n}}} dx =$$

Ans. $$e^x \sqrt{\frac{1 + x^n}{1 - x^n}} + c$$

Sol. $$\int e^x \left(\frac{1 + n x^{n-1} - x^{2n}}{\left(1 - x^n\right)\sqrt{1 - x^{2n}}} \right) dx$$

$$= \int e^x \left(\frac{\left(1 - x^{2n}\right)}{\left(1 - x^n\right)\sqrt{1 - x^{2n}}} + \frac{n x^{n-1}}{\left(1 - x^n\right)\sqrt{1 - x^{2n}}} \right) dx$$

$$= \int e^x \left(\underbrace{\sqrt{\frac{1 + x^n}{1 - x^n}}}_{\downarrow f(x)} + \underbrace{\frac{n x^{n-1}}{\left(1 - x^n\right)\sqrt{1 - x^{2n}}}}_{\downarrow f'(x)} \right) dx$$

$$= e^x \left(\sqrt{\frac{1 + x^n}{1 - x^n}} \right) + C$$

1. $\int \frac{dx}{(2\sin x + \cos x)^2}$

Ans. $-\frac{1}{2}\left(\frac{1}{2\tan x + 1}\right) + C$

Sol. $\int \frac{dx}{(2\sin x + \cos x)^2} = \int \frac{\sec^2 x dx}{(2\tan x + 1)^2}$

let $2\tan x + 1 = t$

$2\sec^2 x\, dx = dt$

$= \int \frac{dt}{2t^2} = -\frac{1}{2t} + C = -\frac{1}{2}\left(\frac{1}{2\tan x + 1}\right) + C$

2. $\int \frac{1}{5 + 7\cos x + \sin x}\, dx$

Ans. $\frac{1}{5}\log_e\left|\frac{\tan(x/2) + 2}{\tan(x/2) - 3}\right| + c$

Sol. $\int \frac{dx}{5 + 7\cos x + \sin x} = \int \frac{\left(1 + \tan^2\frac{x}{2}\right)dx}{5\left(1 + \tan^2\frac{x}{2}\right) + 7\left(1 - \tan^2\frac{x}{2}\right) + 2\tan\left(\frac{x}{2}\right)}$

$= \int \frac{\sec^2\frac{x}{2}}{12 - 2\tan^2\frac{x}{2} + 2\tan\frac{x}{2}}$

let $\tan\frac{x}{2} = t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

$= 2\int \frac{dt}{12 - 2t^2 + 2t} = \frac{-2}{2}\int \frac{dt}{t^2 - t - 6}$

$= -\int \frac{dt}{(t-3)(t+2)} = \frac{1}{5}\ell n\left|\frac{t+2}{t-3}\right| + C$

$$= \frac{1}{5}\ell n\left|\frac{\tan\frac{x}{2}+2}{\tan\frac{x}{2}-3}\right| + C$$

3. $\int \sin^7 x\ dx$

Ans. $-\cos x + \cos^3 x - \frac{3}{5}\cos^5 x + \frac{1}{7}\cos^7 x + c$

Sol. $\int \sin^7 x dx = \int \sin x \cdot \sin^6 x dx$

$$= \int \sin x\left(1-\cos^2 x\right)^3 dx$$

let $\cos x = t$

$-\sin x\, dx = dt$

$$= -\int\left(1-t^2\right)^3 dt = -\left[\int\left(1-t^6-3t^2+3t^4\right)dt\right.$$

$$= -t + \frac{t^7}{7} + t^3 - \frac{3t^5}{5} + C$$

$$= -\cos x + \frac{\cos^7 x}{7} + \cos^3 x - \frac{3}{5}\cos^5 x + C$$

4. $\int \frac{\cos x}{\cos 3x}\ dx$

Ans. $\frac{1}{2\sqrt{3}}\log\left|\frac{1+\sqrt{3}\tan x}{1-\sqrt{3}\tan x}\right| + c$

Sol. $\int \frac{\cos x dx}{4\cos^3 x - 3\cos x} = \int \frac{dx}{4\cos^2 x - 3}$

$$= \int \frac{\sec^2 x dx}{4-3\left(1+\tan^2 x\right)}$$

let $\tan x = t$

$\sec^2 x\, dx = dt$

$$= \int \frac{dt}{1-3t^2} = \int \frac{dt}{1-\left(\sqrt{3}t\right)^2}$$

$$= \frac{1}{2\sqrt{3}} \ell n \left| \frac{1+\sqrt{3}t}{1-\sqrt{3}t} \right| + C = \frac{1}{2\sqrt{3}} \ell n \left| \frac{1+\sqrt{3}\tan x}{1-\sqrt{3}\tan x} \right| + C$$

5. $\int \frac{dx}{5-4\sin x+3\cos x}$

Ans. $\frac{1}{5}\tan\left(\frac{x}{2}+\frac{1}{2}\cos^{-1}\frac{3}{5}\right)+c$

Sol. $\int \frac{dx}{5-4\sin x+3\cos x} = \frac{1}{5}\int \frac{1}{1-\left(\frac{4}{5}\sin x-\frac{3}{5}\cos x\right)}dx$

$$= \frac{1}{5}\int \frac{1}{1+\cos(x+\theta)}dx; \cos\theta = \frac{3}{5}$$

$$= \frac{1}{10}\int \sec^2\left(\frac{x+\theta}{2}\right)dx$$

$$= \frac{1}{10}\tan\left(\frac{x}{2}+\frac{\theta}{2}\right)\cdot 2 + C$$

$$= \frac{1}{5}\tan\left(\frac{x}{2}+\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)\right)+C$$

6. $\int \frac{1}{(\sin x - 2\cos x)(2\sin x + \cos x)}\,dx$

Ans. $\frac{1}{5}\,1n\left|\frac{\tan x-2}{2\tan x+1}\right|+c$

Sol. $\int \frac{dx}{(\sin x-2\cos x)(2\sin x+\cos x)}$

$$= \int \frac{\sec^2 x}{(\tan x-2)(2\tan x+1)}dx$$

let $\tan x = t$

$\Rightarrow \sec^2 x\,dx = dt$

$$= \int \frac{dt}{(t-2)(2t+1)} = \frac{1}{2}\int \frac{dt}{(t-2)\left(t+\frac{1}{2}\right)} = \frac{1}{5}\ell n\left|\frac{t-2}{t+\frac{1}{2}}\right| + C$$

$$= \frac{1}{5}\ell n\left|\frac{\tan x - 2}{\tan x + \frac{1}{2}}\right| + C$$

7. $\int \frac{\sin^8 x - \cos^8 x}{1 - 2\sin^2 x \cos^2 x}\, dx$

Ans. $-\frac{1}{2}\sin 2x + c$

Sol. $\int \frac{\left(\sin^8 x - \cos^8 x\right)}{1 - 2\sin^2 x \cos^2 x} dx = \int \frac{\left(\sin^4 x + \cos^4 x\right)\left(\sin^4 x - \cos^4 x\right)}{1 - 2\sin^2 x \cos^2 x} dx$

$$= -\int \cos 2x dx = -\frac{1}{2}\sin 2x + C$$

8. $\int \frac{3\cos x + 2\sin x}{4\sin x + 5\cos x} dx$

Ans. $\frac{23}{41}x + \frac{2}{41}\log(4\sin x + 5\cos x) + c$

Sol. $\int \frac{3\cos x + 2\sin x}{4\sin x + 5\cos x} dx$

$3\cos x + 2\sin x = A(4\sin x + 5\cos x) + B(4\cos x - 5\sin x)$

$\Rightarrow 3\cos x + 2\sin x = (4A - 5B)\sin x + (5A + 4B)\cos x$

$4A - 5B = 2$

& $5A + 4B = 3$

$$\Rightarrow A = \frac{23}{41} \;\&\; B = \frac{2}{41}$$

$$\Rightarrow \int \frac{3\cos x + 2\sin x}{4\sin x + 5\cos x} dx = \frac{23}{41}\int dx + \frac{2}{41}\int \frac{4\cos x - 5\sin x}{4\sin x + 5\cos x} dx$$

$$= \frac{23}{41}x + \frac{2}{41}\ell n\left(4\sin x + 5\cos x\right) + C$$

9. $\int\left(\sqrt{\tan x}+\sqrt{\cot x}\right)dx$

Ans. $\sqrt{2}\tan^{-1}\frac{1}{\sqrt{2}}\left(\sqrt{\tan x}-\sqrt{\cot x}\right)+c$

Sol. $\int\left(\sqrt{\tan x}+\sqrt{\cot x}\right)dx$

$=\int\frac{\sin x+\cos x}{\sqrt{\cos x\sin x}}dx$

$=\sqrt{2}\int\frac{(\sin x+\cos x)}{\sqrt{\sin 2x+1-1}}dx$

$=\sqrt{2}\int\frac{\sin x+\cos x}{\sqrt{1-(\sin x-\cos x)^2}}dx$

let $\sin x-\cos x=t$

$(\cos x+\sin x)dx=dt$

$=\sqrt{2}\int\frac{dt}{\sqrt{1-t^2}}=\sqrt{2}\sin^{-1}(\sin x-\cos x)+C$

$=\sqrt{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\left(\sqrt{\tan x}-\sqrt{\cot x}\right)\right)+C$

10. $\int(\log_{ex}e)\left(\log_{e^2x}e\right)\left(\log_{e^3x}e\right)\frac{dx}{x}$

Ans. $\frac{1}{2}\ell n\left\{\frac{(1+\ell n x)(3+\ell n x)}{(2+\ell n x)^2}\right\}+c$

Sol. $\int(\log_{ex}e)\left(\log_{e^2x}e\right)\left(\log_{e^3x}e\right)\frac{dx}{x}$

$=\int\frac{1}{1+\ell nx}\cdot\frac{1}{2+\ell nx}\cdot\frac{1}{3+\ell nx}\cdot\frac{1}{x}dx$

let $\ell nx=t\Rightarrow\frac{dx}{x}=dt$

$=\int\frac{dt}{(1+t)(t+2)(t+3)}$

$=\frac{1}{2}\int\frac{dt}{1+t}-\int\frac{dt}{t+2}+\frac{1}{2}\int\frac{dt}{t+3}$

$$=\frac{1}{2}\ell n(t+1)-\ell n(t+2)+\frac{1}{2}\ell n(t+3)+C$$

$$=\frac{1}{2}\ell n\big((t+1)(t+3)\big)-\ell n(t+2)+C$$

$$=\frac{1}{2}\ell n\left(\frac{(t+1)(t+3)}{(t+2)^2}\right)+C$$

$$=\frac{1}{2}\ell n\left\{\frac{(1+\ell nx)(3+\ell nx)}{(2+\ell nx)^2}\right\}+C$$

11. $\int \frac{x^2-8x+7}{(x^2-3x-10)^2}\,dx$

Ans. $\frac{8}{49\,(x-5)}-\frac{27}{49\,(x+2)}+\frac{30}{343}\,ln\left|\frac{x-5}{x+2}\right|+c$

Sol. $\int \frac{x^2-8x-7}{\left(x^2-3x-10\right)^2}dx=\int \frac{(x-1)(x-7)}{(x-5)^2(x+2)^2}dx$

$$\frac{(x-1)(x-7)}{(x-5)^2(x+2)^2}=\frac{A}{x+2}+\frac{B}{(x+2)^2}+\frac{C}{x-5}+\frac{D}{(x-5)^2}$$

$\Rightarrow A(x-5)^2(x+2)+B(x-5)^2+C(x+2)^2(x-5)+D(x+2)^2=(x-1)(x-7)$

$\Rightarrow$ put $x=5$ $\quad x=-2$ $\quad x=1$ $\quad x=0$

$49D=-8$ $\quad 49B=27$ $\quad 2C-6A=\frac{45}{49}$ $\quad 2C-5A=\frac{30}{49}$

$\Rightarrow D=-\frac{8}{49}$ $\quad B=\frac{27}{49}$ $\quad A=-\frac{15}{49}$ $\quad C=-\frac{45}{98}$

$$=-\frac{15}{49}\int\frac{dx}{x+2}+\frac{27}{49}\int\frac{dx}{(x+2)^2}-\frac{8}{49}\int\frac{dx}{(x-5)^2}-\frac{45}{98}\int\frac{dx}{x-5}$$

$$=\frac{8}{49(x-5)}-\frac{27}{49(x+2)}+\frac{30}{343}\ell n\left|\frac{x-5}{x+2}\right|+C$$

12. $\int \frac{x^3-1}{x^3+1}\,dx$

Ans. $x - \log x + \frac{1}{2}\log(x^2+1) - \tan^{-1}x + c$

Sol. $\int \frac{x^3-1}{x^3+x}dx = \int \frac{\left(x^3+x-1-x\right)dx}{x^3+x}$

$= x - \int \frac{(1+x)\,dx}{x\left(x^2+1\right)}$

$= x - \int \frac{1}{x^3+x}dx - \int \frac{dx}{x^2+1}$

$= x - \tan^{-1}x - \int \frac{dx}{x\left(x^2+1\right)}$

$\left\{\frac{1}{x\left(x^2+1\right)} = \frac{A}{x} + \frac{Bx+C}{x^2+1} \Rightarrow A=1, B=-1, C=0\right\}$

$= x - \tan^{-1}x - \left[\int \frac{1}{x}dx + \int \frac{-x\,dx}{x^2+1}\right]$

$= x - \tan^{-1}x - \ell n|x| + \frac{1}{2}\ell n\left|x^2+1\right| + C$

13. $\int \frac{dx}{x^2\left(x^5+1\right)^{4/5}}$

Ans. $c - \frac{\sqrt[5]{1+x^5}}{x}$

Sol. $\int \frac{dx}{x^2\left(x^5+1\right)^{\frac{4}{5}}} = \int \frac{1}{x^2\cdot x^4\left(1+\frac{1}{x^5}\right)^{\frac{4}{5}}}dx$

let $\frac{1}{x^5} = t \Rightarrow \frac{-5x^4}{x^{10}}dx = dt$

$$= -\int \frac{dt}{5\cdot(1+t)^{\frac{4}{5}}} = -\frac{1}{5}\frac{(1+t)^{\frac{1}{5}}}{\frac{1}{5}} + C = -\frac{1}{x}\left(1+x^5\right)^{\frac{1}{5}} + C$$

14. $\int \frac{x^2}{\left(a + bx^2\right)^{5/2}}\, dx$

Ans. $\frac{x^3}{3a\left(a + bx^2\right)^{3/2}} + c$

Sol. $\int \frac{x^2}{\left(a+bx^2\right)^{\frac{5}{2}}} dx = \int \frac{x^2}{x^5\left(\frac{a}{x^2}+b\right)^{\frac{5}{2}}} dx$

let $\frac{1}{x^2} = t$

$-\frac{2}{x^3} dx = dt$

$$= -\int \frac{dt}{2\left(at+b\right)^{\frac{5}{2}}} = -\frac{1}{2a}\frac{\left(a+b\right)^{-\frac{5}{2}+1}}{-\frac{5}{2}+1} + C$$

$$= \frac{1}{2a}\cdot\frac{2}{3}\cdot\frac{1}{\left(at+b\right)^{\frac{3}{2}}} + C$$

$$= \frac{1}{3a\left(\frac{a}{x^2}+b\right)^{\frac{3}{2}}} + C = \frac{x^3}{3a\left(a+bx^2\right)^{\frac{3}{2}}} + C$$

15. $\int \frac{1 + x\cos x}{x\left(1 - x^2 e^{2\sin x}\right)}\, dx$

Ans. $\ln\left(x\, e^{\sin x}\right) - \frac{1}{2}\ln\left(1 - x^2 e^{2\sin x}\right) + c$

Sol. $\int \frac{1+x\cos x}{x\left(1-x^2e^{2\sin x}\right)} dx$

let $xe^{\sin x} = t \Rightarrow e^{\sin x} = \frac{t}{x}$

$\Rightarrow \frac{t}{x}(1 + x\cos x)\,dx = dt$

$\Rightarrow I = \int \frac{dt}{t(1-t^2)}$

$$\left\{\begin{array}{l} \frac{t}{t(t^2-1)} = \frac{A}{t} + \frac{B}{t+1} + \frac{C}{t-1} \\ \Rightarrow A = -1 \\ B = \frac{1}{2} \\ C = \frac{1}{2} \end{array}\right\}$$

$\Rightarrow I = -\ell n|t| + \frac{1}{2}\ell n(t+1)(t-1) + C$

$\Rightarrow I = \ell n\left|xe^{\sin x}\right| - \frac{1}{2}\ell n\left|x^2e^{2\sin x} - 1\right| + C$

16. $\int \frac{dx}{x^4\,(x^3+1)^2}$

Ans. $\frac{2}{3}\ell n\left(\frac{x^3+1}{x^3}\right) - \frac{1}{3x^3} - \frac{1}{3(x^3+1)} + c$

Sol. $\int \frac{dx}{x^4(x^3+1)^2} = \int \frac{dx}{x^{10}\left(1+\frac{1}{x^3}\right)^2}$

let $\left(1+\frac{1}{x^3}\right) = t \Rightarrow \frac{-3}{x^4}dx = dt$

$= -\frac{1}{3}\int \frac{dt}{t^2.\frac{1}{(t-1)^2}} = -\frac{1}{3}\int\left(\frac{t^2-2t+1}{t^2}\right)dt$

$$= -\frac{1}{3}\left[t - 2\ell n|t| - \frac{1}{t} \right] + C$$

$$= -\frac{1}{3}\left[1 + \frac{1}{x^3} - 2\ell n\left|1 + \frac{1}{x^3}\right| - \frac{1}{1+\frac{1}{x^3}} \right] + C$$

$$= -\frac{1}{3x^3} + \frac{2}{3}\ell n\left|\frac{x^3+1}{x^3}\right| - \frac{x^3}{3\left(x^3+1\right)} - \frac{1}{3} + C$$

$$= -\frac{1}{3x^3} + \frac{2}{3}\ell n\left|\frac{x^3+1}{x^3}\right| + \frac{1}{3\left(x^3+1\right)} + C$$

17. $\int \frac{(\sin x - \cos x)\, dx}{(\sin x + \cos x)\sqrt{(\sin x \cos x + \sin^2 x \cos^2 x)}}$

Ans. $\tan^{-1}\left(\sqrt{(\sin x + \cos x)^4 - 1}\right) + c$

Sol. $\int \frac{(\sin x - \cos x)\, dx}{(\sin x + \cos x)\sqrt{\sin x \cos x + \sin^2 x \cos^2 x}}$

let $\sin x + \cos x = t$

$(\cos x - \sin x)\, dx = dt$

$$= \int \frac{-dt}{t\sqrt{\left(\sin x \cos x + \frac{1}{2}\right)^2 - \frac{1}{4}}}$$

$$= \int \frac{-dy}{t\sqrt{\frac{(1+\sin 2x)^2 - 1}{4}}} = -\int \frac{2dt}{t\sqrt{t^4 - 1}}$$

let $\sqrt{t^4 - 1} = v$

$$\Rightarrow \frac{4t^3 dt}{2\sqrt{t^4 - 1}} = dv$$

$$= -\int \frac{dv}{v^2 + 1} = -\tan^{-1}(v) + C$$

$$= -\tan^{-1}\left(\sqrt{t^4-1}\right)+C$$

$$= -\tan^{-1}\left(\sqrt{(\sin x+\cos x)^4-1}\right)+C$$

18. $\int \frac{1}{(x-1)\sqrt{2x+3}}\,dx$

Ans. $\frac{1}{\sqrt{5}}\log\left|\frac{\sqrt{2x+3}-\sqrt{5}}{\sqrt{2x+3}+\sqrt{5}}\right|+c$

Sol. $\int \frac{1}{(x-1)\sqrt{2x+3}}dx$

let $\sqrt{2x+3}=t$

$$\Rightarrow \frac{dx}{\sqrt{2x+3}}=dt$$

$$=\int \frac{2dt}{\left(t^2-5\right)}=\frac{2}{2\cdot\sqrt{5}}\ell n\left|\frac{t-\sqrt{5}}{t+\sqrt{5}}\right|+C$$

$$=\frac{1}{\sqrt{5}}\ell n\left|\frac{\sqrt{2x+3}-\sqrt{5}}{\sqrt{2x+3}+\sqrt{5}}\right|+C$$

19. (i) $\int \frac{dx}{(2-x)\sqrt{x+1}}$ (ii) $\int \sqrt{\frac{x-1}{2-x}}\,dx$

Ans. (i) $\frac{1}{\sqrt{3}}\ln\left(\frac{\sqrt{3}+\sqrt{x+1}}{\sqrt{3}-\sqrt{x+1}}\right)+c$ (ii) $\sin^{-1}\sqrt{x-1}-\sqrt{3x-x^2-2}+c$

Sol. (i) $\int \frac{dx}{(2-x)\sqrt{x+1}}$

let $\sqrt{x+1}=t \Rightarrow x=t^2-1$

$$\frac{dx}{2\sqrt{x+1}}=dt$$

$$=\int \frac{2dt}{\left(3-t^2\right)}=2\int \frac{dt}{\left(\sqrt{3}\right)^2-t^2}$$

$$= \frac{2}{2\cdot\sqrt{3}} \ell n \left| \frac{\sqrt{3}+t}{\sqrt{3}-t} \right| + C$$

$$= \frac{1}{\sqrt{3}} \ell n \left| \frac{\sqrt{3}+\sqrt{x+1}}{\sqrt{3}-\sqrt{x+1}} \right| + C$$

(ii) $\int \sqrt{\frac{x-1}{2-x}} dx = \int \frac{(x-1)\,dx}{\sqrt{(2-x)(x-1)}}$

$$= \int \frac{(x-1)}{\sqrt{3x-x^2-2}} dx = \frac{1}{2}\int \frac{(2x-3+1)}{\sqrt{3x-x^2-2}} dx$$

$$= -\frac{1}{2}\int \frac{(3-2x)\,dx}{\sqrt{3x-x^2-2}} + \frac{1}{2}\int \frac{dx}{\sqrt{3x-x^2-2}}$$

$$= -\frac{1}{2}\int \frac{(3-2x)\,dx}{\sqrt{3x-x^2-2}} + \frac{1}{2}\int \frac{dx}{\sqrt{x-1}\cdot\sqrt{1-\left(\sqrt{x-1}\right)^2}}$$

$$= -\sqrt{3x-x^2-2} + \sin^{-1}\left(\sqrt{x-1}\right) + C$$

OR

M-II :

let $2-x = \cos^2\theta$

$\Rightarrow x-1 = \sin^2\theta$

$$\int \sqrt{\frac{x-1}{2-x}} dx = \int \tan\theta\cdot 2\sin\theta\cos\theta\, d\theta$$

$$= \int (1-\cos 2\theta)\, d\theta = \theta - \frac{\sin 2\theta}{2} + C$$

$$= \theta - \sin\theta\cos\theta + C$$

$$= \sin^{-1}\left(\sqrt{x-1}\right) - \sqrt{3x-x^2-2} + C$$

1. Let $\int_0^1 \frac{dx}{\sqrt{16+9x^2}} + \int_0^2 \frac{dx}{\sqrt{9+4x^2}} = \ln a$. Find a.

Ans. $2^{1/3} \cdot 3^{1/2}$

Sol. $\int_0^1 \frac{dx}{\sqrt{16+9x^2}} + \int_0^2 \frac{dx}{\sqrt{9+4x^2}} = \ln a$

$= \int_0^1 \frac{dx}{\sqrt{4^2+(3x)^2}} + \int_0^2 \frac{dx}{\sqrt{3^2+(2x)^2}}$

$I = \frac{1}{3}\log\left(x+\sqrt{x^2+\frac{16}{9}}\right)_0^1 + \frac{1}{2}\log\left(x+\sqrt{x^2+\frac{9}{4}}\right)_0^2$

$= \frac{1}{3}\log\left(1+\frac{5}{3}\right) - \frac{1}{3}\log\left(\frac{4}{3}\right) + \frac{1}{2}\log\left(2+\frac{5}{2}\right) - \frac{1}{2}\log\left(\frac{3}{2}\right)$

$= \frac{1}{3}\log\frac{\left(\frac{8}{3}\right)}{\left(\frac{4}{3}\right)} + \frac{1}{2}\log\frac{\left(\frac{9}{2}\right)}{\left(\frac{3}{2}\right)}$

$\ell na = \frac{1}{3}\log 2 + \frac{1}{2}\log 3 = \log\left(2^{\frac{1}{3}} \cdot 3^{\frac{1}{2}}\right)$

$a = 2^{\frac{1}{3}} \cdot 3^{\frac{1}{2}}$

2. $\int_0^{\ell n 2} x\ e^{-x}\, dx$

Ans. $\frac{1}{2}\ \ell n\left(\frac{e}{2}\right)$

Sol. $\int_0^{\ell n 2} \underset{I}{x}\, \underbrace{e^{-x}}_{II}\, dx$

$$= \left[-xe^{-x} + \int_0^{\ell n2} e^{-x} dx \right]_0^{\ell n2}$$

$$= \left[-xe^{-x} - e^{-x} \right]_0^{\ell n2}$$

$$= \left[-\ell n2 \times \frac{1}{2} - \frac{1}{2} \right] + [1]$$

$$= -\frac{\ell n2}{2} + \frac{1}{2} = \frac{-\ell n2 + \ell ne}{2} = \frac{1}{2} \ell n \frac{e}{2}$$

3. $\int_1^e \left(\frac{1}{\sqrt{x \ln x}} + \sqrt{\frac{\ln x}{x}} \right) dx$

Ans. $2\sqrt{e}$

Sol. $\int_1^e \left(\frac{1}{\sqrt{x \ln x}} + \sqrt{\frac{\ln x}{x}} \right) dx$

$$= \int_1^e \frac{1 + \ell nx}{\sqrt{x} \cdot \sqrt{\ell nx}} dx$$

let $x\ell nx = t$

$(\ell nx + 1) dx = dt$

$= 2\sqrt{e}$

4. Given $f'(x) = \frac{\cos x}{x}$, $f\left(\frac{\pi}{2}\right) = a$, $f\left(\frac{3\pi}{2}\right) = b$. Find the value of the definite integral $\int_{\pi/2}^{3\pi/2} f(x)\, dx$.

Ans. $2 - \frac{\pi}{2}(a - 3b)$

Sol. $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \underbrace{f(x)}_{I} \cdot \underset{II}{1}\, dx = \left[f(x)x - \int f'(x)x dx \right]_{\frac{\pi}{2}}^{\frac{3\pi}{2}}$

$$= \left[f(x)x - \int \cos x dx \right]_{\frac{\pi}{2}}^{\frac{3\pi}{2}}$$

$$= \left[f\left(\frac{3\pi}{2}\right)\frac{3\pi}{2} - \sin\frac{3\pi}{2} \right] - \left[f\left(\frac{\pi}{2}\right)\frac{\pi}{2} - \sin\frac{\pi}{2} \right]$$

$$= 2 - \frac{\pi}{2}[a - 3b]$$

5. $\int_{-1}^{1} \frac{x\ dx}{\sqrt{5-4x}}$

Ans. $\frac{1}{6}$

Sol. $\int_{-1}^{1} \frac{x\ dx}{\sqrt{5-4x}}$

let $\sqrt{5-4x} = t$

$$\Rightarrow \frac{-4dx}{2\sqrt{5-4x}} = dt$$

$$= \frac{1}{2}\int_{1}^{3} \frac{5-t^2}{4} dt \ = \frac{1}{8}\left[10 - 9 + \frac{1}{3}\right] = \frac{4}{3\times 8} = \frac{1}{6}$$

6. $\int_{2}^{e} \left(\frac{1}{\ell n\, x} - \frac{1}{\ell n^2\, x} \right) dx$

Ans. $e - \frac{2}{\ell n\, 2}$

Sol. $\int_{2}^{e} \left(\frac{1}{\ell n\, x} - \frac{1}{\ell n^2\, x} \right) dx$

put $\ell nx = t$

$$= \int_{\ell n2}^{1} e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dt$$

$$= \left[\frac{e^t}{t} \right]_{\ell n2}^{1} = \left[\frac{-2}{\ell n2} + e \right] = e - \frac{2}{\ell n2}$$

7. $\int_{0}^{\pi/4} \frac{\sin 2x}{\sin^4 x + \cos^4 x}\ dx$

Ans. $\frac{\pi}{4}$

Sol. $\int_0^{\pi/4} \frac{\sin 2x}{\sin^4 x + \cos^4 x}\, dx$

$= \int_0^{\frac{\pi}{4}} \frac{\sin 2x}{1-2\sin^2 x\cos^2 x} dx$

$= \int_0^{\frac{\pi}{4}} \frac{\sin 2x}{1-2\sin^2 x\left(1-\sin^2 x\right)} dx = \int_0^{\frac{\pi}{4}} \frac{\sin 2x dx}{1-2\sin^2 x+2\sin^4 x}$

let $\sin^2 x = t$

$= \int_0^{\frac{1}{2}} \frac{dt}{2t^2-2t+1} = \frac{1}{2}\int_0^{\frac{1}{2}} \frac{dt}{t^2-t+\frac{1}{2}}$

$= \frac{1}{2}\int_0^{\frac{1}{2}} \frac{dt}{\left(t-\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2} = \left(\tan^{-1}(2t-1)\right)_0^{\frac{1}{2}}$

$= \frac{\pi}{4}$

8. $\int_0^{\pi/2} \frac{\cos x\, dx}{(1+\sin x)(2+\sin x)}$

Ans. $\ell n \frac{4}{3}$

Sol. put $\sin x = t$

$dx = \frac{dt}{\cos x}$

$= \int_0^1 \frac{dt}{(1+t)(t+2)} = \ell n\left|\frac{t+1}{t+2}\right|_0^1 = \ell n\frac{2}{3} - \ell n\left(\frac{1}{2}\right)$

$= \ell n\left(\frac{4}{3}\right)$

9. $\int_0^{\pi/4} \frac{\sin^2 x \cdot \cos^2 x}{\left(\sin^3 x + \cos^3 x\right)^2} dx$

Ans. $\frac{1}{6}$

Sol. $\int_0^{\pi/4} \frac{\sin^2 x \cdot \cos^2 x}{\left(\sin^3 x + \cos^3 x\right)^2} dx$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin^2 x \cdot \cos^2 x}{\cos^6 x \left(1 + \tan^3 x\right)^2} dx = \int_0^{\frac{\pi}{4}} \frac{\tan^2 x \sec^2 x}{\left(1 + \tan^3 x\right)^2} dx$$

$1 + \tan^3 x = t$

$(3 \tan^2 x \sec^2 x)\, dx = dt$

$$= \frac{1}{3}\int_1^2 \frac{1}{t^2} dt = \frac{1}{3}\left(-\frac{1}{t}\right)_1^2 = \frac{1}{3}\left(-\frac{1}{2} + 1\right) = \frac{1}{6}$$

10. $\int_{1/3}^{3} \frac{\sin^{-1} \frac{x}{\sqrt{1+x^2}}}{x} dx$

Ans. $\frac{\pi \ln 3}{2}$

Sol. $\int_{\frac{1}{3}}^{3} \frac{\sin^{-1} \frac{x}{\sqrt{1+x^2}}}{x} dx = I$ ...(1)

put $x \to \frac{1}{x}$

$$I = -\int_3^{\frac{1}{3}} \frac{\sin^{-1} \frac{1}{\sqrt{1+x^2}}}{x} dx = -\int_3^{\frac{1}{3}} \frac{\cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)}{x} dx$$

$$I=\int_{\frac{1}{3}}^{3}\frac{\cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)}{x}dx \qquad ...(2)$$

$(1)+(2)$

$$2I=\frac{\pi}{2}\int_{\frac{1}{3}}^{3}\frac{1}{x}dx=\frac{\pi}{2}\left(\ell n3+\ell n3\right)\Rightarrow 2I=\frac{2\pi}{2}\ell n3$$

$$I=\frac{\pi}{2}\ell n3$$

11. $\int_{2}^{3}\frac{dx}{\sqrt{(x-1)\,(5-x)}}$

Ans. $\frac{\pi}{6}$

Sol. $\int_{2}^{3}\frac{dx}{\sqrt{(x-1)(5-x)}}=\int_{2}^{3}\frac{dx}{\sqrt{4-(x-3)^2}}$

$$=\sin^{-1}\left(\frac{x-3}{2}\right)_2^3$$

$$=-\sin^{-1}\left(-\frac{1}{2}\right)=\frac{\pi}{6}$$

12. $\int_{3/2}^{2}\left(\frac{x-1}{3-x}\right)^{1/2}dx$

Ans. $\frac{\sqrt{3}}{2}-1+\frac{\pi}{6}$

Sol. $\int_{3/2}^{2}\left(\frac{x-1}{3-x}\right)^{1/2}dx$

$$=\int_{\frac{3}{2}}^{2}\sqrt{\frac{(x-1)}{(3-x)}\times\frac{(x-1)}{(x-1)}}=\int_{\frac{3}{2}}^{2}\frac{(x-1)dx}{\sqrt{-x^2+4x-3}}$$

$$= -\frac{1}{2}\int_{\frac{3}{2}}^{2}\frac{-2x-2+4-4}{\sqrt{-x^2+4x-3}}$$

$$= -\frac{1}{2}\int_{\frac{3}{2}}^{2}\frac{-2x+4-6}{\sqrt{-x^2+4x-3}}dx$$

$$= -\frac{1}{2}\int_{\frac{3}{2}}^{2}\frac{-2x+4}{\sqrt{-x^2+4x-3}}dx + 3\int_{\frac{3}{2}}^{2}\frac{dx}{\sqrt{-x^2+4x-3}}$$

$$= \left[-\frac{1}{2}\cdot 2\sqrt{-x^2+4x-3}\right]_{\frac{3}{2}}^{3} + 3\int_{\frac{3}{2}}^{3}\frac{dx}{\sqrt{1-(x-2)^2}} = \frac{\sqrt{3}}{2} - 1 + \frac{\pi}{6}$$

13. $\int_0^{\pi/4} x\cos x\ \cos 3x\,dx$

Ans. $\frac{\pi - 3}{16}$

Sol. $\int_0^{\frac{\pi}{4}} x\cos x\cos 3x dx = \frac{1}{2}\int_0^{\frac{\pi}{4}}\left(x\cos(4x) + x\cos 2x\right)dx$

$$= \frac{1}{2}\left[\frac{x}{4}\sin 4x + \frac{\cos 4x}{16} + \frac{x}{2}\sin 2x + \frac{\cos 2x}{4}\right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2}\left[0 - \frac{1}{16} + \frac{\pi}{8} + 0 - \frac{1}{16} - \frac{1}{4}\right]$$

$$= \frac{1}{2}\left[\frac{\pi}{8} - \frac{1}{8} - \frac{1}{4}\right] = \frac{\pi - 3}{16}$$

14. $\int_0^{\pi/2} \frac{dx}{5 + 4\sin x}$

Ans. $\frac{2}{3}\tan^{-1}\frac{1}{3}$

Sol. $I = \int_0^{\pi/2} \frac{dx}{5+4\sin x}$

$$I = \int_0^{\frac{\pi}{2}} \frac{dx}{5+\frac{8\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}$$

$$= \int_0^{\frac{\pi}{2}} \frac{\left(1+\tan^2\frac{x}{2}\right)}{5+5\tan^2\frac{x}{2}+8\tan\frac{x}{2}} dx$$

let $\tan\frac{x}{2} = t$

$$= \int_0^1 \frac{2dt}{5+5t^2+8t} = 2\int_0^1 \frac{dt}{5\left(t^2+\frac{8}{5}t+1\right)}$$

$$= \frac{2}{5}\int_0^1 \frac{dt}{\left(t+\frac{4}{5}\right)^2+\frac{9}{25}} = \frac{2}{3}\left[\tan^{-1}\left(\frac{t+\frac{4}{5}}{\frac{3}{5}}\right)\right]_0^1$$

$$= \frac{2}{3}\left[\tan^{-1}(3) - \tan^{-1}\left(\frac{4}{3}\right)\right] = \frac{2}{3}\tan^{-1}\left(\frac{1}{3}\right)$$

15. $\int_2^3 \frac{dx}{(x-1)\sqrt{x^2-2x}}$

Ans. $\frac{\pi}{3}$

Sol. $\int_2^3 \frac{dx}{(x-1)\sqrt{x^2-2x}}$

$$= \int_2^3 \frac{dx}{(x-1)\sqrt{(x-1)^2-1}}$$

$$= \left[\sec^{-1}(x-1)\right]_2^3 = \sec^{-1}2 - \sec^{-1}(1)$$

$$= \frac{\pi}{3}$$

16. $\int_0^{\pi/2} \frac{dx}{1+\cos\theta\cdot\cos x}$ $\theta\in(0,\pi)$

Ans. $\frac{\theta}{\sin\theta}$

Sol. $I = \int_0^{\frac{\pi}{2}} \frac{dx}{1+\cos x\cos\theta}; \theta \in (0,\pi)$

$$= \int_0^{\frac{\pi}{2}} \frac{dx}{1+\cos\theta\cdot\left(\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)} = \int_0^{\frac{\pi}{2}} \frac{\sec^2\frac{x}{2}}{1+\tan^2\frac{x}{2}+\cos\theta\left(1-\tan^2\frac{x}{2}\right)}dx$$

let $\tan\frac{x}{2} = t$

$$= \int_0^1 \frac{2dt}{1+t^2+\cos\theta\left(1-t^2\right)} = 2\int_0^1 \frac{dt}{(1+\cos\theta)+t^2(1-\cos\theta)}$$

$$= \frac{2}{1-\cos\theta}\int_0^1 \frac{dt}{\cot^2\frac{\theta}{2}+t^2}$$

$$= \frac{2}{1-\cos\theta}\frac{1}{\cot\frac{\theta}{2}}\left[\tan^{-1}\left(\frac{t}{\cot\frac{\theta}{2}}\right)\right]_0^1 = \frac{2}{\sin\theta}\frac{\theta}{2} = \frac{\theta}{\sin\theta}$$

17. $\int_0^{\frac{\ln 3}{2}} \frac{e^x+1}{e^{2x}+1}dx$

Ans. $\frac{1}{2}\left(\frac{\pi}{6}+\ln 3-\ln 2\right)$

Sol. $I=\int_0^{\frac{\ln 3}{2}} \frac{e^x+1}{e^{2x}+1}dx$

let $e^x=t$

$$I=\int \frac{t+1}{\left(t^2+1\right)t}dt$$

$$\frac{t+1}{\left(t^2+1\right)t}=\frac{A}{t}+\frac{Bt+C}{t^2+1}=\frac{At^2+A+Bt^2+Ct}{t\left(t^2+1\right)}$$

$\Rightarrow A+B=0$ & $C=1, A=1$

$A=-B$ $A=1, B=-1$

$$=\int_1^{\sqrt{3}}\frac{1}{t}dt-\int_1^{\sqrt{3}}\frac{t-1}{t^2+1}dt=\left(\ell nt\right)_1^{\sqrt{3}}-\frac{1}{2}\int_1^{\sqrt{3}}\frac{2tdt}{t^2+1}+\int_1^{\sqrt{3}}\frac{dt}{t}$$

$$=\frac{1}{2}\ell n3+\frac{1}{2}\frac{\pi}{6}-\frac{\ell n2}{2}$$

18. $\int_0^{\pi/4}\cos 2x\ \sqrt{1-\sin 2x}\ dx=$

Ans. $\frac{1}{3}$

Sol. $I=\int_0^{\pi/4}\cos 2x\ \sqrt{1-\sin 2x}\ dx$

Let $\left(1-\sin 2x\right)=t$

$$\Rightarrow dx=\frac{-1}{2}\frac{dt}{\cos 2x}$$

$$=\frac{-1}{2}\int_1^0\sqrt{t}dt$$

$$=+\frac{1}{3}(1)$$

$$=\frac{1}{3}$$

19. $\int_0^3 \sqrt{\frac{x}{3-x}}\,dx =$

Ans. $\frac{3\pi}{2}$

Sol. $\int_0^3 \sqrt{\frac{x}{3-x}}\,dx$

$$= \int_0^3 \frac{x}{\sqrt{3x-x^2}}\,dx = \int_0^3 \frac{x\,dx}{\sqrt{-\left(x^2-3x\right)}} = \frac{3\pi}{2}$$

20. $\int_0^{1/2} \frac{dx}{\left(1-2x^2\right)\sqrt{1-x^2}}$

Ans. $\frac{1}{2}\,\ell n\left(2+\sqrt{3}\right)$

Sol. $\int_0^{1/2} \frac{dx}{\left(1-2x^2\right)\sqrt{1-x^2}}$

let $x = \sin^2\theta$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin 2\theta}{\cos 2\theta \cos\theta}\,d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{2\sin\theta\cos\theta}{\left(2\cos^2\theta-1\right)\cos\theta}\,d\theta$$

let $\cos\theta = t$

$$= -2\int_1^{\frac{1}{\sqrt{2}}} \frac{1}{\left(\sqrt{2}t\right)^2-1}\,dt = +2\ell n\left(\frac{\sqrt{2}t+1}{\sqrt{2}t+1}\right)_1^{\frac{1}{\sqrt{2}}}$$

$$= 2\ell n\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right) = 2\ell n\left(2+\sqrt{3}\right)$$

21. $\int_1^2 \frac{dx}{x\left(x^4+1\right)}$

Ans. $\frac{1}{4}\ln\frac{32}{17}$

Sol. $\int_1^2 \frac{dx}{x\left(x^4+1\right)} = \int_1^2 \frac{dx}{x^5\left(1+\frac{1}{x^4}\right)}$

let $1+\frac{1}{x^4} = t$

$\Rightarrow -\frac{4}{x^5}dx = dt$

$\Rightarrow dx = -\frac{x^5}{4}dt$

$= -\frac{1}{4}\int_2^{\frac{17}{16}} \frac{1}{t}dt = -\frac{1}{4}\left[\ell n\left(\frac{17}{16}\right) - \ell n(2)\right]$

$= \frac{1}{4}\ell n\left(\frac{32}{17}\right)$

22. $\int_0^{\pi/2} \sin\phi\cos\phi\sqrt{\left(a^2\sin^2\phi + b^2\cos^2\phi\right)}\,d\phi \quad a\neq b\ (a>0, b>0)$

Ans. $\frac{1}{3}\frac{a^3-b^3}{a^2-b^2}$

Sol. $\int_0^{\pi/2} \sin\phi\cos\phi\sqrt{\left(a^2\sin^2\phi + b^2\cos^2\phi\right)}\,d\phi \quad a\neq b\ (a>0, b>0)$

let $a^2\sin^2\phi + b^2\cos^2\phi = t^2$

$\Rightarrow 2\sin\phi\cos\phi\left(a^2-b^2\right)d\phi = 2tdt$

$= \frac{1}{a^2-b^2}\int_b^a t^2dt = \frac{1}{a^2-b^2}\left(\frac{t^3}{3}\right)_b^a$

$$= \frac{1}{3}\frac{a^3 - b^3}{a^2 - b^2}$$

23. (a) $\int_0^{3\pi/4} \left((1+x)\sin x + (1-x)\cos x\right)dx$, (b) $\int_{\pi/2}^{\pi} x^{\sin x}\,(1 + x\cos x \cdot \ln x + \sin x)\,dx$

Ans. (a) $2\left(\sqrt{2}+1\right)$; (b) $\left(\pi - \frac{\pi^2}{4}\right)$

Sol. (a) $I = \int_0^{3\pi/4} (\sin x + \cos x)\,dx + \int_0^{3\pi/4} \underbrace{x}_{I}\,(\underbrace{\sin x - \cos x}_{II})\,dx$

$$= \int_0^{3\pi/4} (\sin x + \cos x)\,dx + x(-\cos x - \sin x)\Big|_0^{3\pi/4} + \int_0^{3\pi/4} (\sin x + \cos x)\,dx$$

$$= 2\int_0^{3\pi/4} (\sin x + \cos x)dx - \left[\underbrace{x[\sin x + \cos x]}_{\text{zero}}\right]_0^{\frac{3\pi}{4}} = 2\left(\sqrt{2}+1\right)$$

(b) $\int_{\frac{\pi}{2}}^{\pi} x^{\sin x+1}\left(\frac{1}{x} + \cos x\,\ell n x + \frac{\sin x}{x}\right)dx$

$x^{\sin x + 1} = t$

$\ell n\, x + \sin x\, \ell n\, x = \ell n\, t$

$$= \left(\frac{1}{x} + \frac{\sin x}{x} + \ell n \cos x\right)dx = \frac{1}{t}dt$$

$$\int_{\frac{\pi^2}{4}}^{\pi} \frac{t}{t}dt = \pi - \frac{\pi^2}{4}$$

24. $\int_0^1 x\,(\tan^{-1}x)^2\,dx$

Ans. $\frac{\pi}{4}\left(\frac{\pi}{4} - 1\right) + \frac{1}{2}\ell n 2$

Sol. $\int_0^1 \underset{I}{x}\underbrace{\left(\tan^{-1}x\right)^2}_{II}dx$

$$=\left(\left(\tan^{-1}x\right)^2\frac{x^2}{2}\right)_0^1\left(-\int_0^1\frac{2}{x^2+1}\tan^{-1}x\frac{x^2}{2}dx\right)$$

$$=\frac{\pi}{4}\left(\frac{\pi}{4}-1\right)+\frac{1}{2}\ell n2$$

25. Suppose that f, f ' and f '' are continuous on [0, ln 2] and that f (0) = 0, f ' (0) = 3, $f(l\ln 2)=6, f'(l\ln 2)=4$ and $\int_0^{l\ln 2}e^{-2x}\cdot f(x)dx=3$. Find the value of $\int_0^{l\ln 2}e^{-2x}\cdot f''(x)dx$.

दिया है f, f' and f'' अन्तराल [0, ln 2] में सतत् हैं तथा $f(0)=0, f'(0)=3, f(l\ln 2)=6, f'(l\ln 2)=4$ और $\int_0^{l\ln 2}e^{-2x}\cdot f(x)dx$ = 3 है। निश्चित समाकलन $\int_0^{l\ln 2}e^{-2x}\cdot f''(x)dx$ का मान ज्ञात कीजिए।

Ans. 13

Sol. $\int_0^{\ln 2}e^{-2x}\cdot f''(x)dx=e^{-2x}f'(x)+2\int_0^{\ell n2}\underset{I}{e^{-2x}}\underbrace{f'(x)}_{II}dx$

$$=\left[e^{-2x}f'(x)+2\left[e^{-2x}f(x)\right]\right]_0^{\ell n2}+\int_0^{\ell n2}4e^{-2x}f(x)dx$$

= 13

1. $\int_0^1 \frac{dx}{x^2+2x\cos\alpha+1}$ where $-\pi<\alpha<\pi$

Ans. $\begin{cases} \frac{1}{2} & \alpha = 0 \\ \frac{\alpha}{2\sin\alpha} & \alpha \neq 0 \end{cases}$

Sol. $I = \int_0^1 \frac{dx}{x^2+2x\cos\alpha+1}$

$$= \begin{cases} \int_0^1 \frac{dx}{x^2+2x+1} = \frac{1}{2} & \alpha = 0 \\ \int_0^1 \frac{dx}{(x+\cos\alpha)^2+\sin^2\alpha} & \alpha \neq 0 \end{cases}$$

$$= \begin{cases} \frac{1}{2} & \alpha = 0 \\ \left[\frac{1}{\sin\alpha}\tan^{-1}\left(\frac{x+\cos\alpha}{\sin\alpha}\right)\right]_0^1 & \alpha \neq 0 \end{cases}$$

$$= \begin{cases} \frac{1}{2} & \alpha = 0 \\ \frac{1}{\sin\alpha}\left[\tan^{-1}\cot\left(\frac{\alpha}{2}\right)-\tan^{-1}\cot\alpha\right] & \alpha \neq 0 \end{cases}$$

$$= \begin{cases} \frac{1}{2} & \alpha = 0 \\ \frac{\alpha}{2\sin\alpha} & \alpha \neq 0 \end{cases}$$

2. $\int_a^b \frac{dx}{\sqrt{1+x^2}}$ where $a = \frac{e-e^{-1}}{2}$ and $b = \frac{e^2-e^{-2}}{2}$

Ans. 1

Sol. $= \ell n \left| x + \sqrt{1+x^2} \right|_a^b$

$= \ell n \left| \dfrac{b+\sqrt{1+b^2}}{a+\sqrt{1+a^2}} \right|$

Now, $b = \dfrac{e^4-1}{2e^2}$

$b + \sqrt{1+b^2} = \dfrac{e^4-1}{2e^2} + \sqrt{1+\left(\dfrac{e^4-1}{2e^2}\right)^2}$

$= \dfrac{e^4-1}{2e^2} + \sqrt{\dfrac{4e^4+e^8+1-2e^4}{4e^4}} = \dfrac{e^4-1+e^4+1}{2e^2} = e^2$

$a + \sqrt{1+a^2} = \dfrac{e^2-1}{2e} + \sqrt{1+\left(\dfrac{e^2-1}{2e}\right)^2} = e$

$\Rightarrow \ell n \left| \dfrac{b+\sqrt{1+b^2}}{a+\sqrt{1+a^2}} \right| = \ell ne = 1$

3. $\displaystyle\int_{0^+}^{1} \frac{x^x (x^{2x}+1)(\ell n\, x+1)}{x^{4x}+1} dx =$

Ans. 0

Sol. Put $x^x = t$

$\displaystyle\int_1^1 \frac{t^2+1}{t^4+1} dt = 0$

4. $\displaystyle\int_0^1 x^5 \sqrt{\frac{1+x^2}{1-x^2}}\, dx =$

Ans. $\dfrac{3\pi+8}{24}$

Sol. $I = \displaystyle\int_0^1 x^5 \sqrt{\frac{1+x^2}{1-x^2}} dx$

let $x^2 = \cos 2\theta$

$$I = -\int_{\frac{\pi}{4}}^{0} \cot\theta\cdot\left(\cos^2 2\theta\right)\sin 2\theta d\theta = -2\int_{\frac{\pi}{4}}^{0} \cos^2\theta\cos^2 2\theta d\theta$$

$$= \frac{1}{2}\int_{0}^{\frac{\pi}{4}} \left(\cos 3\theta + \cos\theta\right)^2 d\theta$$

$$= \frac{1}{2}\int_{0}^{\frac{\pi}{4}} \left(\cos^2 3\theta + \cos^2\theta + \cos 4\theta + \cos 2\theta\right) d\theta$$

$$= \frac{1}{4}\left[\theta + \frac{\sin 6\theta}{6} + \theta + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{4}} + \frac{1}{2}\left[\frac{\sin 4\theta}{4} + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{8} + \frac{1}{3} - \frac{1}{8} + \frac{1}{8} = \frac{3\pi + 8}{24}$$

5. Suppose that the function f, g, f ' and g ' are continuous over [0, 1], $g(x) \neq 0$ for $x \in [0, 1]$, $f(0) = 0$, $g(0) = \pi$, $f(1) = \frac{2009}{2}$ and $g(1) = 1$. Find the value of the definite integral,

$$\int_0^1 \frac{f(x)\cdot g'(x)\left\{g^2(x)-1\right\} + f'(x)\cdot g(x)\left\{g^2(x)+1\right\}}{g^2(x)} dx .$$

माना फलन f, g, f ' और g ' अन्तराल [0, 1] में सतत् हैं, $x \in [0, 1]$ के लिए $g(x) \neq 0$, $f(0) = 0$, $g(0) = \pi$, $f(1) = \frac{2009}{2}$ और $g(1) = 1$. निश्चित समाकलन $\int_0^1 \frac{f(x)\cdot g'(x)\left\{g^2(x)-1\right\} + f'(x)\cdot g(x)\left\{g^2(x)+1\right\}}{g^2(x)} dx$ का मान ज्ञात कीजिए।

Ans. 2009

Sol. $$I = \int_0^1 \frac{f(x)\cdot g'(x)\left\{g^2(x)-1\right\} + f'(x)\cdot g(x)\left\{g^2(x)+1\right\}}{g^2(x)} dx$$

$$= \int_0^1 \frac{d}{dx}\left(f(x)g(x)\right)dx + \int_0^1 \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right)dx$$

$$\left[f(x)g(x)\right]_0^1+\left[\frac{f(x)}{g(x)}\right]_0^1$$

$$f(1).g(1)-f(0).g(0)+\frac{f(1)}{g(1)}-\frac{f(0)}{g(0)}$$

$$\frac{2009}{2}+\frac{2009}{2}=2009$$

6. $\int_0^{\pi/4}\frac{\sin\theta+\cos\theta}{9+16\sin 2\theta}\,d\theta=$

Ans. $\frac{1}{20}\,\ell n\,3$

Sol. $I=\int_0^{\pi/4}\frac{\sin\theta+\cos\theta}{25-16(1-\sin 2\theta)}.d\theta$

$$I=\int_0^{\frac{\pi}{4}}\frac{\sin\theta+\cos\theta}{25-16(\sin\theta-\cos\theta)^2}d\theta$$

let $\sin\theta-\cos\theta=t$

$$=\int_{-1}^{0}\frac{dt}{5^2-(4t)^2}=\frac{1}{40}\ell n\left|\frac{5-4t}{5+4t}\right|_{-1}^{0}$$

$$=\frac{1}{40}\log\left|\frac{9}{1}\right|=\frac{1}{20}\log 3$$

7. $\int_0^{\pi}\theta\,\sin^2\theta\cos\theta\,d\theta=$

Ans. $-\frac{4}{9}$

Sol. $I=\int_0^{\pi}\theta\sin^2\theta\cos\theta d\theta=\left[\frac{\theta\sin^3\theta}{3}\right]_0^{\pi}-\int_0^{\pi}\frac{\sin^3\theta}{3}d\theta$

$$=-\frac{1}{3}\int_0^{\pi}\sin^3\theta d\theta=-\frac{1}{3}\int_0^{\pi}(1-\cos^2\theta)\sin\theta d\theta$$

let $\cos\theta=t$

$$= \frac{1}{3}\int_{1}^{-1}\left(1-t^2\right)dt = \frac{1}{3}\left[t-\frac{t^3}{3}\right]_{1}^{-1} = -\frac{4}{9}$$

8. $\int_{0}^{\pi/2} \frac{1+2\cos x}{(2+\cos x)^2}\, dx =$

Ans. $\frac{1}{2}$

Sol. $I = \int_{0}^{\frac{\pi}{2}} \frac{1+2\cos x}{\left(2+\cos x\right)^2} dx$

$$I = \int_{0}^{\frac{\pi}{2}} \frac{\sin^2 x+\cos^2 x+2\cos x}{\left(2+\cos x\right)^2} dx$$

$$= \int_{0}^{\frac{\pi}{2}} \frac{\cos x}{\left(2+\cos x\right)} dx + \int_{0}^{\frac{\pi}{2}} \frac{\sin^2 x}{\left(2+\cos x\right)^2} dx$$

$$= \int_{0}^{\frac{\pi}{2}} \underbrace{\cos x}_{II} \cdot \underbrace{\frac{1}{2+\cos x}}_{I} dx + \int_{0}^{\frac{\pi}{2}} \frac{\sin^2 x}{\left(1+\cos x\right)^2} dx$$

$$= \left[\frac{1}{2+\cos x}\sin x\right]_{0}^{\frac{\pi}{2}} - \int_{0}^{\frac{\pi}{2}} \frac{\sin^2 x dx}{\left(2+\cos x\right)^2} + \int_{0}^{\frac{\pi}{2}} \frac{\sin^2 x}{\left(2+\cos x\right)^2} dx$$

$$= \frac{1}{2} - 0$$

$$= \frac{1}{2}$$

9. $\int_{0}^{\pi/2} \frac{x+\sin x}{1+\cos x}\, dx =$

Ans. $\frac{\pi}{2}$

Sol. Put $\sin x = \dfrac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}$

$$\cos x = \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}$$

$$I = \int_0^{\pi/2} \frac{x+\sin x}{1+\cos x}dx = \int_0^{\frac{\pi}{2}} \frac{x+\dfrac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}{1+\dfrac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}}dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{x\left(\sec^2\frac{x}{2}\right)+2\tan\frac{x}{2}}{2}dx = \left[x\tan\frac{x}{2}\right]_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

10. Let $A = \int_{3/4}^{4/3} \frac{2x^2+x+1}{x^3+x^2+x+1}dx$ then find the value of e^A.

माना $A = \int_{3/4}^{4/3} \frac{2x^2+x+1}{x^3+x^2+x+1}dx$ हो, तो e^A का मान ज्ञात कीजिए।

Ans. $\frac{16}{9}$

Sol. $I = \int \frac{(x^3+x^2+x+1)+(x^2-x^3)}{x^3+x^2+x+1}dx$

$$= x + \int \frac{(x^2+1)-(x^3+1)}{(x^2+1)\cdot(x+1)}dx = x + \ln(x+1) - \int \frac{x^2+1-x}{x^2+1}dx$$

$= x + \ell n\,(x+1) - \left(x - \frac{1}{2}\ell n\,(x^2+1) \right)$

$= \ell n\,(x+1) + \ell n\,\sqrt{x^2+1}$;

hence $A = \left[\ell n\left(x+1\right) + \ell n\sqrt{x^2+1} \right]_{\frac{3}{4}}^{\frac{4}{3}} = \ell n\left(\frac{\frac{35}{9}}{\frac{35}{16}} \right) = \ell n\left(\frac{16}{9} \right)$

$\Rightarrow e^A = \frac{16}{9}$

11. $\int_0^1 \frac{2 - x^2}{(1+x)\sqrt{1-x^2}}\,dx$

Ans. $\frac{\pi}{2}$

Sol. $\int_0^1 \frac{1+\left(1-x^2\right)}{\left(1+x\right)\left(\sqrt{1-x^2}\right)}dx = \underbrace{\int_0^1 \frac{1}{\left(1+x\right)\sqrt{1-x^2}}dx}_{I_1} + \underbrace{\int_0^1 \frac{\sqrt{1-x^2}}{x+1}dx}_{I_2}$

$I_1 = \int_0^1 \frac{1}{\left(1+x\right)\sqrt{1-x^2}}dx$

let $x = \cos\theta$

$I_1 = -\int_{\frac{\pi}{2}}^0 \frac{1}{1+\cos\theta}d\theta = -\int_{\frac{\pi}{2}}^0 \frac{1}{2\cos^2\frac{\theta}{2}}d\theta = 1$

$I_2 = \int_0^1 \sqrt{\frac{1-x}{1+x}}dx$

let $x = \cos 2\theta$

$I_2 = 2\int_0^{\frac{\pi}{4}} \sin 2\theta \tan\theta d\theta = 4\int_0^{\frac{\pi}{4}} \sin^2\theta d\theta$

$$= 2\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{4}} = \frac{\pi}{2} - 1$$

$$I_1 + I_2 = \frac{\pi}{2}$$

12. $\int_{-1}^{1}\left(\frac{d}{dx}\left(\frac{1}{1+e^{1/x}}\right)\right)dx$

Ans. $\frac{1-e}{1+e}$

Sol. $\int_{-1}^{1}\left(\frac{d}{dx}\left(\frac{1}{1+e^{1/x}}\right)\right)dx = \left(\frac{1}{1+e^{1/x}}\right)_{-1}^{1} = \frac{1}{1+e} - \frac{1}{1+e^{-1}} = \frac{1-e}{1+e}$

13. $\int_{1}^{e}\frac{dx}{\ln(x^x e^x)}$

Ans. $\ell n\, 2$

Sol. $I = \int_{1}^{e}\frac{dx}{x(1+\ln x)}$

Let $1 + \ell nx = t$

$$I = \int_{1}^{2}\frac{dt}{t} = \ell n 2$$

14. $\int_{0}^{\pi}\left[\cos^2\left(\frac{3\pi}{8} - \frac{x}{4}\right) - \cos^2\left(\frac{11\pi}{8} + \frac{x}{4}\right)\right]dx =$

Ans. $\sqrt{2}$

Sol. $\int_{0}^{\pi}\left(\sin\left(\frac{7\pi}{4}\right).\sin\left(\pi + \frac{x}{2}\right)\right)dx$

$$= \frac{1}{\sqrt{2}}\int_{0}^{\pi}\sin\frac{x}{2}dx = -\sqrt{2}\left[\cos\left(\frac{x}{2}\right)\right]_0^{\pi}$$

$$= \sqrt{2}$$

15. If $f(\pi)=2$ and $\int_0^{\pi}\left(f(x)+f''(x)\right)\sin x\ dx=5$, then find f(0)

यदि $f(\pi)=2$ तथा $\int_0^{\pi}\left(f(x)+f''(x)\right)\sin x\ dx=5$, है तो f(0) ज्ञात किजिये

Ans. 3

Sol. $\int_0^{\pi}f(x).\sin x.dx+\int_0^{\pi}f''(x).\sin x dx=5$

$$\left[-f(x)\cos x\right]_0^{\pi}+\int_0^{\pi}f'(x).\cos x dx+\int_0^{\pi}f''(x).\sin x dx=5$$

$$\Rightarrow f(\pi)+f(0)=5\Rightarrow f(0)=3$$

16. $\int_a^b\frac{|x|}{x}dx=$

Ans. $|b|-|a|$

Sol. $$\int_a^b\frac{|x|}{x}dx=\begin{cases}\int_a^b dx=b-a & a>0,b>0\\ -\int_a^b dx=a-b & a<0,b<0\\ -\int_a^0 dx+\int_0^b dx=a+b & a<0,b>0\\ -\int_b^a\frac{|x|}{x}=-b-a & a>0,b<0\end{cases}=|b|-|a|$$

17. $\int_{\ell n2}^{\ell n3}f(x)dx$, where $f(x)=e^{-x}+2e^{-2x}+3e^{-3x}+..\infty$

$\int_{\ell n2}^{\ell n3}f(x)dx$, जहां $f(x)=e^{-x}+2e^{-2x}+3e^{-3x}+..\infty$

Ans. $\frac{1}{2}$

Sol. $I=\int_{\ell n2}^{\ell n3}f(x)dx$

$$= -1\left[e^{-x} + e^{-2x} + e^{-3x} + ... \right]_{\ell n2}^{\ell n3}$$

$$= -1\left[\frac{e^{-x}}{1-e^{-x}} \right]_{\ell n2}^{\ell n3} = -1\left[\frac{1}{e^{x}-1} \right]_{\ell n2}^{\ell n3} = \frac{1}{2}$$

18. $\int_0^{\pi/2} \sqrt{\frac{\sec x - \tan x}{\sec x + \tan x}} \frac{\operatorname{cosec} x}{\sqrt{1+2\operatorname{cosec} x}}\, dx$

Ans. $\frac{\pi}{3}$

Sol. $I = \int_0^{\pi/2} \sqrt{\frac{\sec x - \tan x}{\sec x + \tan x}} \frac{\cos ecx}{\sqrt{1+2\cos ecx}} dx$

$$= \int_0^{\pi/2} (\sec x - \tan x) \frac{dx}{\sqrt{\sin^2 x + 2\sin x}} = \int_0^{\pi/2} \frac{1-\sin x}{\cos x} \frac{dx}{\sqrt{(\sin x+1)^2 - 1}}$$

$$= \int_0^{\pi/2} \frac{\cos x}{(1+\sin x)} \frac{dx}{\sqrt{(1+\sin x)^2 - 1}} = \int_1^2 \frac{dt}{t\sqrt{t^2-1}} = \sec^{-1} t \Big]_1^2 = \sec^{-1}2 - \sec^{-1}1 = \frac{\pi}{3}$$

Alternative : put $1 + 2\operatorname{cosec} x = t^2$

19. $\int_0^1 x\, f''(x)\, dx$, where $f(x) = \cos(\tan^{-1}x)$

$\int_0^1 x\, f''(x)\, dx$, जहां $f(x) = \cos(\tan^{-1}x)$

Ans. $1 - \frac{3}{2\sqrt{2}}$

Sol. $I = \int_0^1 xf''(x)dx = \left[xf'(x) \right]_0^1 - \int_0^1 f'(x)\, dx$

$$= -\frac{1}{2\sqrt{2}} - \left[f(x) \right]_0^1$$

$$= -\frac{1}{2\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} = 1 - \frac{3}{2\sqrt{2}}$$

20. (a) If g (x) is the inverse of f (x) and f (x) has domain $x \in [1, 5]$, where f (1) = 2 and

$f(5) = 10$ then find the value of $\int_1^5 f(x)\,dx + \int_2^{10} g(y)\,dy$.

(b) Suppose f is continuous, $f(0) = 0, f(1) = 1, f'(x) > 0$ and $\int_0^1 f(x)\,dx = \frac{1}{3}$. Find the value of the definite integral $\int_0^1 f^{-1}(y)\,dy$.

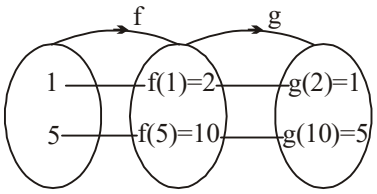
(a) यदि g (x), f (x) का व्युत्क्रम है तथा x $\in$ [1, 5], f (x) का प्रान्त है जहाँ f (1) = 2 तथा f (5) = 10 हो, तो $\int_1^5 f(x)\,dx + \int_2^{10} g(y)\,dy$ का मान ज्ञात कीजिए।

(b) यदि f सतत् फलन है, $f(0) = 0, f(1) = 1, f'(x) > 0$ तथा $\int_0^1 f(x)\,dx = \frac{1}{3}$ हो तो निश्चित समाकलन $\int_0^1 f^{-1}(y)\,dy$ का मान ज्ञात करो।

Ans. (a) 48 (b) 2/3

Sol. $y = f(x) \quad \Rightarrow \quad x = f^{-1}(y) = g(y)$

$dy = f'(x)\,dx$



$\therefore \quad I = \int_1^5 f(x)\,dx + \int_1^5 x\,f'(x)\,dx$

where y is 2 then x = 1

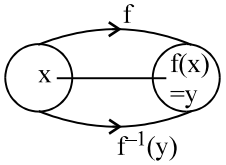
y is 10 then x = 5

$\therefore \quad I = \int_1^5 \big(f(x) + x\,f'(x)\big)\,dx = x\,f(x)\big|_1^5 = 5\,f(5) - f(1) = 5 \cdot 10 - 2 = 48$

(b) $f(0) = 0 \quad \Rightarrow \quad f^{-1}(0) = 0$

and $f(1) = 1 \quad \Rightarrow \quad f^{-1}(1) = 1$

now $I = \int_0^1 f^{-1}(y)\,dy$



$y = f(x) \quad \Rightarrow \quad x = f^{-1}(y)$. Also $dy = f'(x)\,dx$

$= \int_0^1 x\,f'(x)\,dx = \; x\,f(x)\big]_0^1 - \int_0^1 f(x)\,dx = 1 - \frac{1}{3} = \frac{2}{3}$

21. $\int_0^{\infty} \frac{dx}{(x+\sqrt{x^2+1})^3} =$

(A) 3/8 (B) 1/8 (C) –3/8 (D) None of these

Ans. A

Sol. $I = \int_0^{\infty} \frac{dx}{(x+\sqrt{x^2+1})^3}$

let $x = \tan\theta$

$$I = \int_0^{\pi/2} \frac{\sec^2\theta d\theta}{(\tan\theta+\sec\theta)^3}$$

$$I = \int_0^{\pi/2} \frac{\cos\theta.d\theta}{(1+\sin\theta)^3}$$

let $1+\sin\theta = P$

$$I = \int_1^2 \frac{dp}{p^3} = \frac{-1}{2}\left[\frac{1}{p^2}\right]_1^2 = \frac{3}{8}$$

22. $\int_1^{e^{37}} \frac{\pi\sin(\pi\ell nx)}{x}dx =$

(A) 1 (B) 2 (C) e (D) 37

Ans. B

Sol. $I = \int_1^{e^{37}} \frac{\pi\sin(\pi\ell nx)}{x}dx$

Let $\ell nx = t$

$$I = \int_0^{37} \pi\sin(\pi t).dt = \left[-\cos\pi t\right]_0^{37}$$

$= 2$

23. $\int_{ka}^{kb} f(x)dx =$

(A) $k^2\int_a^b f(x)dx$ (B) $k\int_a^b f(x)dx$ (C) $k\int_a^b f(kx)\,dx$ (D) $k^3\int_a^b f(k\,x)dx$

Ans. C

Sol. $\int_{ka}^{kb} f(x)dx$

Let $x = kt$

$dx = kdt$

$= k\int_{a}^{b} f(kt)dt$

$= k\int_{a}^{b} f(kx)\,dx$

24. $\int_{a-c}^{b-c} f(x+c)\,dx =$

(A) $\int_{a}^{b} f(x+c)dx$ (B) $\int_{a}^{b} f(x)dx$ (C) $\int_{a-2c}^{b-2c} f(x)dx$ (D) $\int_{a}^{b} f(x+2c)dx$

Ans. B

Sol. $I = \int_{a-c}^{b-c} f(x+c)dx$

Let $x + c = t$

$I = \int_{a}^{b} f(t)\,dt = \int_{a}^{b} f(x)\,dx$

25. The value of $\int_{0}^{2} \frac{dx}{(1-x)^2}$ is :

(A) –2 (B) 0 (C) 15 (D) indeterminate

$\int_{0}^{2} \frac{dx}{(1-x)^2}$ का मान होगा :

(A) –2 (B) 0 (C) 15 (D) अनिर्धाय

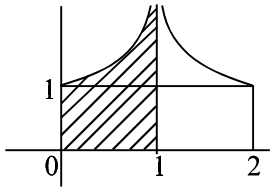
Ans. D

Sol. $\int_{0}^{2} \frac{dx}{(1-x)^2} = \int_{0}^{1^-} \frac{dx}{(1-x)^2} + \int_{1^+}^{2} \frac{dx}{(1-x)^2}$

$$= \frac{1}{1-x}\Big]_0^{1^-} + \frac{1}{1-x}\Big]_{1^+}^{2}$$

$= (\infty - 1) + (-1) - (-\infty) \Rightarrow$ indeterminant

Note that the shaded area is divergent



26. If $\int_{-1}^{4} f(x)dx = 4$ and $\int_{2}^{4}[3-f(x)]dx = 7$, then $\int_{2}^{-1} f(x)dx$ is equal to :

(A) 2 (B) –2 (C) –5 (D) None of these

यदि $\int_{-1}^{4} f(x)dx = 4$ तथा $\int_{2}^{4}[3-f(x)]dx = 7$, तो $\int_{2}^{-1} f(x)dx$ बराबर है :

(A) 2 (B) –2 (C) –5 (D) इनमें से कोई नही

Ans. C

Sol. $3\int_{2}^{4} dx - \int_{2}^{4} f(x).dx = 7$

$\Rightarrow \int_{2}^{4} f(x)\,dx = -1$

Now, $\int_{-1}^{2} f(x)dx + \int_{2}^{4} f(x)dx = 4$

$\int_{-1}^{2} f(x).dx = 5$

$\int_{2}^{-1} f(x)dx = -5$

27. Given $\int_{1}^{2} e^{x^2}\,dx = a$, then the value of $\int_{e}^{e^4} \sqrt{\ell n(x)}\,dx$ is :

दिया है कि $\int_{1}^{2} e^{x^2}\,dx = a$, तब $\int_{e}^{e^4} \sqrt{\ell n(x)}\,dx$ का मान है :

(A) $e^4 - e$ (B) $e^4 - a$ (C) $2e^4 - a$ (D) $2e^4 - e - a$

Ans. D

Sol. Given : $\int_1^2 e^{x^2}\,dx = a$

$\Rightarrow I = \int_e^{e^4} \sqrt{\ell n(x)}dx$

let $\ell nx = t^2$

$\Rightarrow dx = e^{t^2}\cdot 2tdt$

$\Rightarrow I = \int_1^2 e^{t^2}\cdot 2t^2 dt = \left[te^{t^2} \right]_1^2 - \int_1^2 e^{t^2} dt$

$= 2e^4 - e - a$

28. The true solution set of the inequality, $\sqrt{5x - 6 - x^2} + \left(\frac{\pi}{2}\int_0^x dz \right) > x\int_0^{\pi} \sin^2 x\,dx$ is :

असमिका $\sqrt{5x - 6 - x^2} + \left(\frac{\pi}{2}\int_0^x dz \right) > x\int_0^{\pi} \sin^2 x\,dx$ का हल होगा :

(A) R (B) (1, 6) (C) (–6, 1) (D) (2, 3)

Ans. D

Sol. $\sqrt{5x - 6 - x^2} + \frac{\pi x}{2} > x\frac{\pi}{2} \Rightarrow 5x - 6 - x^2 > 0$

$\Rightarrow x \in (2,3)$

1. If $f(x)=\begin{cases}x^2, & \text{when } 0\le x<1\\ \sqrt{x}, & \text{when } 1\le x<2\end{cases}$, then $\int_0^2 f(x)\,dx$ equals :

(A) $\frac{1}{3}(4\sqrt{2}-1)$ (B) $\frac{1}{3}(4\sqrt{2}+1)$ (C) 0 (D) does not exist

यदि $f(x)=\begin{cases}x^2, & \text{when } 0\le x<1\\ \sqrt{x}, & \text{when } 1\le x<2\end{cases}$, तो $\int_0^2 f(x)\,dx$ बराबर है :

(A) $\frac{1}{3}(4\sqrt{2}-1)$ (B) $\frac{1}{3}(4\sqrt{2}+1)$ (C) 0 (D) विद्यमान नहीं है

Ans. A

Sol. $\int_0^2 f(x)dx=\int_0^1 f(x)dx+\int_1^2 f(x)dx=\int_0^1 x^2dx+\int_1^2 \sqrt{x}dx$

$$\left(\frac{x^3}{3}\right)_0^1+\left(\frac{2}{3}x^{\frac{3}{2}}\right)_1^2=\frac{\left(4\sqrt{2}-1\right)}{3}$$

2. $\int_{1/e}^{e}|\log x|\,dx$ is :

(A) $e^{-1}-1$ (B) $2\left(1-\frac{1}{e}\right)$ (C) $1-\frac{1}{e}$ (D) None of these

Ans. B

Sol. $\int_{1/e}^{e}|\log x|dx=\int_{1/e}^{1}-\ln x dx+\int_1^e \ln x dx$

$$=-\left[x\left(\ln x-1\right)\right]_{1/e}^{1}+\left[x\left(\ln x-1\right)\right]_1^e$$

$$=1-\frac{2}{e}+1=2\left(1-\frac{1}{e}\right)$$

3. $\int_0^1 |\sin 2\pi x| dx$ is equal to :

$\int_0^1 |\sin 2\pi x| dx$ का मान बराबर होगा :

(A) 0 (B) $-1/\pi$ (C) $1/\pi$ (D) $2/\pi$

Ans. D

Sol. $\int_0^1 |\sin 2\pi x| dx = \int_0^{1/2} \sin 2\pi x dx - \int_{1/2}^1 \sin 2\pi x dx$

$= -\frac{\cos 2\pi x}{2\pi}\Big)_0^{1/2} + \frac{1}{2\pi}(\cos 2\pi x)_{\frac{1}{2}}^1$

$= -\frac{1}{2\pi}[-1-1] + \frac{1}{2\pi}[2] = \frac{2}{\pi}$

4. $\int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$ is equal to :

$\int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx$ बराबर है :

(A) $\frac{\pi}{2} + \log 2$ (B) $\frac{\pi}{2} - \log 2$ (C) $\frac{\pi}{4} + \log 2$ (D) $\frac{\pi}{4} - \log 2$

Ans. B

Sol. Let $x = \tan\theta$

$\int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx = \int_0^{\pi/4} \sin^{-1}(\sin 2\theta) \times \sec^2\theta . d\theta$

$= 2\int_0^{\pi/4} \theta . \sec^2\theta d\theta$

$= 2\left[[\theta \tan\theta]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan\theta d\theta \right]$

$= \frac{\pi}{2} - \ell n 2$

5. $\int_{-1}^{3/2} |x \sin \pi x| dx$ is :

(A) $(3/\pi) + 1/\pi^2$ (B) $(3/\pi) - 1/\pi^2$ (C) $1/\pi\ (\pi + 1)$ (D) None of these

Ans. A

Sol. $\int_{-1}^{3/2} |x \sin \pi x| dx = \int_{-1}^{1} x \sin \pi x\ dx - \int_{1}^{\frac{3}{2}} x \sin \pi x\, dx$

$= 2\int_{0}^{1} x \sin \pi x\, dx - \int_{1}^{\frac{3}{2}} x \sin \pi x\, dx$

$= 2\left[\left(\frac{x \cos \pi x}{-\pi}\right)_0^1 + \frac{1}{\pi}\int_0^1 \cos \pi x dx\right] - \left[\left(\frac{x \cos \pi x}{-\pi}\right)_1^{\frac{3}{2}} + \frac{1}{\pi}\int_1^{\frac{3}{2}} \cos \pi x dx\right]$

$= 2\left[\frac{1}{\pi} + 0\right] - \left[-\frac{1}{\pi} - \frac{1}{\pi^2}\right]$

$= \left(\frac{3}{\pi}\right) + \frac{1}{\pi^2}$

6. $\int_{-1}^{1} \left(\sqrt{1+x+x^2} - \sqrt{1-x+x^2}\right) dx$ equals :

$\int_{-1}^{1} \left(\sqrt{1+x+x^2} - \sqrt{1-x+x^2}\right) dx$ बराबर है :

(A) 1 (B) 0 (C) $\sqrt{2}$ (D) 2

Ans. B

Sol. $I = \int_{-1}^{1} \left(\sqrt{1+x+x^2} - \sqrt{1-x+x^2}\right) dx$

Apply king property

$I = \int_{-1}^{1} \left(\sqrt{1-x+x^2} - \sqrt{1+x+x^2}\right) dx$

$\Rightarrow 2I = 0$

$\Rightarrow I = 0$

7. The value of the integral $\int_{-2}^{2}\left|1-x^2\right|dx$ is :

(A) 0 (B) 4 (C) 2 (D) None of these

समाकल $\int_{-2}^{2}\left|1-x^2\right|dx$ का मान है :

(A) 0 (B) 4 (C) 2 (D) इनमें से कोई नहीं

Ans. B

Sol. $=2\int_{0}^{2}|1-x^2|\,dx=2\int_{0}^{2}|x^2-1|\,dx$

$=2\int_{0}^{1}-\left(x^2-1\right)dx+2\int_{1}^{2}\left(x^2-1\right)dx$

$=2\left(x-\frac{x^3}{3}\right)_0^1+2\left(\frac{x^3}{3}-x\right)_1^2$

$=\frac{4}{3}+2\left[\left(\frac{8}{3}-2\right)-\left(\frac{1}{3}-1\right)\right]=\frac{4}{3}+\frac{8}{3}=4$

8. If $f:R\to R$ and $g:R\to R$ are two continuous functions, then the value of the integral

$\int_{-\pi/2}^{\pi/2}\left[f(x)+f(-x)\right]\left[g(x)-g(-x)\right]dx$ is :

यदि $f:R\to R$ तथा $g:R\to R$ दो संतत फलन हों, तो समाकल $\int_{-\pi/2}^{\pi/2}\left[f(x)+f(-x)\right]\left[g(x)-g(-x)\right]dx$ का मान है :

(A) π (B) 1 (C) – 1 (D) 0

Ans. D

Sol. $I=\int_{-\pi/2}^{\pi/2}\left[f(x)+f(-x)\right]\left[g(x)-g(-x)\right]dx$

Apply king property

$I=\int_{-\pi/2}^{\pi/2}\left[f(-x)+f(x)\right]\left[g(-x)-g(x)\right]dx$

$\Rightarrow 2I = 0$

$\Rightarrow I = 0$

9. $\int_{-1}^{1} x^{17} \cos^4 x \, dx$ is :

(A) –2 (B) 2 (C) 0 (D) 1

Ans. C

Sol. $I = \int_{-1}^{1} x^{17} \cos^4 x \, dx$

Apply king property

$I = -\int_{-1}^{1} x^{17} \cos^4 x \, dx$

$\Rightarrow 2I = 0$

$\Rightarrow I = 0$

10. $\int_{-a}^{a} \sin x \, f(\cos x) dx =$

(A) f(a) (B) – f(a) (C) 2 f(a) (D) 0

Ans. D

Sol. $\int_{-a}^{a} \underbrace{\sin x f(\cos x)}_{\text{odd function}} dx = 0$

11. $\int_{-\pi}^{\pi} (1 - x^2) \sin x \cos^2 x \, dx =$

(A) 0 (B) $\pi - \frac{\pi^3}{3}$ (C) $2\pi - \pi^3$ (D) $\frac{7}{4} - 2\pi^3$

Ans. A

Sol. $\int_{-\pi}^{\pi} \underbrace{\left(1 - x^2\right) \sin x \cos^2 x}_{\text{odd function}} dx = 0$

12. $\int_{-2}^{2} (ax^3 + bx + c) dx$, value depends upon :

(A) Value of a (B) Value of b (C) Value of c (D) Value of a and b

$\int_{-2}^{2}(ax^3+bx+c)dx$, का मान निम्न पर निर्भर करता है :

(A) a का मान (B) b का मान (C) c का मान (D) a तथा b का मान

Ans. C

Sol. $\int_{-2}^{2}\underbrace{\left(ax^3+bx\right)}_{\text{odd function}}dx+2c\int_{0}^{2}dx$

$= 0 + 2c\,(2) = 4c$

13. $\int_{-1}^{1}\sin^{11}x\,dx =$

(A) $\frac{10}{11}\cdot\frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}$ (B) $\frac{10}{11}\cdot\frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}\cdot\frac{\pi}{3}$ (C) 1 (D) 0

Ans. D

Sol. $I=\int_{-1}^{1}\underbrace{\sin^{11}x}_{\text{odd function}}\,dx=0$

14. The value of $\int_{0}^{\pi/2}\frac{\sin x}{\sin x+\cos x}\,dx$ is :

$\int_{0}^{\pi/2}\frac{\sin x}{\sin x+\cos x}\,dx$ का मान है :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) π (D) 2π

Ans. B

Sol. $I=\int_{0}^{\frac{\pi}{2}}\frac{\sin x}{\sin x+\cos x}dx$

Apply King Property

$I=\int_{0}^{\frac{\pi}{2}}\frac{\cos x}{\cos x+\sin x}dx$

$\Rightarrow 2I=\frac{\pi}{2}\Rightarrow I=\frac{\pi}{4}$

15. $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\, dx$ equals :

$\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\, dx$ बराबर है :

(A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi^2}{4}$ (D) $\frac{\pi^2}{2}$

Ans. C

Sol. $I = \int_0^{\pi} \frac{x.\sin x}{1+\cos^2 x} dx$ .....(1)

Apply King Property

$I = \int_0^{\pi} \frac{(\pi - x)\sin x}{1+\cos^2 x} dx$ .....(2)

(1) + (2)

$2I = \int_0^{\pi} \frac{\pi \sin x}{1+\cos^2 x} dx \Rightarrow I = \pi \int_0^{\pi/2} \frac{\sin x}{1+\cos^2 x} dx$

$I = -\pi \left[\tan^{-1}(\cos x) \right]_0^{\frac{\pi}{2}}$

$= \frac{\pi^2}{4}$

16. $\int_0^{\pi/2} \frac{dx}{1+\cot x}$ equals :

$\int_0^{\pi/2} \frac{dx}{1+\cot x}$ बराबर है :

(A) 1 (B) $\frac{\pi}{4}$ (C) $\frac{1}{2}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx$

Apply king property

$= \int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x} dx$

$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} dx$

$\Rightarrow I = \frac{\pi}{4}$

17. $I = \int_0^1 f(x)\, dx$ equals :

(A) $\int_0^1 f(1-x)dx$ (B) $\int_0^1 f(-x)dx$ (C) $2\int_0^{1/2} f(x)dx$ (D) None of these

$\int_0^1 f(x)\, dx$ बराबर है :

(A) $\int_0^1 f(1-x)dx$ (B) $\int_0^1 f(-x)dx$ (C) $2\int_0^{1/2} f(x)dx$ (D) इनमें से कोई नही

Ans. A

Sol. Apply King Property

$I = \int_0^1 f(1-x)dx$

18. The value of $\int_0^{\pi} \frac{x}{1+\sin\ x}\ dx$ is :

$\int_0^{\pi} \frac{x}{1+\sin\ x} dx$ बराबर है :

(A) π (B) $\pi/2$ (C) $\pi/4$ (D) 2π

Ans. A

Sol. $\int_0^{\pi} \frac{x}{1+\sin x}\,dx = I$ .....(1)

Apply King Property

$I = \int_0^{\pi} \frac{\pi - x}{1+\sin x}\,dx$ .....(2)

(1) + (2)

$2I = \int_0^{\pi} \frac{\pi}{1+\sin x} = 2\pi \int_0^{\pi/2} \frac{1}{1+\sin x}\,dx$

$$I = \pi \int_0^{\pi/2} \frac{1}{1+\sin x}\,dx = \pi \int_0^{\frac{\pi}{2}} \frac{\sec^2 \frac{x}{2}\,dx}{\left(1+\tan\frac{x}{2}\right)^2} = \left[-\frac{2\pi}{\tan\frac{x}{2}+1} \right]_0^{\pi/2} = -2\pi\left[\frac{1}{2}-1\right]$$

$= \pi$

19. $\int_0^{a} \frac{dx}{x+\sqrt{a^2-x^2}}$ equals :

$\int_0^{a} \frac{dx}{x+\sqrt{a^2-x^2}}$ बराबर है :

(A) $\pi/2$ (B) $\pi/3$ (C) $\pi/4$ (D) π

Ans. C

Sol. $I = \int_0^{a} \frac{dx}{x+\sqrt{a^2-x^2}}$

let $x = a \sin\theta \Rightarrow dx = a\cos\theta\, d\theta$

$$\Rightarrow I = \int_0^{\pi/2} \frac{a\cos\theta}{a\sin\theta + a\cos\theta} d\theta = \frac{\pi}{4}$$

20. $\int_0^{\pi/2} \frac{\cos^2 x}{2+\sin x+\cos x}\,dx$ is equal to :

(A) $\frac{1}{\sqrt{2}}\left(\tan^{-1}\sqrt{2}+\cot^{-1}\sqrt{2}\right)$ (B) $\frac{1}{\sqrt{2}}\left(\tan^{-1}\sqrt{2}-\cot^{-1}\sqrt{2}\right)$

(C) $\frac{1}{2}\left(\tan^{-1}\sqrt{2}-\cot^{-1}\sqrt{2}\right)$ (D) None of these

$\int_0^{\pi/2}\frac{\cos^2 x}{2+\sin x+\cos x}dx$ बराबर है :

(A) $\frac{1}{\sqrt{2}}\left(\tan^{-1}\sqrt{2}+\cot^{-1}\sqrt{2}\right)$ (B) $\frac{1}{\sqrt{2}}\left(\tan^{-1}\sqrt{2}-\cot^{-1}\sqrt{2}\right)$

(C) $\frac{1}{2}\left(\tan^{-1}\sqrt{2}-\cot^{-1}\sqrt{2}\right)$ (D) इनमें से कोई नहीं

Ans. B

Sol. $I=\int_0^{\pi/2}\frac{\cos^2 x}{2+\sin x+\cos x}dx$ ...(1)

Apply King Property

$I=\int_0^{\pi/2}\frac{\sin^2 x}{2+\sin x+\cos x}dx$ ...(2)

(1) + (2)

$2I=\int_0^{\pi/2}\frac{1}{2+\sin x+\cos x}dx$

$I=\int_0^{\pi/2}\frac{\frac{1}{2}\sec^2\frac{x}{2}}{2+2\tan^2\frac{x}{2}+2\tan\frac{x}{2}+1-\tan^2\frac{x}{2}}dx$

(let $\tan\frac{x}{2}=t$)

$=\int_0^1\frac{dt}{t^2+2t+3}$

$=\left[\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t+1}{\sqrt{2}}\right)\right]_0^1=\frac{1}{\sqrt{2}}\left[\tan^{-1}\left(\sqrt{2}\right)-\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)\right]$

$$= \frac{1}{\sqrt{2}}\left(\tan^{-1}\sqrt{2} - \cot^{-1}\sqrt{2}\right)$$

21. If $f(x) = f(a+b-x)$, then $\int_a^b x\, f(x)\,dx$ is equal to :

यदि $f(x) = f(a+b-x)$, तो $\int_a^b x\, f(x)\,dx$ बराबर है :

(A) $(a+b)\int_a^b f(x)dx$ (B) $\frac{1}{2}(a+b)\int_a^b f(x)dx$ (C) $(b-a)\int_a^b f(x)dx$ (D) $\frac{1}{2}(b-a)\int_a^b f(x)dx$

Ans. B

Sol. $I = \int_a^b x.f(x)dx$ ...(1)

Apply King Property

$$I = \int_a^b (a+b-x) f(a+b-x)\,dx = \int_a^b (a+b-x) f(x)\,dx \quad ...(2)$$

$$2I = \int_a^b (a+b) f(x)\,dx$$

$$I = \frac{a+b}{2}\int_a^b f(x)\,dx$$

22. $\int_0^{\pi} \cos\ mx\ \sin\ nx\ dx = 0$, $m,n \in N$ " :

(A) always (B) when $(n-m) \in N$ (C) when $(n-m)$ is even (D) when $(n-m)$ is odd

$\int_0^{\pi} \cos\ mx\ \sin\ nx\ dx = 0$, $m,n \in N$:

(A) सर्वदा (B) जब $(n-m) \in N$ (C) जब $(n-m)$ सम है (D) जब $(n-m)$ विषम है

Ans. C

Sol. $f(x) = \cos mx.\sin nx$

$$f(\pi - x) = \cos m(\pi - x).\sin(n\pi - nx) = \begin{cases} -f(x) & m, n \text{ even} \\ -f(x) & m, n \text{ odd} \\ f(x) & m \text{ even } n \text{ odd} \\ f(x) & m \text{ odd } n \text{ even} \end{cases}$$

$\int_0^{\pi} f(x)\,dx = 0$ if m & n both are even or odd.

23. $\int_0^{\pi} x \sin x \cos^4 x\,dx$ is equal to :

(A) $3\pi/5$ (B) $2\pi/5$ (C) $\pi/5$ (D) None of these

$\int_0^{\pi} x \sin x \cos^4 x\,dx$ बराबर है :

(A) $3\pi/5$ (B) $2\pi/5$ (C) $\pi/5$ (D) इनमें से कोई नही

Ans. C

Sol. $I = \int_0^{\pi} x \sin x.\cos^4 x dx$ ...(1)

Apply King Property

$I = \int_0^{\pi} (\pi - x)\sin x.\cos^4 x dx$ .....(2)

by (1) + (2)

$2I = \pi\int_0^{\pi} \sin x.\cos^4 x dx$

$I = \pi\int_0^{\pi/2} \sin x.\cos^4 x dx$

$= \frac{\pi}{5}$

24. Let $I_1 = \int_0^{\pi} x \log \sin x\,dx$, $I_2 = \int_0^{\pi} \log \sin x\,dx$, then :

माना $I_1 = \int_0^{\pi} x \log \sin x \, dx, I_2 = \int_0^{\pi} \log \sin x \, dx$, तो :

(A) $I_1 = I_2$ (B) $I_1 = \pi I_2$ (C) $2I_1 = \pi I_2$ (D) $\pi I_1 = I_2$

Ans. C

Sol. $I_1 = \int_0^{\pi} x \log \sin x$ ...(1)

Apply King Property

$I_1 = \int_0^{\pi} (\pi - x) \log \sin x dx$ ...(2)

by (1) + (2)

$2I_1 = \pi \int_0^{\pi} \log \sin x \, dx$

$2I_1 = \pi I_2$

25. If [x] stands for the greatest integer function, the value of $\int_4^{10} \frac{[x^2]}{[x^2 - 28x + 196] + [x^2]} dx$ is :

(A) 0 (B) 1 (C) 3 (D) None of these

यदि [x] महत्तम पूर्णाक फलन दर्शाता है, तब $\int_4^{10} \frac{[x^2]}{[x^2 - 28x + 196] + [x^2]} dx$ का मान है :

(A) 0 (B) 1 (C) 3 (D) इनमें से कोई नही

Ans. C

Sol. $I = \int_4^{10} \frac{[x^2]}{\left[x^2 - 28x + 196\right] + \left[x^2\right]} dx$ ...(1)

Apply King Property

$I = \int_4^{10} \frac{\left[x^2 - 28x + 196\right]}{\left[x^2 - 28x + 196\right] + \left[x^2\right]} dx$ ...(2)

by (1) + (2)

$2I = \int_4^{10} dx$

$\Rightarrow I = 3$

26. If $\int_0^{\pi} \log \sin x dx = k$,then the value of $\int_0^{\pi/4} \log (1+\tan x)\, dx$ is :

यदि $\int_0^{\pi} \log \sin x dx = k$, तो $\int_0^{\pi/4} \log (1+\tan x)\, dx$ का मान है :

(A) $-\frac{k}{4}$ (B) $\frac{k}{4}$ (C) $-\frac{k}{8}$ (D) $\frac{k}{8}$

Ans. C

Sol. $\int_0^{\pi} \log \sin x dx = 2\int_0^{\pi/2} \log \sin x dx$

$= 2\left(-\frac{\pi}{2}\ln 2\right) = -\pi \log 2 = k$

Now $I = \int_0^{\pi/4} \log(1+\tan x)\, dx$ ...(1)

Apply King Property

$I = \int_0^{\pi/4} \log\left(\frac{2}{1+\tan x}\right) dx$ ...(2)

Equation (1) + (2)

$2I = \int_0^{\pi/4} \log 2 dx$

$I = \frac{1}{2}\left(\frac{\pi}{4}\right)\log 2 = \frac{\pi \log 2}{8} = -\frac{k}{8}$

27. If f(x) and g(x) are continuous functions satisfying $f(x) = f(a-x)$ and $g(x) + g(a-x) = 2$, then $\int_0^a f(x)\ g(x)\, dx$ is equal to :

(A) $\int_0^a g(x)\, dx$ (B) $\int_0^a f(x)\, dx$ (C) 0 (D) none of these

यदि f (x) तथा g(x) दो सतंत् फलन, f(x) = f(a – x) तथा g (x) + g(a – x) = 2 संतुष्ठ है, तो $\int_0^a f(x)\ g\ (x)\, dx$ बराबर है :

(A) $\int_0^a g(x)\, dx$ (B) $\int_0^a f(x)\, dx$ (C) 0 (D) इनमें से कोई नही

Ans. B

Sol. $I = \int_0^a f(x).g(x)dx$ ...(1)

Apply King Property

$I = \int_0^a f(x)\left[2 - g(x)\right]dx$ ...(2)

Equation (1) + (2)

$2I = 2\int_0^a f(x)dx$

$I = \int_0^a f(x)dx$

28. The value of the integral $\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4}\cos^{-1}\frac{2x}{1+x^2}\, dx$:

(A) 0 (B) $\frac{\pi}{\sqrt{3}} + \log\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (C) $\frac{1}{\sqrt{3}} + \frac{\pi}{2}\log\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (D) None of these

निम्न समाकलन का मान है $\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4}\cos^{-1}\frac{2x}{1+x^2}\, dx$

(A) 0 (B) $\frac{\pi}{\sqrt{3}} + \log\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (C) $\frac{1}{\sqrt{3}} + \frac{\pi}{2}\log\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (D) इनमें से कोई नहीं

Ans. D

Sol. $I = \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4}\cos^{-1}\frac{2x}{1+x^2}\, dx$ .....(1)

Apply King Property

$$I=\int_{-\frac{1}{\sqrt{3}}}^{1/\sqrt{3}}\frac{x^4}{1-x^4}\left[\pi-\cos^{-1}\left(\frac{2x}{1+x^2}\right)\right]dx \qquad \text{.....(2)}$$

by (1) + (2)

$$2I=\int_{-\frac{1}{\sqrt{3}}}^{1/\sqrt{3}}\frac{\pi x^4}{1-x^4}dx \Rightarrow I=\pi\int_0^{1/\sqrt{3}}\frac{x^4}{1-x^4}dx$$

$$I=\pi\int_0^{1/\sqrt{3}}\frac{1-\left(1-x^4\right)}{1-x^4}dx$$

$$=\pi\int_0^{1/\sqrt{3}}\frac{1}{\left(1-x^2\right)\left(1+x^2\right)}dx-\pi\int_0^{\frac{1}{\sqrt{3}}}dx$$

$$=\frac{\pi}{2}\int_0^{1/\sqrt{3}}\left(\frac{1}{1-x^2}+\frac{1}{1+x^2}\right)dx-\frac{\pi}{\sqrt{3}}=\frac{\pi}{2}\left[\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|_0^{\frac{1}{\sqrt{3}}}\right]-\frac{\pi}{\sqrt{3}}$$

$$=\frac{\pi}{2}\left[\frac{\pi}{6}+\frac{1}{2}\ln\left|\frac{\sqrt{3}+1}{\sqrt{3}+1}\right|\right]-\frac{\pi}{\sqrt{3}}$$

$$=\frac{\pi^2}{12}+\frac{\pi}{4}\ln\left|\frac{\sqrt{3}+1}{\sqrt{3}-1}\right|-\frac{\pi}{\sqrt{3}}$$

29. If f(y) is a non negative continuous function such that $f(y)+f\left(y+\frac{5}{2}\right)=7 \ \forall y\in\left(0,\frac{5}{2}\right)$, then find the value of $\int_0^5 f(y)\,dy$.

यदि f(y) एक अऋणात्मक सतत् फलन है इस प्रकार कि $f(y)+f\left(y+\frac{5}{2}\right)=7 \ \forall y\in\left(0,\frac{5}{2}\right)$, तो $\int_0^5 f(y)\,dy$ का मान ज्ञात कीजिये।

Ans. 35/2

Sol. $$\int_0^5 f(y)dy=\int_0^{5/2} f(y)dy+\int_{5/2}^5 f(y)dy$$

$$y = \frac{5}{2} + t$$

$$= \int_0^{5/2} f(y)dy + \int_0^{5/2} f\left(t + \frac{5}{2}\right)dt$$

$$= \int_0^{5/2} f(y)dy + \int_0^{5/2} f\left(y + \frac{5}{2}\right)dy$$

$$= \int_0^{5/2} \left(f(y) + f\left(y + \frac{5}{2}\right)\right)dy$$

$$= 7\int_0^{5/2} dy = 7\left(\frac{5}{2}\right) = \frac{35}{2}$$

1. $\int_0^{\infty} \frac{\log x}{1+x^2}\, dx$

(A) π (B) 0 (C) $\log 2$ (D) $\pi \log 2$

Ans. B

Sol. Let $x = \tan\theta \Rightarrow dx = \sec^2\theta.d\theta$

$$\int_0^{\infty} \frac{\log x}{1+x^2} dx = \int_0^{\pi/2} \frac{\ell n(\tan\theta).\sec^2\theta.d\theta}{(1+\tan^2\theta)} = \int_0^{\pi/2} \ell n(\tan\theta).d\theta$$

$$= \int_0^{\pi/2} \ell n \sin\theta.d\theta - \int_0^{\pi/2} \ell n \cos\theta.d\theta$$

Apply king property

$$= \int_0^{\pi/2} \ell n \cos\theta.d\theta - \int_0^{\pi/2} \ell n \cos\theta.d\theta$$

$= 0$

2. If $I_1 = \int_0^{3\pi} f(\cos^2 x)\, dx$ and $I_2 = \int_0^{\pi} f(\cos^2 x)\, dx$, then :

यदि $I_1 = \int_0^{3\pi} f(\cos^2 x)\, dx$ तथा $I_2 = \int_0^{\pi} f(\cos^2 x)\, dx$, तो :

(A) $I_1 = I_2$ (B) $I_1 = 3I_2$ (C) $I_1 = 5I_2$ (D) $3\, I_1 = I_2$

Ans. B

Sol. As period of $\cos^2 x$ *is* π

$$I_1 = \int_0^{3\pi} f(\cos^2 x).dx = 3\int_0^{\pi} f(\cos^2 x).dx \quad ___(1)$$

$$I_2 = \int_0^{\pi} f(\cos^2 x).dx \quad ____(2)$$

from equ. (1) and (2)

$I_1 = 3I_2$

3. The value of $\int_0^{[x]} \frac{2^t}{2^{[t]}}\,dt$ is :

(A) $[x]\log 2$ (B) $\frac{[x]}{\log 2}$ (C) $\frac{1}{2}\frac{[x]}{\log 2}$ (D) none of these

$\int_0^{[x]} \frac{2^t}{2^{[t]}}\,dt$ का मान है :

(A) $[x]\log 2$ (B) $\frac{[x]}{\log 2}$ (C) $\frac{1}{2}\frac{[x]}{\log 2}$ (D) इनमें से कोई नही

Ans. B

Sol. $I = \int_0^{[x]} 2^{t-[t]}dt$

$= \int_0^{[x]} 2^{\{t\}}dt$

$= [x]\int_0^1 2^{\{t\}}dt$

$= \frac{[x]}{\log 2}$

4. The value of $\int_0^{16\pi/3} |\sin x|\,dx$ is :

$\int_0^{16\pi/3} |\sin x|\,dx$ का मान है :

(A) 21 (B) 21/2 (C) 10 (D) 11

Ans. B

Sol. $I = \int_0^{5\pi} |\sin x|.dx + \int_{5\pi}^{5\pi+\frac{\pi}{3}} |\sin x|.dx$

$$= 5\int_0^{\pi} |\sin x| .dx + \int_0^{\pi/3} \sin x.dx$$

$$= 10 - (\cos x)_0^{\pi/3} = 10 - \left(\frac{1}{2} - 1\right)$$

$$= 10 + \frac{1}{2} = \frac{21}{2}$$

5. The value of the integral $\int_0^{100} \sin\ (x - [x])\pi\, dx$ is :

समाकल $\int_0^{100} \sin (x - [x])\pi\, dx$ का मान है :

(A) $100/\pi$ (B) $200/\pi$ (C) 100π (D) 200π

Ans. B

Sol. $I = \int_0^{100} \sin(\{x\}\pi).dx = 100\int_0^1 \sin \pi x.dx$

$$= -\frac{100}{\pi}(\cos \pi x)_0^1 = \frac{-100}{\pi}(\cos \pi - \cos 0)$$

$$= \frac{-100}{\pi}(-1-1) = \frac{200}{\pi}$$

6. The value of the integral $\int_0^{n\pi+t} (|\cos x| + |\sin x|)\, dx$ is :

समाकलन $\int_0^{n\pi+t} (|\cos x| + |\sin x|)\, dx$ का मान है :

(A) 0 (B) $2\pi + \sin t + \cos t$ (C) $\cos 1$ (D) $4n + \sin t - \cos t + 1$

Ans. D

Sol. $I = \int_0^{n\pi} (|\cos x| + |\sin x|).dx + \int_{n\pi}^{n\pi+t} (|\cos x| + |\sin x|)\, dx$

$$I = n\int_0^{\pi} |\cos x| + |\sin x| .dx + \int_0^{t} (\cos x + \sin x)dx$$

$$I = n\left[\int_0^{\frac{\pi}{2}}(\cos x + s\,in\,x)\,dx + \int_{\frac{\pi}{2}}^{\pi}(\sin x - \cos x)\,dx\right] + \int_0^t(\cos x + \sin x)dx$$

$$I = 4n + (\sin x - \cos x)_o^t$$

$$= 4n + (\sin t - \cos t) + (\sin 0 - \cos 0)$$

$$= 4n + \sin t - \cos t - 1$$

7. The integral, $\int_{\pi/4}^{5\pi/4}(|\cos t|\sin t + |\sin t|\cos t)\,dt$ has the value equal to :

$\int_{\pi/4}^{5\pi/4}(|\cos t|\sin t + |\sin t|\cos t)\,dt$ का मान है :

(A) 0 (B) 1/2 (C) $1/\sqrt{2}$ (D) 1

Ans. A

Sol. $I = \int_{\pi/4}^{\pi/2} 2\sin t\cos t\,dt + \underbrace{\int_{\pi/2}^{\pi}(-\sin t\cos t) + (\sin t\cos t)dt}_{\text{zero}} + \int_{\pi}^{5\pi/4} -2\sin t\cos t\,dt$

$$= \int_{\pi/4}^{\pi/2}\sin 2t\,dt - \int_{\pi}^{5\pi/4}\sin 2t\,dt$$

$$= 0$$

8. If $f(x) = \int_{x^2}^{x^3}\log t\,dt\,(x > 0)$, then $f'(x)$ is equal to :

यदि $f(x) = \int_{x^2}^{x^3}\log t\,dt\,(x > 0)$, तो $f'(x)$ बराबर है :

(A) $(4x^2 - 9x)\log x$ (B) $(9x^2 + 4x)\log x$ (C) $(9x^2 - 4x)\log x$ (D) $(x^2 + x)\log x$

Ans. C

Sol. $f'(x) = 3x^2\log(x^3) - 2x\log(x^2)$

$= \left(9x^2 - 4x\right)\log x$

9. The derivative of $F(x) = \int_{x^2}^{x^3} \frac{1}{\log t}\,dt\ (x > 0)$ is :

$F(x) = \int_{x^2}^{x^3} \frac{1}{\log t}\,dt\ (x > 0)$ का अवकलज है :

(A) $\frac{1}{3\log x} - \frac{1}{2\log x}$ (B) $\frac{1}{3\log x}$ (C) $\frac{3x^2}{3\log x}$ (D) $(\log x)^{-1}.x\,(x-1)$

Ans. D

Sol. $F'(x) = \frac{3x^2}{3\log x} - \frac{2x}{2\log x} = \frac{\left(x^2 - x\right)}{\log x} = \left(\log x\right)^{-1} x\left(x-1\right)$

10. $\lim_{n\to\infty}\left[\frac{1^2}{1^3+n^3} + \frac{2^2}{2^3+n^3} + ... + \frac{r^2}{r^3+n^3} + ... + \frac{1}{2n}\right]$ is :

(A) $(1/2)\log 3$ (B) $(1/3)\log 2$ (C) $3\log 2$ (D) $(1/2)\log 2$

Ans. B

Sol. $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^2}{r^3+n^3} = \lim_{n\to\infty}\sum_{r=1}^{n}\frac{\left(\frac{r^2}{n^2}\right)}{n\left(1+\frac{r^3}{n^3}\right)}$

$= \frac{1}{3}\int_0^1 \frac{3x^2}{1+x^3}.dx = \frac{1}{3}\left(\log\left|1+x^3\right|\right)_0^1$

$= \frac{1}{3}\log 2$

11. $\lim_{n\to\infty}\frac{1^{99} + 2^{99} + ... + n^{99}}{n^{100}}$ is :

(A) $\frac{99}{100}$ (B) $\frac{1}{100}$ (C) $\frac{1}{99}$ (D) $\frac{1}{101}$

Ans. B

Sol. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{r^{99}}{n^{100}}=\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\left(\frac{r}{n}\right)^{99}\times\frac{1}{n}$

$=\int\limits_0^1 x^{99}.dx=\left(\frac{x^{100}}{100}\right)_0^1=\frac{1}{100}$

12. $\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=1}^{2n}\frac{r}{\sqrt{n^2+r^2}}$ is :

(A) $-1+\sqrt{2}$ (B) $-1+\sqrt{5}$ (C) $1+\sqrt{5}$ (D) $1+\sqrt{2}$

Ans. B

Sol. $\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=1}^{2n}\frac{r}{\sqrt{n^2+r^2}}=\int\limits_0^2\frac{x}{\sqrt{1+x^2}}dx$

let $1+x^2=t$ $\Rightarrow xdx=t.dt$

$\int\limits_1^{\sqrt{5}}\frac{t.dt}{t}=\sqrt{5}-1$

13. $\lim\limits_{n\to\infty}\left[\frac{1}{2n}+\frac{1}{\sqrt{4n^2-1}}+\frac{1}{\sqrt{4n^2-4}}+...+\frac{1}{\sqrt{3n^2+2n-1}}\right]$ is :

(A) $\pi/4$ (B) $\pi/3$ (C) $\pi/2$ (D) $\pi/6$

Ans. D

Sol. $\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n-1}\frac{1}{\sqrt{4n^2-(r)^2}}=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n-1}\frac{1}{n}\sqrt{\frac{1}{4-\left(\frac{r}{n}\right)^2}}$

$\int\limits_0^1\frac{dx}{\sqrt{4-x^2}}=\left(\sin^{-1}\left(\frac{x}{2}\right)\right)_0^1=\sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{6}$

14. $\lim\limits_{n\to\infty}\left(\frac{n!}{n^n}\right)^{1/n}$ is equal to :

(A) e (B) 1/e (C) $\sqrt{e}$ (D) None of these

$\lim\limits_{n\to\infty}\left(\frac{n!}{n^n}\right)^{1/n}$ बराबर है :

(A) e (B) 1/e (C) $\sqrt{e}$ (D) इनमें से कोई नही

Ans. B

Sol. $\lim\limits_{n\to\infty}\left(\dfrac{1\times2\times3\times.....n}{n\times n\times n\times.....\times n}\right)^{1/n}=\lim\limits_{n\to\infty}\prod\limits_{r=1}^{n}\left(\dfrac{r}{n}\right)^{1/n}=A\left(let\right)$

Taking ℓn both side

$$\ell nA=\frac{1}{n}\lim_{n\to\infty}\sum_{r=1}^{n}\ell n\left(\frac{r}{n}\right)$$

$$\ell nA=\int_0^1\ell nx.dx=\left(x\ell nx-x\right)_0^1$$

$$\ell nA=\left(1.\ell nx1-1\right)-0$$

$$\ell nA=-1$$

$$A=\frac{1}{e}$$

15. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{r=4n}\dfrac{\sqrt{n}}{\sqrt{r}\left(3\sqrt{r}+4\sqrt{n}\right)^2}$

(A) $\dfrac{1}{35}$ (B) $\dfrac{1}{14}$ (C) $\dfrac{1}{10}$ (D) $\dfrac{1}{5}$

Ans. C

Sol. $T_r=\dfrac{1}{\sqrt{\dfrac{r}{n}}\cdot n\left(3\sqrt{\dfrac{r}{n}}+4\right)^2}$

$$S=\frac{1}{n}\sum_{r=1}^{4n}\frac{1}{\left(3\sqrt{\frac{r}{n}}+4\right)^2\cdot\sqrt{\frac{r}{n}}}=\int_0^4\frac{dx}{\sqrt{x}(3\sqrt{x}+4)^2}$$

let $3\sqrt{x}+4=t$

$$\Rightarrow\ \frac{3}{2}\frac{1}{\sqrt{x}}dx=dt$$

$$=\frac{2}{3}\int_4^{10}\frac{dt}{t^2}=\frac{2}{3}\left[\frac{1}{t}\right]_{10}^{4}=\frac{2}{3}\left[\frac{1}{4}-\frac{1}{10}\right]=\frac{2}{3}\cdot\frac{6}{40}=\frac{1}{10}$$

16. Let $S_n = \frac{n}{(n+1)(n+2)} + \frac{n}{(n+2)(n+4)} + \frac{n}{(n+3)(n+6)} + ... + \frac{1}{6n}$ then $\lim\limits_{n\to\infty} S_n$ has the value equal to:

माना $S_n = \frac{n}{(n+1)(n+2)} + \frac{n}{(n+2)(n+4)} + \frac{n}{(n+3)(n+6)} + ... + \frac{1}{6n}$ तो $\lim\limits_{n\to\infty} S_n$ का मान होगा :

(A) $\ell n\frac{3}{2}$ (B) $\ell n\frac{9}{2}$ (C) $2\,\ell n\frac{3}{2}$ (D) $\frac{1}{2}\,\ell n\frac{3}{2}$

Ans. A

Sol. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{n}{(n+r)(n+2r)} = \lim\limits_{n\to\infty}\frac{n}{\left(1+\frac{r}{n}\right)\left(1+2\frac{r}{n}\right).n^2}$

$$\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n\left(1+\frac{r}{n}\right)\left(1+2\frac{r}{n}\right)} = \int_0^1\frac{dx}{(1+x)(1+2x)} = \left(-\ell n|1+x| + \frac{2}{2}\ell n|2x+1|\right)_0^1$$

$$= -\ell n2 + \ell n3 = \ell n\frac{3}{2}$$

17. Let $I_1 = \int_1^2 \frac{dx}{\sqrt{1+x^2}}$ and $I_2 = \int_1^2 \frac{dx}{x}$, then :

माना $I_1 = \int_1^2 \frac{dx}{\sqrt{1+x^2}}$ and $I_2 = \int_1^2 \frac{dx}{x}$, then :

(A) $I_1 = I_2$ (B) $I_2 > I_1$ (C) $I_1 > I_2$ (D) $I_1 > 2\,I_2$

Ans. B

Sol. $I_1 = \left(\tan^{-1}x\right)_1^2 = \tan^{-1}2 - \frac{\pi}{4}$

$I_2 = \ell n2 = 0.693$

Cleary $I_2 > I_1$

18. Let $I_1 = \int_0^{\pi/2} e^{-x^2}\sin(x)dx$; $I_2 = \int_0^{\pi/2} e^{-x^2}dx$; $I_3 = \int_0^{\pi/2} e^{-x^2}(1+x)\,dx$ and consider the statements

I $I_1 < I_2$

II $I_2 < I_3$

III $I_1 = I_3$

Which of the following is(are) true?

(A) I only (B) II only

(C) Neither I nor II nor III (D) Both I and II

माना $I_1 = \int_0^{\pi/2} e^{-x^2} \sin(x)dx$; $I_2 = \int_0^{\pi/2} e^{-x^2} dx$; $I_3 = \int_0^{\pi/2} e^{-x^2} (1+x)\,dx$

तथा माना कथन

I $I_1 < I_2$

II $I_2 < I_3$

III $I_1 = I_3$

तो निम्न में से कौनसा कथन सत्य है?

(A) केवल I (B) केवल II

(C) न तो I ना II और न हीं III (D) I तथा II दोनों

Ans. D

Sol. since $0 < \sin x < 1$ and $1 + x > 1$ in $(0, \pi/2)$

hence $I_3 > I_2 > I_1$

19. $\int_0^{\pi/2} \sin^7 x \cos x\, dx$ is :

(A) 1/7 (B) 1/8 (C) $\pi/16$ (D) $\pi/14$

Ans. B

Sol. $\int_0^{\pi/2} \sin^7 x \cos x\, dx = 1 \times \frac{6 \times 4 \times 2}{8 \times 6 \times 4 \times 2} = \frac{1}{8}$

20. $\int_0^{\pi/2} \sin^2\theta \cos^6\theta\, d\theta$ is :

(A) $-\pi/16$ (B) $\pi/16$ (C) $5\pi/256$ (D) $-5\pi/256$

Ans. C

Sol. $\int_0^{\pi/2} \sin^2\theta \cos^6\theta\, d\theta = \frac{\pi}{2} \times \frac{5 \times 3 \times 1}{8 \times 6 \times 4 \times 2} = \frac{5\pi}{256}$

21. $\int_0^3 \sqrt{\frac{x^3}{3-x}}\, dx$ is :

(A) $3\pi/16$ (B) $27\pi/8$ (C) $3\pi/32$ (D) $9\pi/8$

Ans. B

Sol. Let $x = 3\sin^2\theta$

$$\int_0^3 \sqrt{\frac{x^3}{3-x}}\,dx = \int_0^{\pi/2} \sqrt{\frac{27.\sin^6\theta}{3.\cos^2\theta}} \times 6.\sin\theta.\cos\theta.d\theta$$

$$= 18\int_0^{\pi/2} \sin^4\theta.d\theta = 18 \times \frac{\pi}{2} \times \frac{3\times 1}{4\times 2}$$

$$= \frac{27\pi}{8}$$

22. $\int_0^{\pi/2} \sin^2 x \cos^2 x\,dx$

(A) $\pi//16$ (B) $\pi/8$ (C) 1/8 (D)1/16

Ans. A

Sol. $\int_0^{\pi/2} \sin^2 x \cos^2 x\,dx = \frac{\pi}{2}.\frac{1}{4\times 2} = \frac{\pi}{16}$

23. $\int_0^{\pi/4} \sin^4 2x dx$

(A) $2\pi/32$ (B) $3\pi/32$ (C) $\pi/32$ (D) $3\pi/16$

Ans. B

Sol. $I = \int_0^{\pi/4} \sin^4 2x dx$

let $2x = t$

$$= \frac{1}{2}\int_0^{\pi/2} \sin^4 t.dt = \frac{1}{2} \times \frac{\pi}{2} \times \frac{3\times 1}{4\times 2} = \frac{3\pi}{32}$$

1. If the value of the definite integral $\int_{\pi/6}^{\pi/4} \frac{1+\cot x}{e^x \sin x}dx$, is equal to $ae^{-\pi/6} + be^{-\pi/4}$ then $(a+b)$ equals :

यदि निश्चित समाकलन $\int_{\pi/6}^{\pi/4} \frac{1+\cot x}{e^x \sin x}dx$ का मान $ae^{-\pi/6} + be^{-\pi/4}$ तो $(a+b)$ का मान होगा :

(A) $2-\sqrt{2}$ (B) $2+\sqrt{2}$ (C) $2\sqrt{2}-2$ (D) $2\sqrt{3}-\sqrt{2}$

Ans. A

Sol. $\int_{\pi/6}^{\pi/4} e^{-x}\left(\text{cosec}\, x + \cot x \,\text{cosec}\, x\right)dx$; put $-x = t$; $dx = -dt$

$$-\int_{-\pi/6}^{-\pi/4} e^{t}\left(-\text{cosec}(t) + \cot(t)\cdot\text{cosec}(t)\right)dt = \int_{-\pi/6}^{-\pi/4} e^{t}\left(\text{cosec}(t) - \cot(t)\cdot\text{cosec}(t)\right)dt$$

$$= e^t \,\text{cosec}(t)\Big]_{-\pi/6}^{-\pi/4} = -\sqrt{2}e^{-\frac{\pi}{4}} + 2e^{-\frac{\pi}{6}} \Rightarrow 2e^{-\frac{\pi}{6}} - \sqrt{2}e^{-\frac{\pi}{4}} \Rightarrow a+b = 2-\sqrt{2}$$

2. Number of solution of the equation $\frac{d}{dx}\int_{\cos x}^{\sin x} \frac{dt}{1-t^2} = 2\sqrt{2}$ in $[0, \pi]$ is :

समीकरण $\frac{d}{dx}\int_{\cos x}^{\sin x} \frac{dt}{1-t^2} = 2\sqrt{2}$ का अन्तराल $[0, \pi]$ में हलों कि संख्या होगी :

(A) 4 (B) 3 (C) 2 (D) 1

Ans. C

Sol. $\frac{d}{dx}\int_{\cos x}^{\sin x} \frac{dt}{1-t^2} = \sec x + \text{cosec}\, x = 2\sqrt{2}$; $x \in [0\ \pi]$

$\Rightarrow x = \frac{\pi}{4}$ and $\frac{11\pi}{12}$

3. Let $f(x) = \frac{\sin x}{x}$, then $\int_0^{\pi/2} f(x)\, f\left(\frac{\pi}{2}-x\right)dx =$

माना $f(x)=\frac{\sin x}{x}$ है तो $\int_0^{\pi/2} f(x)\, f\left(\frac{\pi}{2}-x\right)dx =$

(A) $\frac{2}{\pi}\int_0^{\pi} f(x)\,dx$ (B) $\int_0^{\pi} f(x)\,dx$ (C) $\pi\int_0^{\pi} f(x)\,dx$ (D) $\frac{1}{\pi}\int_0^{\pi} f(x)\,dx$

Ans. A

Sol. $I=\int_0^{\pi/2}\frac{\sin x\cos x}{x\left(\frac{\pi}{2}-x\right)}dx = \int_0^{\pi/2}\frac{\sin 2x}{x(\pi-2x)}dx$

Let $2x=t$

$$I=\int_0^{\pi}\frac{\sin t}{t(\pi-t)}dt = \frac{1}{\pi}\int_0^{\pi}\left(\frac{\sin t}{t}+\frac{\sin t}{(\pi-t)}\right)dt = \frac{1}{\pi}\int_0^{\pi}\frac{\sin t}{t}dt + \frac{1}{\pi}\int_0^{\pi}\frac{\sin t}{\pi-t}dt$$

Apply King Property

$$=\frac{1}{\pi}\int_0^{\pi}\frac{\sin t}{t}dt+\frac{1}{\pi}\int_0^{\pi}\frac{\sin t}{t}dt = \frac{2}{\pi}\int_0^{\pi}\frac{\sin t}{t}dt$$

4. Let $u=\int_0^1\frac{ln(x+1)}{x^2+1}dx$ and $v=\int_0^{\pi/2} ln(\sin 2x)\,dx$ then :

माना $u=\int_0^1\frac{ln(x+1)}{x^2+1}dx$ तथा $v=\int_0^{\pi/2} ln(\sin 2x)\,dx$ तो :

(A) $u=4v$ (B) $4u+v=0$ (C) $u+4v=0$ (D) $2u+v=0$

Ans. B

Sol. $u=\int_0^1\frac{ln(x+1)}{x^2+1}dx$ let $x=\tan\theta$

$$=\int_0^{\pi/4} ln(1+\tan\theta)\,d\theta = \int_0^{\pi/4} ln\left(1+\frac{1-\tan\theta}{1+\tan\theta}\right)d\theta = \int_0^{\pi/4} ln\frac{2}{1+\tan\theta}d\theta = \frac{\pi}{4}\,ln\,2-u$$

$\therefore\quad u=\frac{\pi}{8}\,ln\,2 \quad\Rightarrow\quad 4u=\frac{\pi}{2}\,ln\,2$ ....(1)

Now $v=\int_0^{\pi/2} ln(\sin 2x)\,dx$ (let $2x=t$) ; $v=\frac{1}{2}\int_0^{\pi} ln(\sin t)\,dt = \int_0^{\pi/2} ln(\sin t)\,dx$

$$v = -\frac{\pi}{2} \ln 2 \qquad ....(2)$$

by (1) + (2)

$\Rightarrow \; 4u + v = 0$

5. Let $a_n = \int_0^{\pi/2} (1-\sin t)^n \sin 2t \, dt$ then $\lim\limits_{n\to\infty} \sum\limits_{n=1}^{n} \frac{a_n}{n}$ is equal to :

माना $a_n = \int_0^{\pi/2} (1-\sin t)^n \sin 2t \, dt$ तो $\lim\limits_{n\to\infty} \sum\limits_{n=1}^{n} \frac{a_n}{n}$ का मान होगा :

(A) 1/2 (B) 1 (C) 4/3 (D) 3/2

Ans. A

Sol. $a_n = \int_0^{\pi/2} (1-\sin t)^n \sin 2t \, dt$

Let $1 - \sin t = u \qquad \Rightarrow \; -\cos t \, dt = du$

$$= 2\int_0^1 u^n (1-u)\, du = 2\left(\int_0^1 u^n \, du - \int_0^1 u^{n+1} \, du\right) = 2\left(\frac{1}{n+1} - \frac{1}{n+2}\right)$$

hence $$\frac{a_n}{n} = 2\left(\frac{1}{n(n+1)} - \frac{1}{n(n+2)}\right)$$

$$\lim_{n\to\infty} \sum_{n=1}^{n} \frac{a_n}{n} = 2\left(\sum\left(\frac{1}{n} - \frac{1}{n+1}\right) - \frac{1}{2}\sum\left(\frac{1}{n} - \frac{1}{n+2}\right)\right) = 2\sum_{n=1}^{n}\left(\frac{1}{n} - \frac{1}{n+1}\right) - \sum_{n=1}^{n}\left(\frac{1}{n} - \frac{1}{n+2}\right)$$

$$= 2(1) - \left[\left(1-\frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + ...\right] = 2 - \frac{3}{2} = \frac{1}{2}$$

6. Let $G(x) = \int e^x \left(\int_0^x f(t)\, dt + f(x)\right) dx$ where $f(x)$ is continuous on R. If $f(0) = 1, G(0) = 0$ then $G'(0)$ equals:

माना $G(x) = \int e^x \left(\int_0^x f(t)\, dt + f(x)\right) dx$ जहाँ $f(x)$, R में सतत् है। यदि $f(0) = 1, G(0) = 0$ तो $G'(0)$ का मान होगा :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. A

Sol. $G'(x)= e^x\left(\int_0^x f(t)\,dt + f(x)\right)$

$G'(0)= f(0)=1$

7. $\int_{\frac{1}{2}}^{3\frac{1}{2}}\left\{\frac{1}{2}(|x-3|+|1-x|-4)\right\}dx$ equals : (Where {} denotes the fractional part function)

$\int_{\frac{1}{2}}^{3\frac{1}{2}}\left\{\frac{1}{2}(|x-3|+|1-x|-4)\right\}dx$ का मान होगा : (जहाँ {} भिन्नात्मक फलन है)

(A) $-\frac{3}{2}$ (B) $\frac{9}{8}$ (C) $\frac{1}{4}$ (D) $\frac{3}{2}$

Ans. C

Sol. $\int_{\frac{1}{2}}^{3\frac{1}{2}}\left\{\frac{1}{2}(|x-3|+|1-x|-4)\right\}dx$

$\int_{\frac{1}{2}}^{1}\left\{\frac{1}{2}(3-x+1-x-4)\right\}dx + \int_{1}^{3}\left\{\frac{1}{2}(3-x+x-1-4)\right\}dx + \int_{3}^{\frac{7}{2}}\left\{\frac{1}{2}(x-3+x-1-4)\right\}dx$

$=\int_{\frac{1}{2}}^{1}\{-x\}dx+\int_{3}^{\frac{7}{2}}\{x-4\}dx = \int_{\frac{1}{2}}^{1}(1-x)dx+\int_{3}^{\frac{7}{2}}\{x\}dx = \int_{\frac{1}{2}}^{1}(1-x)dx+\int_{3}^{\frac{7}{2}}(x-3)dx$

$=\frac{1}{2}-\frac{1}{2}\left(\frac{3}{4}\right)+\frac{1}{2}\left(\frac{49}{4}-9\right)-3\left(\frac{1}{2}\right)$

$=\frac{1}{4}$

8. $\lim\limits_{t\to 0}\int\limits_0^{2\pi}\frac{|\sin(x+t)-\sin x|}{|t|}dx =$

(A) 0 (B) 1 (C) 2 (D) 4

Ans. D

Sol. $I=\lim\limits_{t\to 0}\int\limits_0^{2\pi}\left|\frac{\sin(x+t)-\sin x}{t}\right|dx=\int\limits_0^{2\pi}\left(\lim\limits_{t\to 0}\left|\frac{2\cos\left(x+\frac{t}{2}\right)\sin\frac{t}{2}}{t}\right|\right)dx$

$=\int\limits_0^{2\pi}\lim\limits_{t\to 0}\left|\cos\left(x+\frac{t}{2}\right)\frac{\sin\frac{t}{2}}{\frac{t}{2}}\right|=\int\limits_0^{2\pi}|\cos x|dx=4$ Ans.

9. Suppose that F (x) is an antiderivative of $f(x)=\frac{\sin x}{x}$, $x>0$ then $\int\limits_1^3\frac{\sin 2x}{x}dx$ can be expressed as :

माना F (x), फलन $f(x)=\frac{\sin x}{x}$ का, $x>0$ के लिए, समाकलन है, तो $\int\limits_1^3\frac{\sin 2x}{x}dx$ को निम्न प्रकार लिखा जा सकता है :

(A) F (6) – F (2) (B) $\frac{1}{2}$ (F (6) – F (2)) (C) $\frac{1}{2}$ (F (3) – F (1)) (D) 2(F (6) – F (2))

Ans. A

Sol. Let $I=\int\limits_1^3\frac{\sin 2x}{x}dx$, also let $2x=t, dx=\frac{dt}{2}\Rightarrow I=\frac{2}{2}\int\limits_2^6\frac{\sin t}{t}dt=\int\limits_2^6\frac{\sin t}{t}dt$

But $\int\frac{\sin x}{x}dx=F(x)\Rightarrow\int\limits_2^6\frac{\sin t}{t}dt=F(6)-F(2)$

10. The value of the definite integral $\int\limits_0^{2\left(2+\sqrt{3}\right)}\frac{16}{(4+x^2)^2}dx$ is equal to :

$\int\limits_0^{2\left(2+\sqrt{3}\right)}\frac{16}{(4+x^2)^2}dx$ का मान होगा :

(A) $\frac{5+3\pi}{12}$ (B) $\frac{3+5\pi}{6}$ (C) $\frac{5+3\pi}{6}$ (D) $\frac{3+5\pi}{12}$

Ans. D

Sol. Let $I=\int_0^{2(2+\sqrt{3})}\frac{16}{(4+x^2)^2}dx$

put $x=2\tan\theta \quad \Rightarrow \quad dx=2\sec^2\theta\, d\theta$

$\therefore \quad I=\int_0^{\frac{5\pi}{12}}\frac{16\cdot 2\sec^2\theta}{16\sec^2\theta\cdot\sec^2\theta}d\theta=\int_0^{\frac{5\pi}{12}}2\cos^2\theta\, d\theta=\int_0^{\frac{5\pi}{12}}(1+\cos 2\theta)\,d\theta$

$=\theta+\frac{1}{2}\sin 2\theta\Big|_0^{\frac{5\pi}{12}}=\frac{5\pi}{12}+\frac{1}{4}=\frac{3+5\pi}{12}$

11. The value of the definite integral $\int_0^{\pi/3}\ln(1+\sqrt{3}\tan x)\,dx$ equals :

$\int_0^{\pi/3}\ln(1+\sqrt{3}\tan x)\,dx$ का मान होगा :

(A) $\frac{\pi}{3}\ln 2$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi^2}{6}\ln 2$ (D) $\frac{\pi}{2}\ln 2$

Ans. A

Sol. $I=\int_0^{\pi/3}\ln\left(\frac{\cos x+\sqrt{3}\sin x}{\cos x}\right)dx=\int_0^{\pi/3}\underbrace{\ln 2\cos\left(x-\frac{\pi}{3}\right)}_{\text{use King}}dx-\int_0^{\pi/3}\ln\cos x\,dx$

$=\int_0^{\pi/3}\ln(2\cos x)\,dx-\int_0^{\pi/3}\ln(\cos x)\,dx=\int_0^{\pi/3}\ln 2\,dx+\int_0^{\pi/3}\ln(\cos x)\,dx-\int_0^{\pi/3}\ln(\cos x)\,dx=\frac{\pi}{3}\ln 2$

Alternatively: $I=\int_0^{\pi/3}\ln\left(1+\sqrt{3}\tan\left(\frac{\pi}{3}-x\right)\right)dx=\int_0^{\pi/3}\ln\left(1+\sqrt{3}\left(\frac{\sqrt{3}-\tan x}{1+\sqrt{3}\tan x}\right)\right)dx$

$=\int_0^{\pi/3}\ln\left(\frac{1+\sqrt{3}\tan x+\sqrt{3}-\sqrt{3}\tan x}{1+\sqrt{3}\tan x}\right)dx=\int_0^{\pi/3}\ln\left(\frac{4}{(1+\sqrt{3}\tan x)}\right)dx$

$$2I = \left(\frac{\pi}{3}\right) \cdot 2\, ln\, 2 \quad \Rightarrow \quad I = \frac{\pi}{3} ln\, 2$$

12. $\int_0^\infty f\left(x + \frac{1}{x}\right) \cdot \frac{\ell n\, x}{x} dx$:

(A) is equal to zero (B) is equal to one (C) is equal to $\frac{1}{2}$ (D) can not be evaluated

$\int_0^\infty f\left(x + \frac{1}{x}\right) \cdot \frac{\ell n\, x}{x} dx$:

(A) शून्य (B) एक (C) $\frac{1}{2}$ (D) ज्ञात नहीं किया जा सकता

Ans. A

Sol. put $ln\, x = t \quad \Rightarrow \quad x = e^t \quad \Rightarrow \quad dx = e^t\, dt$

$$I = \int_{-\infty}^{\infty} f(e^t + e^{-t}) \frac{t}{e^t} e^t\, dt = \int_{-\infty}^{\infty} f(e^t + e^{-t}) t\, dt = 0 \text{ (as the function is odd)}$$

Alternatively-1: put $x = \tan\theta$; $\int_0^{\pi/2} f(\tan\theta + \frac{1}{\tan\theta}) \frac{ln \tan\theta}{\tan\theta} \cdot \sec^2\theta\, d\theta = \int_0^{\pi/2} f(\tan\theta + \frac{1}{\tan\theta}) \frac{ln \tan\theta}{\sin\theta \cos\theta} d\theta$

Alternatively-2: put $x = 1/t \quad \Rightarrow \quad I = -I \quad \Rightarrow \quad 2I = 0 \quad \Rightarrow \quad I = 0$

13. Positive value of 'a' so that the definite integral $\int_a^{a^2} \frac{dx}{x + \sqrt{x}}$ achieves the smallest value is :

'a' का धनात्मक मान जिसके लिए $\int_a^{a^2} \frac{dx}{x + \sqrt{x}}$ का मान न्यूनतम हैं, होगा :

(A) $\tan^2\left(\frac{\pi}{8}\right)$ (B) $\tan^2\left(\frac{3\pi}{8}\right)$ (C) $\tan^2\left(\frac{\pi}{12}\right)$ (D) 0

Ans. A

Sol. Let $f(a) = \int_a^{a^2} \frac{dx}{x + \sqrt{x}}$

$$f'(a) = \frac{2a}{a + a^2} - \frac{1}{a + \sqrt{a}} = 0 \quad \Rightarrow \quad 2a^2 + 2a\sqrt{a} = a^2 + a \quad \Rightarrow \quad a^2 + 2a\sqrt{a} - a = 0$$

$a + 2\sqrt{a} - 1 = 0 \quad \Rightarrow \quad (\sqrt{a}+1)^2 = 2 \qquad \Rightarrow \quad \sqrt{a} = \sqrt{2} - 1 \qquad = \tan\frac{\pi}{8}$

$a = (\sqrt{2}-1)^2 = \tan^2\left(\frac{\pi}{8}\right)$ Ans.

14. The value of the definite integral $\int_e^{e^{2010}} \frac{1}{x}\left(1+\frac{1-\ell n\, x}{\ell n\, x\ \ell n\left(\frac{x}{\ell n\, x}\right)}\right)dx$, equals :

का मान होगा :

(A) $2009 - \ln(2010 - \ln 2010)$ (B) $2010 - \ln(2009 - \ln 2009)$

(C) $2009 - \ln(2010 - \ln 2009)$ (D) $2010 - \ln(2010 - \ln 2010)$

Ans. A

Sol. $I = \int_e^{e^{2010}} \frac{1}{x}\left(1+\frac{1-\ln x}{\ln x\ \ln\left(\frac{x}{\ln x}\right)}\right)dx$

Let $\ell nx = t$, we get

$$\int_1^{2010}\left(1+\frac{1-t}{t\left(t-(\ell nt)\right)}\right)dt = 2009 + \int_1^{2010-\ell n2010}\frac{-1}{u}du$$

let $u = t - \ell nt$

$= 2009 - \ell n(2010 - \ell n\, 2010)$

15. For $U_n = \int_0^1 x^n\,(2-x)^n\,dx$; $V_n = \int_0^1 x^n\,(1-x)^n\,dx$ $n \in N$, which of the following statement(s) is/are true :

$U_n = \int_0^1 x^n\,(2-x)^n\,dx$; $V_n = \int_0^1 x^n\,(1-x)^n\,dx$ $n \in N$, निम्न में से कौनसा कथन सत्य है :

(A) $U_n = 2^n V_n$ (B) $U_n = 2^{-n} V_n$ (C) $U_n = 2^{2n} V_n$ (D) $U_n = 2^{-2n} V_n$

Ans. C

Sol. Given $U_n = \int_0^1 x^n .(2-x)^n \, dx$; $V_n = \int_0^1 x^n .(1-x)^n \, dx$

in U_n put $x = 2t$ $\Rightarrow dx = 2dt$

$\therefore \quad U_n = 2\int_0^{1/2} 2^n .t^n \, 2^n (1-t)^n \, dt$ ....(1)

Now $V_n = 2\int_0^{1/2} x^n \, (1-x)^n \, dx$ (Using Queen) .....(2)

From (1) and (2)

$U_n = 2^{2n}. V_n \Rightarrow$ (C)

16. Let $S(x) = \int_{x^2}^{x^3} \ln t \, dt \; (x > 0)$ and $H(x) = \frac{S'(x)}{x}$. Then H(x) is :

(A) continuous but not derivable in its domain (B) derivable and continuous in its domain
(C) neither derivable nor continuous in its domain (D) derivable but not continuous in its domain

माना $S(x) = \int_{x^2}^{x^3} \ln t \, dt \; (x > 0)$ तथा $H(x) = \frac{S'(x)}{x}$ तो H(x) का मान है :

(A) प्रान्त में सतत् परन्तु अवकलनीय नहीं है (B) प्रान्त में सतत् तथा अवकलनीय है
(C) न तो सतत् और नहीं अवकलनीय है (D) प्रान्त में अवकलनीय परन्तु सतत् नहीं है

Ans. B

Sol. $S(x) = \int_{x^2}^{x^3} \ell n t dt$

Apply Leibnitz rule

$S'(x) = \ln x^3 . 3x^2 - \ln x^2 . 2x = 9x^2 \ln x - 4x \ln x$

$= x \, l n \, x \, (9x - 4)$

$\Rightarrow \frac{S'(x)}{x} = \ln x \, (9x - 4)$

Now it is obvious that $\frac{S'(x)}{x}$ is continuous and derivable in its domain.

17. Let $J = \int_0^\infty \frac{\ln x}{1+x^3}dx$ and $K = \int_0^\infty \frac{x\ln x}{1+x^3}dx$ then :

माना $J = \int_0^\infty \frac{\ln x}{1+x^3}dx$ तथा $K = \int_0^\infty \frac{x\ln x}{1+x^3}dx$ तो :

(A) $J + K = 0$ (B) $J - K = 0$ (C) $J + K < 0$ (D) $J + K > 0$

Ans. A

Sol. $J + K = \int_0^\infty \frac{(x+1)\ln x}{1+x^3}dx = \int_0^\infty \frac{\ln x\,dx}{x^2-x+1}dx$

let $x = \frac{1}{t}$

$\Rightarrow J + K = -(J + K) \Rightarrow J + K = 0$

Note: $J = -\frac{2\pi^2}{27}$ and $K = \frac{2\pi^2}{27}$ (this note is not for display to students)

18. Consider the polynomial $f(x) = ax^2 + bx + c$. If $f(0) = 0, f(2) = 2$ then the minimum value of $\int_0^2 |f'(x)|dx$ equals :

माना बहुपद $f(x) = ax^2 + bx + c$ यदि $f(0) = 0, f(2) = 2$ तो $\int_0^2 |f'(x)|dx$ का मान होगा :

(A) 0 (B) 1 (C) 2 (D) None

Ans. C

Sol. $\int_0^2 |f'(x)|dx \geq \left|\int_0^2 f'(x)dx\right|$

$\int_0^2 |f'(x)|dx \geq |f(2) - f(0)| = 2$

19. Let $f(x)$ be a continuous function on $[0,4]$ satisfying $f(x)\,f(4-x) = 1$. The value of the definite integral $\int_0^4 \frac{1}{1+f(x)}\,dx$ equals :

माना $f(x)$ अन्तराल $[0, 4]$ में सतत् है तथा $f(x)\ f(4-x) = 1$ को सन्तुष्ट करता है, तो $\int_0^4 \frac{1}{1+f(x)}\,dx$ का मान होगा :

(A) 0 (B) 1 (C) 2 (D) 4

Ans. C

Sol. Let $I = \int_0^4 \frac{1}{1+f(x)}\,dx$ .... (1)

Now on applying king property in (1), we get

$I = \int_0^4 \frac{1}{1+f(4-x)}\,dx$, put $f(4-x) = \frac{1}{f(x)}$ $\Rightarrow I = \int_0^4 \frac{f(x)}{f(x)+1}\,dx$ ....(2)

Now (1) + (2) $\Rightarrow 2I = \int_0^4 dx$ $\Rightarrow I = 2$